

Global Facilities - Design & Construction Standard - Chemical - Slurry Distribution System Specification

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1.0 Purpose

This Standard Specification defines the minimum requirements for design, engineering, fabrication, installation, supply, test & commission, certify, validate the performance, train the Owner's personal in the operation and maintenance of systems, and guarantee the pressure/flow/purity/safety performance for the Slurry distribution system. The Slurry Distribution System includes drum/tote/IBC module enclosure, Slurry Transfer Units, Slurry Transfer & Blend Unit, Slurry Dispense Unit, Slurry Blend & Dispense Unit and other associated modules such as day tanks, distribution piping network, VMBs, MHEB (expansion box), JB etc. As process tools are sensitive to required pressure, flow and quality, dispense modules are required to operate within the parameter specifications.

This document is part of Micron's general specification standards developed to meet the process requirements while requirements for safety is duly considered. TD may decide to have a scaled down version of the Slurry distribution system but shall generally follow the design concepts where possible as the consumption and capacity requirement is much smaller and downtime impact is not a major concern as compared to a HVM site.

Note: If any sites practice (or require) higher standards/ specifications/ regulations on compliance/etc., those sites may have to follow their existing (or better) standards/ specifications/ regulations on compliance/ etc. in order to meet site specific requirements, standards and specifications.

2.0 Scope

2.1 Work Scope:

- a) Design and installation of Slurry Distribution Systems in Slurry Transfer and Slurry Dispense room/ area as specified during initial design. Based on requirements, drawings for location of the rooms will be provided.
- b) Interconnecting piping/tubing between Slurry Distribution systems and other system components with required secondary containment.
- c) Hook-up lines from other utilities such as UPW, PCW, ICA, CDA, PN2, UN2, drain, etc. to different system modules and other required components.
- d) Network Distribution tubing, secondary containment, leak detection, and Slurry valve manifold boxes (including MHEB) located in the Sub-fab areas to serve the process tools.
- e) Sump pump stations, power distribution panels and electric wiring.
- f) Control, Instrumentation, dedicated monitoring system SCADA (based on requirement) and Data concentrator. If there is existing control system network, required additions/modifications shall be made.
- g) FAT, SAT, installation inspection, testing, start-up, commissioning, validation and training

2.2 General Scope

This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects. However, based on requirements and applicability with existing design, this document can also be used for Brownfield and other small expansion projects.

3.0 Definitions and Acronyms

- **APV** – New PTFE APV (**A**dvanced **P**erformance **V**essel), aka Modified PTFE
- **ATEX** – **EX**plosive **At**mospheres
- **AOV** – Air Operated Valve
- **BKM** – Best Known Method
- **CDA** – Clean Dry Air.
- **CMP** – Chemical Mechanical Polishing
- **CFD** – Computational Fluid Dynamics
- **STU** – Slurry Transfer Unit
- **STDU** – Slurry Transfer Dispense Unit
- **SBDU** – Slurry Blend & Dispense Unit
- **COHE** – Control of Hazardous Energy

- **EPCM** - Engineering Procurement and Construction Management
 - **ET**- Etch: chemical etched sheet on the back of lining material for bonding to shell
 - **ERT** – Emergency Response Team
 - **EHS** – Environmental, Health & Safety
 - **FAT** – Factory Acceptance Test
 - **FRP** – Fiber-reinforced plastic
 - **FEP** - Fluorinated Ethylene Propylene
 - **FCT** – Facilities Central Team
 - **FIOP** – Field Input Output Panel
 - **FM** – Factory Mutual
 - **FCT** – Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
 - **GN2** – General Nitrogen
 - **GB** – Glass Backing, intermediary sheet bonded between tank liner and shell
 - **GFTT** – Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
 - **GFC** – Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
 - **Global Facilities Central Team** – This is used interchangeably with Innovation & Integration Team
 - **Global Facilities Construction Team** – This is used interchangeably with Global Construction Team
 - **Global Facilities Planning Team** – This is used interchangeably with Global Facilities Line Planning Team
 - **Global Facilities Operations Support Team** – This is used interchangeably with Global Facilities Operations Team
 - **HAZOP** – Hazardous Operation
 - **HAZMAT** – Hazardous Material
 - **HMI** – Human Machine Interface
 - **HDPE** - High Density Polyethylene
 - **HCDS** – Hexachlorodisilane
 - **I&C** – Instrumentation and Control
 - **ICA** – Instrument Control Air
 - **JB** – Junction Box with no valves
 - **LSS** – Life Safety System
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- **LPP** – Low **P**ermeation **P**roperties, two-layer PFA tube
- **MHEB** – Main Header End Box
- **MTBF** – Mean Time Between Failure
- **MTTR** – Mean Time To Repair
- **MV** – Manual Valve
- **N2** – Nitrogen
- **NPW** – Non-Portable Water
- **NFPA** – National Fire Protection Association
- **PCW** – Process Cooling Water
- **PLC** - Programmable Logic Controller
- **PN2** – Process Nitrogen
- **PTFE** - Polytetrafluoroethylene
- **PTFM** – Modified PTFE
- **P&ID** – Plumbing and Instrumentation Diagram
- **PSM** – Process Safety Management
- **PFA** - Perfluoroalkoxy Alkane
- **PP**- Polypropylene
- **PVC**- Polyvinyl Chloride
- **PVDF** - Polyvinylidene Difluoride
- **RE** – Relative Error
- **RFP** – Request for Proposal
- **RRA** – Risk Ranking Analysis
- **RIO** – Remote Input / Output Panel
- **RPM** – Revolution Per Minute
- **SAT** – Site Acceptance Test
- **SDS** – Safety Data Sheet
- **SPOF** – Single Point of Failure
- **SS** – Stainless Steel
- **SOO** – Sequence of Operation
- **TD** – Technology Development
- **TGCM** – Total Gas & Chemical Management
- **TMAH** – Tetramethyl Ammonium Hydroxide
- **UN2** – Utility Nitrogen
- **UPW** – Ultra Pure Water
- **UHP** – Ultra High Purity
- **VMB** – Valve Manifold Box
- **VFD** – Variable Frequency Drive

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead	• Develop and maintain System Standard document • Incorporate Best Known Methods (BKM), lessons learned, and emerging technologies in the System Standard • Perform periodic document review and update all relevant components in System Standard • Support the Site teams by providing knowledge and technical expertise
FCT Lead	• Review and provide technical feedback on System Standard • Update BKMs and Develop relevant BKMs • Support the Site teams by providing knowledge and technical expertise in operation and maintenance
Site Project Construction Team	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard • Consult with Global/Site PSM on the Relative Ranking Analysis (RRA) and ensure the RRA is less than 30. Otherwise carry out mitigation till RRA comes below 30.
Site Facilities Team	• Use System Standard as technical reference for new system designs
System Supplier or other contractors	• Comply or exceed the requirements mentioned in relevant Micron standards
Global/Site PSM	• Provide guidance for Relative Ranking Analysis rating for systems • Provide consultancy on the required mitigation

5.0 Safety

5.1 Project Safety Requirements

- a) The EPCM contractor or system vendor's written EHS program shall contain all the necessary elements to administer the program in accordance with all applicable EHS Laws, Micron corporate and site-specific safety policies as well as local regulatory requirements. Compliance with the below mentioned link to Micron safety standards shall also be adhered.
- i. [Global EHS - Construction Contractor EHS Minimum Performance Requirements](#)
 - ii. [Global EHS - Construction EHS Management Strategy](#)
 - iii. [Global EHS - Contractor and Vendor EHS Standard](#)
 - iv. [Global EHS - Contractor EHS Requirements](#)
 - v. [Global EHS - Cranes and Rigging Standard](#)
 - vi. [Global EHS - Confined Space Program Standard](#)
 - vii. [Global EHS - Control of Hazardous Energy \(CoHE\) Standard](#)
 - viii. [Global EHS - Electrical Safety Standard](#)
 - ix. [Global EHS - Work At Heights Standard](#)
 - x. [Global EHS - Lone Worker Standard](#)
 - xi. [Global EHS - Standard for Storing Flammables and Combustibles](#)
 - xii. [Global EHS - TMAH Safety Standard](#)
 - xiii. [Global EHS - Electrical Safety Standard](#)
 - xiv. [Global EHS - Safety Permit To Work Standard](#)
- b) Contractors / system vendors and Micron site project team are expected to assess all related safety risk factors during Slurry /Additives receiving, handling, flushing, commissioning and sampling activities. Preventive and contingency plans shall be created to address all safety concerns (Job Hazard Analysis (JHA) or Risk Assessments shall be conducted as needed).
- c) Personal Protective Equipment (PPE) – All contractor's /system vendor's personnel shall adhere to their own mandated PPE requirements for slurry/additives handling and shall be reviewed with Micron site's Environmental Health, and Safety, (EHS) for adequacy.
- d) Some examples of major safety concerns include, but are not limited to:
- General area safety (e.g. tripping hazards)
 - Potential chemical exposure / leaks / spill
 - Waste management (e.g. Hazardous)
- e) Contractors / system vendors are expected to ensure that their Team Members and sub-contractors involved in Slurry/Additives handling activities are adequately trained (with valid certification if applicable) for conducting the task. This should include all necessary in-house and specialized training specific to equipment or system (if applicable).
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- f) Slurry/Additives Information – Slurry/Additives encountered during execution of commissioning procedure may vary as such refer to chemical information labels affixed to the equipment or specific tool and the Safety Data Sheet (SDS).
- g) Descriptions of Danger, Warning, Caution, Notice, and Important.
 - **Danger:** Indicates a hazardous situation that, if not avoided, will result in death or a permanent disabling injury.
 - **Warning:** Indicates a hazardous situation which, if not avoided, could result in death or severe injury.
 - **Caution:** Indicates a hazardous situation that, if not avoided, may result in moderate or minor injury.
 - **Notice:** Indicates a hazardous situation that, if not avoided, may result in property damage.
 - **Important:** Alerts the user to information that is crucial to a step.

6.0 General Requirements and Design Concept

6.1 General Requirements:

- a) The Slurry Distribution System (and sub-systems) shall be provided, installed, and commissioned by selected Contractor. The Contractor shall have sole responsibility for compliance with the project Contract documents, and shall coordinate with other sub-contractors, the Owner and/or EPCM to complete the System according to the project schedule.
- b) Complete system includes but not limited to delivery and installation of all required materials, components, auxiliaries, fittings, fastenings, supports, labelling, training of the operation personnel; temporary provisions; vertical/horizontal transportation, utilities for commissioning like temporary hoses for drains, etc. This means that everything is included from design to fabrication to installation to testing & commissioning to hand over in obtaining correct and perfect operating systems, even if such items are not specifically mentioned in the specifications or indicated in the drawings.
- c) The System shall be designed, fabricated, installed, and validated in compliance with the requirements stated in this document and associated standards.
- d) Review the make-up air and room exhaust system design, including distribution ducting, inside the designated areas.
- e) The System shall be suitable for continuous operation and therefore, it shall be designed with minimal maintenance and operator intervention. The facility shall consistently supply Slurry quality within specification and shall operate 365 days per year, 7 days per week, 24 hours per day, with no unscheduled down time.

- f) Unit quantities, system sizing, flow rates, pressures, temperatures, etc. stated in this standard (or specification related to chemistry will be provided during design review) are general guidelines. However, the contractor must make his own evaluation in order to achieve the required pressure at the specified quantity and quality for each point of use. Piping distribution drawings, equipment layout drawings, and design matrixes shall be provided to enable pressure and flow calculations. The contractor shall be liable for guaranteeing that the systems, including all sub-systems, can meet all requirements within the RFP including required flow rates.
- g) All systems and related equipment near forklift paths to be protected from potential damage from forklift, pallet lifters, or payload impacts. Walls and wall corners, or anything protruding from walls that may be susceptible to damage should also be protected (e.g. steel plates, etc.).
- h) Locate all slurry/additives Drum / Tote locations in specified common areas, and close to door, if possible, to minimize forklift / motorized pallet jacks travel in room. This will also minimize any potential damage to equipment cabinets.
- i) Depending on the country of system installation, all local code requirements shall be met and therefore contractor shall be responsible for all associated costs/fees to obtain the necessary permits/written approvals. In addition, all documents submissions/clearance to/from authority for written approvals for construction/installation/operation approvals where applicable shall be included. Including all necessary PE endorsements for Micron to obtain their P&FM license.
- j) Contractor shall be fully responsible for understanding what a local PE requires, when he reviews a sub-system that is a listed chemical in the P&FM list, in regards to assuring system integrity and/or compliance with the latest revision of the local Import of Petroleum & Flammable Materials (P&FM: Fire Safety Regulations).
- k) Work included consists of the furnishing of all labour, materials, tools, consumables process slurry/additives will be provided by Micron in a quantity as proposed by the suppliers in their quotations, transportation, lifting devices, services, permits, fees, utility connection fees, professional fees, taxes etc., which are applicable, required, and/or necessary for a complete and proper installation of slurry distribution systems as defined within this standard.
- l) This standard represents the minimum requirements but does not include specific design details for the operational and performance requirements. It is the responsibility of the Contractor to provide a complete and functional System and achieve proper system performance.
- m) Micron provides only typical P&ID and therefore, contractor needs to provide a complete design, including detailed P&ID's, process flow diagrams, 3D drawings/models, pipe and equipment plans and details, equipment skid general arrangements, storage vessel fabrication drawings, equipment support, pipe support and restraint details for Systems.

- n) Systems are to be designed based on the given specifications and contractor shall use either AFT Fathom 9.0 hydraulic or PIPE-FLO®, or an agreed software for modeling to validate flow requirement. Predicted Flow, Loop Pressure, and Pressure Drop across Filters, Valves, other components and along the loop shall be provided for approval before 3D modelling. 3D models for each system shall be provided for approval before construction.
- o) If other software such as CFD (Computational Fluid Dynamics), etc. to be used for blending, tank net volume calculations (Vortex and LL level setting), any mixer application, etc. Validated information shall be passed to Micron upon completion of software modelling.
- p) Contractor to submit source calculations to Micron to support design review.
- q) If Micron provides any drawings (such as P&ID, Layout, network architecture, single line diagrams, etc.) it is the scope of work of Contractor to take ownership of these drawings, update them as required, and provide additional drawings as needed following the project Legend Sheet requirements as well as standards for layering, symbology, and other criteria described in the project documentation standards and CAD standards.
- r) Relevant 3D drawings (or BIM) for rooms, Sub-Fabs and other areas shall be submitted so that these drawings will be superimposed to BIM to check the feasibility of installation without having conflicts with other utilities or equipment.
- s) Electronic copies of required documents shall be submitted by the contractor upon award.
- t) Space Management: Based on the drawings developed by the contractor to meet the requirements, contractor shall identify any additional and designated space requirements in the Bid.
- u) The Contractor shall consider repeated handling of the equipment, especially if he intends to store his equipment on-site. Storage of material/equipment up-to 6 weeks also forms part of the scope of the contractor. Prior to material/equipment move-in to site, Contractor shall provide a detail method statement & risk assessment on delivery and installation.
- v) The Contractor shall provide all necessary means for continuous protection of the equipment during construction, handling, testing, start-up, commissioning, and certification phases. Contractor shall provide fences/barriers and appropriate PPE during the qualification of the slurry systems. The contractor shall also provide a permanent PPE cabinet with appropriate PPE for chemical handling and spill control in each of the slurry room.
- w) The Contractor shall be responsible for the appropriate disposal of all waste generated during construction and commissioning. If mutual agreement between Micron and Contractor is not available with respect to cost on disposal of waste, the purchaser shall co-ordinate the disposal on a chargeable basis.

- x) if tender drawing design is not strictly followed, contractor is required to provide drawings to support alternative design in the bid and indicate deviation or exclusion.
- y) Any deviations from design and/or critical Micron requirements stated in this standard, either temporary or permanent shall be documented in the "Deviation Form" in **Appendix 9** and shall be approved prior to implementation.

6.1.1 System Robustness and Reliability Requirements

System robustness is a very critical factor in the design of these systems and therefore, the due importance shall be given by the contractors. As it is known, the Fab operates continuously and therefore, supply of consistent quality slurry/additives with consistent pressure & flow is required 24 hours per day, 365 days per year. Thus, contractors shall ensure the followings:

- a) To provide Mean Time Between Failure (MTBF) analysis data for equipment and components being proposed.
- b) To provide Mean Time to Repair (MTTR < 3 Hours) guarantee for both response and repair time, 1.5 hour + 1.5 hour (or as per site requirements).
- c) Operational requirement: capable of fully automatic operation; periodic maintenance and calibration activities without any impact to Fab operations. Maximize water, Instrument air, CDA, Nitrogen and energy conservation and minimize generation of slurry to waste streams.

Likewise, reliability and functionality of the system are also very critical to ensure operating under different circumstances. Thus, contractors shall ensure the followings:

- a) To provide control system including instrumentation, redundant or highly reliable PLC's, software, source code, operator interface terminals, supervisory control and data acquisition (SCADA) software. Interface to and integrate with site facility management and control system (FMCS) and life safety system (LSS).
- b) To provide tubing, piping, conduit, insulation, wiring, exhaust ductwork, structural supports, equipment restraints, vibration isolation, electrical equipment, instrumentation and controls plus other components and materials needed to provide complete and fully functioning systems.
- c) Thermal Movement: Design piping to accommodate thermal expansion and contraction i.e. no degradation shall be observed on the mechanical lifetime of the piping and connected equipment.

- d) Seismic Performance: Contractor to provide equipment anchoring in conformance to the requirements of the site specification and the applicable local code.
- e) N+1 redundancy shall be provided for all major system components that require routine servicing and /or repair. This includes Pumps.
- f) Power distribution design to address electrical redundancy and single-point-of-failure mitigation. Systems shall be directly supplied from the Dual Feed AC Main Power Board having dual feed power supplies.

6.1.2 Performance Requirements

The Contractor shall be liable for the overall performance of the complete system as per contract, including guarantee on quality and flow. As mentioned in general requirements, the system shall be suitable for continuous operation, requiring minimal maintenance and operator intervention. The facility shall consistently supply slurry quality within specification and shall operate 365 days per year, 7 days per week, 24 hours per day, with no unscheduled down time. Thus, performance requirements shall also include the followings:

- a) The Contractor shall be liable delivery and installation of all required materials, components, auxiliaries, fittings, fastenings, supports, labelling, training of the operation personnel; temporary provisions; vertical/horizontal transportation, utilities for commissioning like temporary hoses for drains, etc. This means that everything is included in obtaining correct and perfect operating systems, even if such items are not specifically mentioned in the specifications or indicated in the drawings, however essential for the correct operation.
- b) Provide a detailed qualification plan and milestone schedule within 30 days of the contract award. Qualification plan and sampling plan along with Micron and/or EPCM shall be finalized within 45 days of contract award.
- c) Site Acceptance Tests (SATs) and Functional Acceptance Tests (FATs) shall be performed and documented. FATs shall include, but are not limited to, PLC interlock performance, HMI functionality, alarm functionality, and instrument and sequence of operation compliance. Evidence of test e.g. SCADA, HMI alarm, trend data shall be attached as a proof of testing for SAT. Any software bugs identified during system commissioning or after handover to Micron operations shall be rectified without any cost impact.
- d) The SCADA system shall be fully tested to be aligned to the system controller, and each limit and alarm documented.

Analog values shall have a maximum deviation of 0.1 UOM from the local reading at the HMI.

- e) All SCADA setting / Alarm tags shall be user configurable with high level password.
- f) System vendor shall provide the final system settings with justification.
- g) All timers and parameters shall be user configurable with different level password and shall not be hard coded.
- h) System vendor shall provide system alarm list for review and approval before implementation.

Initial system acceptance test (or FAT) shall be carried out at contractor's premises in order to demonstrate operational and functional testing of the system. Though it is carried out at the factory, detailed testing of the I/O list, I/O Loop test, assigned tagging both on component and on HMI, I/O firing test, instrument scaling test, etc. shall be performed. Detailed FAT requirement is explained under Testing and Commissioning section.

- i) All qualification instruments used online shall be with required calibration certificates. Soft copy or hard copy of the instrument measurement values shall be used as the proof for the qualification.
- j) Initial qualification of the Systems shall be performed with Ultrapure Water wherever applicable. It is important to meet the criteria for qualification with UPW at points-of-use (furthest VMB) and at the return tie in point in the associated slurry system. Demonstrate system performance for all slurry systems using Ultrapure Water (unless otherwise specified).
- k) After successful performance testing with Ultrapure Water and the acceptance from Micron or EPCM company, contractor is responsible for sampling and final qualification of slurry systems with actual slurry/additives. Samples to be analyzed at an approved Laboratory whose laboratory for analytical services is acceptable to Micron. Other Laboratories can be proposed and quoted as an alternative in the contractors offer.
- l) Contractor to demonstrate system performance of blending, mixing and dilution systems with actual slurry additives wherever is applicable.
- m) Deliver slurry/additives to point-of-use with no degradation outside of the acceptance criteria specification of slurry/additives quality. Quality is determined by collecting samples of each slurry/additives as fed to and supplied by the system and testing the collected samples for large particle size, conductivity, density, and metals (and assay if

required). Systems are not accepted until acceptance criteria are met.

- n) Sampling by Micron (or appointed TGCM Contractor) will be conducted weekly/fortnightly/monthly during warranty period following initial acceptance of System in conformance with the same acceptance criteria specification to evaluate the consistency of the slurry supplied to the Fab.
- o) It is very essential to note that prior to chemical filling, contractor has to provide and comply with Micron safety requirements. This shall include but not be limited to:
 - i. Only chemical handling trained workers are allowed to work with slurry/additives. The employees handling chemical shall be HAZMAT trained.
 - ii. Safe operating procedure - e.g. spill chemical handling and cleaning procedure, chemical/slurry receiving procedure, safety standing order, safety training and emergency response team and procedure
 - iii. PPE – hard hat, safety shoes, safety glass, arm guard, aprons, acid/solvent gloves, and first aid kit
 - iv. Spill control facilities - spill absorbent pads, spill absorbent socks, spill absorbent pillows, acid/base/HF neutralizer, pH paper, barricade cones and barricade tape.
 - v. Safety signs - correct signage(s) and notice(s) on door(s), proper slurry labelling on tanks and pipes
 - vi. Fire protection - fire extinguisher
 - vii. Ensure all the required commissioning activities such as electrical testing, mechanical installation inspections and other integrity checking, SAT, flushing with UPW, etc. are completed before slurry/additives charging.
 - viii. Seismic Performance: depending on the location of the Fab required importance on compliance to local and international regulations shall be conformed by the contractor. Contractor shall provide equipment anchoring in conformance to the requirements of the site specification and the applicable local jurisdiction
 - ix. A checklist for first charge of chemical into system & loop and a checklist for Chemical Receiving is attached in **Appendix 3 & 4**. These checklist can be used for slurry & additives charging as well.

6.1.3 General Civil, Structural and Other Utility Requirements:

Though Slurry Distribution system builder is not involved directly in building the slurry rooms, docks, move-in paths, entrance size and other civil /structural works, system contractor has certain responsibilities and upon not fulfilling the required responsibilities, contractor shall face the consequences.

- a) The Contractor shall be responsible for checking and monitoring the building with regard to conformance to contractor's civil brief and specifications. If contractor fails in notifying Micron about deviations found in time, he will be responsible for absorbing any resulting costs and schedule delays, except costs for the necessary correction. All construction activities shall be coordinated within the Slurry rooms once the rooms are handed over from Micron to the contractor.
- b) The Contractor shall check the structural drawings before ordering equipment. The Contractor shall prepare plinth and penetration drawings and room layout drawings three weeks after award of the contract. As part of the tender, a wall penetration drawing may be provided for common wall penetrations for the contractors to utilize. However, contractor shall confirm/provide the required wall penetration locations and sizes.
- c) Sump pit location & size and trench sizes for required drainpipe size & slope are critical to avoid flooding, drainpipe choking and backflow to systems. Contractor shall study civil drawings with respect to drain and collection and notify the Micron if changes are needed. However, as the changes in sump pit size and trench sizes will affect the construction plan, consensus to be reached between Micron and the contractor.
- d) Detailed size and location information of the of the slurry systems and related equipment shall be confirmed during the contractor's detail design stage with input from Micron.

Note: If it is an expansion project, civil/structural work may also have to be covered by the contractor based on requirements.

6.1.4 Temporary Utilities and Provisions

- a) Based on the contractual agreement with Contractor, Micron (or appointed EPCM) shall provide all necessary containers for pre-fabrication.
- b) Contractor shall supply and install temporary floor covering in each room to protect floor and walls from any damage
- c) Temporary power will be provided by Micron or EPCM for low amperage requirements, i.e. grinders, drills, etc. This temporary power is not guaranteed to be without fluctuations or interruptions and is not suitable for orbital welding. Contractor shall take necessary precautions by installing provisions to keep the power supply stable. The Location of temporary power panel to be advised once temporary power supply is established.
- d) Micron or EPCM will provide Low level lighting for Safety purposes throughout the building. Contractor to provide task lighting in each of the Slurry rooms and/or other required areas. Code compliance for task lighting in all required areas is a necessity as needed basis.
- e) Contractor shall provide mini clean room (or flow box with proper laminar flow) with utilities for pre-fabrication of ultra-high purity piping assemblies, etc.
- f) Based on the requirement for site office and quantity of material to be handled, contractor shall provide on-site 20-40 ft containers for material storage and site offices. Provide utilities, telephones, internet, etc. as required.
- g) Contractor shall provide high quality welding Argon per SEMI F78 for SS UHP piping system installation.
- h) Contractor shall provide high quality Nitrogen 5.0 for drying and pressure test
- i) Due to delay in any utility system installation, required utilities may not be available and temporary utility may have to be arranged for testing and commissioning activities. It is the obligation of the contractor to ensure to bring up the relevant issues with construction team and get it rectified with Micron site team (or appointed EPCM contractor).

6.1.5 House Keeping

Construction management shall form part of the scope of the Contractor. The Contractor shall be responsible for maintaining good housekeeping in all areas where the Contractor personnel works, e.g. storage and prefabrication areas.

- a) Cleaning team shall always be in place to maintain the Slurry rooms and other work areas in a clean and orderly manner. Cleanliness will be closely monitored by Micron for compliance.
- b) If anyone need to enter cleanroom, shall comply with Micron Clean-room protocol specification.
- c) Based on the area of work and based on protocol requirements, contractor shall provide the shoe covers, clean room shoes, facemask, gloves and cleanroom garments to the contractor employees.

6.1.6 Failure Mode and Effect Analysis and HAZOP

Failure Mode and Effect Analysis (FMEA) will have to be carried out by the system supplier in separate FMEAs for Process step functions. If a separate FMEA to be done for controls for having better understanding of different failure modes due to controls, it shall be done. Process FMEA shall be carried out in a structured approach that all potential failures that may exist in the system to be clearly discovered.

As minimum requirement, Process FMEA shall cover the followings:

- a) All different processes such as rinsing, filling, stand-by, ready, etc.
- b) Target set points, Specifications, control limits.
- c) Process qualifications based on different online parameters
- d) Process parameters such as tank levels, loop pressures, differential pressure, conductivity, concentration, density or Specific Gravity, temperature, etc.
- e) Deviation limits between two instruments when two instruments are used to do monitoring of each other
- f) Safety parameters such as leakages, rated pressure, exhaust, air exchange rates, etc.
- g) Other safety parameters such as UV/IR detection, etc.
- h) Bubbles in suction line
- i) NPSH and vapor pressure of the liquids
- j) Degassing nature of the liquids

- k) Vapor pressure of the liquids and the required N2 blanketing pressure
- l) Utility requirements such as UPW flow and pressure, Instrument air & CDA flow and pressure, UN2 flow and pressure, Process N2 flow and Pressure, etc.
- m) Based on FMEA (Link: [Facilities Process FMEA and Control Plan Business Rules](#)) outcomes, if there are any routine activities to be carried out such as preventive maintenance of equipment, calibration/validation of instruments, health check of any components, etc., these activities have to be specifically stated with required procedures.
- n) Before any installation of any system, Relative Risk Assessment (Link: [RRA Procedure](#)) according to Micron's Process Safety Management shall be carried out and when the Relative Risk Rating falls in dangerous or Hazardous categories according to Micron's Process Safety Management, HAZOP (Link: [HAZOP Study Business Process](#)) shall be carried out. When HAZOP is carried out, it is the duty of the system supplier to arrange facilitators, experts of different disciplines, Micron representatives, etc. to drive the whole required processes of HAZOP.

6.2 Design Concept and Requirements:

6.2.1 General Design Requirements:

- a) Slurry room sizes and layouts shall be coordinated with Slurry Distribution System designs to maximize the number of systems installed in any one room. Space allocation for future systems shall be evaluated during the design phase.
- b) All parts of the systems made of plastic materials working under pressure shall be protected by double containment. Double containment shall be able to contain the slurry / additives with proper sealing until a proper response is taken.
- c) Based on risk assessment, code analysis, and safety requirements, sprinkler systems shall be installed in all Slurry system rooms.
- d) Slurry Distribution modules in closed-cabinet configuration shall be equipped with exhaust according to the slurry/additive's properties and safety requirements. Materials for the exhaust ducts shall be identified based on compatibility, risk assessment, and cost comparison. These materials are typically Teflon-lined S.S. duct work or other FM 4922-compliant materials. All N2 blanketed vessels shall

be connected to the exhaust, and related vents shall be routed to the appropriate exhaust system.

- e) Material routing and equipment move-in path shall be designated during layout planning.
- f) Blend/Dispense tanks shall be routed to the appropriate exhaust system with some type of check valve system to prevent back contamination from the exhaust system.
- g) Space for future expansion and additional Slurry Distribution Systems shall be planned during the initial room design and therefore, system footprint and space between systems shall be optimized to maintain space for future systems. Also, ease of removal of systems shall be taken into consideration where possible during room layout planning to minimize land locking.

6.2.2 Slurry System Design Parameters:

- a) Slurry and additives are supplied in drums, totes.
- b) All Slurry/Additives shall be distributed to the fab under pressure by dispense systems, which consist of all equipment required to transfer slurry/additives from original containers to distribution pipelines routing through the Fab and be equipped with valves to supply tools. During the entire transfer path, the slurry/additives must be protected from contamination and/or change from the original properties.
- c) Slurry/Additives dispense systems shall be designed to operate continuously, with the theoretical capability to operate 100% of the time. Redundant components shall be provided to guarantee the supply of slurry/additive even if part of the system shuts down due to a failure or maintenance.
- d) The delivery pressure and expected fluctuations of slurry distributions shall comply with the limits and parameters provided by Micron.
- e) The delivery pressure shall be evaluated at the farthest Point of Connection (POC) and the end of the global loop return line and mid-loop.
- f) Slurry/Additives dispense system piping shall be sized based on system flow requirements and design parameters.

- g) To ensure safety, the N2 blanketing design must avoid the Explosion or Implosion of the drum/tote, blend/dispense tank and day tank.
 - i. To prevent explosion, the N2 blanketing system shall not exceed the maximum pressure of the drum, day tank and bulk tank at all times. Safety relief valve burst disk shall be sized and installed according to the type of drum, day tanks and blend/dispense tanks and in accordance with all applicable codes and industry standards.
 - ii. To prevent implosion during a transfer mode, blanketing N2 flow shall compensate the maximum liquid flow.
- h) To prevent contamination, the blanketing system must guarantee that the N2 flow compensate the liquid flow during a transfer operation.
 - i. The drum/totes shall be prevented from back contamination from exhaust (due to insufficient N2 flow), by designing the system based on the maximum liquid flow transferred.
 - ii. Day tanks shall be routed to the appropriate exhaust system with a check valve system and/or filter to prevent back contamination from the exhaust system.
- i) If relief valves are utilized for N2 blanketing for drums/totes/tanks, it shall be sized in accordance with all applicable codes and industry standards.
- j) Day tanks shall be designed into the system when a minimum volume of additives is required as intermediate storage.
- k) Slurry/Additives dispense modules shall be constructed as enclosed system in which the pumps and pressurized components are contained within a chemical compatible enclosure based upon local codes and regulations, safety requirements, principles of functionality, slurry/additives properties, available space, and design criteria.
- l) Non-wetted materials shall be identified according to safety requirements, material compatibility, risk assessment analysis, and cost comparison.
- m) The maximum rated pressure of all components shall be considered during the initial design. Design shall consider all local codes and regulations, safety requirements, and operating conditions.

- n) When systems are required to operate with a specific temperature range, appropriate temperature control and monitoring devices shall be installed.
- o) Each module shall have the appropriate facility utilities made available within 3m of the system. Typical required facilities are: Power (Normal Power and UPS (Uninterrupted Power Supply)), UPW, Instrument Air, CDA, UN2, NPW, drains, and exhaust. Refer to **Appendix 5** – Utilities Interfaces and Requirements.
- p) The following parameters shall be evaluated and combined to model the most efficient Slurry/Additives dispense system with defined over-capacity:
 - i. System Design Considerations:
 - Total quantity and type of tools to be supplied by the distribution system.
 - The elevation of the slurry/additives dispense systems, relative to production tools.
 - The length of the distribution line(s) from slurry/additives dispense systems to the farthest point of connection and back for supply/return recirculating supply loops.
 - Slurry room space and positioning.
 - ii. Tools flow requirements:
 - Average flow: is based on Tool Service Requirements (TSR) data for each tool. Average flow is the amount of slurry/additives required by the tool to flow from the POC to the Point of Use (POU). The sum of the average flows of all tools is used in the calculation to determine the blend/dispense tank and day tank volume.
 - Peak flow: is based on Tool Service Requirements (TSR) data for each tool. Peak flow is the amount of chemical required by the tool to flow from the POC to the Point of Use (POU). The sum of the peak flows of all tools is used in the calculation to find the system diversity factor and ultimately the system maximum design capacity, distribution loop size and design flow.
 - iii. Diversity factor: is a derived number typically between 0.01 and 0.99 based on the tool utilization rate of slurry/additives compared to actual consumption. To find the diversity factor for a particular system, the following items need to be taken into consideration: peak flow of all related tools connected to a slurry/additives system and tool cycle time, wafer

- throughputs, tool utilization percentage, and other factors.
- iv. Design flow to be guaranteed at the required pressure. It is calculated by multiplying the peak flow by the diversity factor and shall guarantee to correctly supply POC at the required pressure and flow.
 - q) System modeling, using PIPE-FLO®, AFT Fathom 9.0 hydraulic or an equivalent software program, shall be used during distribution network design phase to review system dispense and distribution loads, sizing, and minimum performance requirements. CFD or an equivalent software program can be used for other modelling such as mixing, tank Low-Low level analysis, pump suction head analysis, and other appropriate design analysis.
 - i. When a new tool is added to the distribution system, the system piping model shall be re-engaged to ensure that the impacted slurry/additives dispense system and subsequent distribution system can support the new addition. The model shall also be utilized to understand any system impacts due to the changes, and how to mitigate those impacts.
 - r) Distribution pressure loss is the maximum pressure loss as the distribution is performed throughout the fab. This value shall be calculated to correctly size the dispense system delivery pressure.
 - s) Maximum pressure loss shall be calculated at the last POC on the return line of a loop system to verify minimum flow requirements.
 - t) When modifications or additions are required to be made to an existing operational system, those modifications shall be tested and qualified prior to installation and shall not be physically connected until they are ready for operation.
 - u) Construction Material Shall be followed as below: When contractor realizes the material, specification is not appropriate for the specific slurry/additive, it is the duty of the contractor to inform Micron in writing. Thus, contractor also takes ownership of material selection.
 - v) To prevent excursion and minimize response time for operation validation, refer to Global Facilities - O&M BKM - Chemical - System interlock Design and Improvement Guideline that shall be incorporated into system interlock designs.

6.2.3 Slurry System Design Configurations and Specifications

The process slurry distribution systems generally comprises of Slurry Blending and Dispense Unit (SBDU) - typically for slurry but can also be utilized for additives, Slurry Transfer and Blend Unit (STBU) - typically for additives like KP3100+STI2910, Slurry Dispense Unit (SDU) with day tanks - typically for additives, Slurry Transfer Unit (STU), drum cabinets or IBC stations, distribution piping networks, valve manifold boxes (VMB), Mid/End Loop Boxes with pressure (and or temperature) transmitter and sump pumps and a dedicated SCADA control and monitoring system equipped with the supplier's own architecture.

Slurry and additives shall have global re-circulation loop to keep the slurry/additives moving and eliminate stagnation anywhere in the distribution network system. The PFA tube in the loop and the pump shall be sized to obtain a minimum velocity of 2 foot per second or based on each slurry requirement as stipulated by the supplier. If the velocity of the slurry in the distribution piping is not sufficient, the particles in the slurry may fall out of suspension (resulting in settling), and eventually clog the piping network. The flow rate required to achieve the linear velocity is a function of the tube diameter. The contractor as part of the tender for each Slurry sub-system will maintain the required Slurry sub-system flow rates based on each unique slurry requirement.

Vortex generation in either raw slurry drum or blend tank shall be avoided as vortex creates particle homogeneity issues with smaller particles in the center and larger particles on the outer.

6.2.3.1 Slurry Drum Pre-Mix Methods

If there is a requirement from slurry supplier for raw slurry in drum to be pre-mixed for the purpose homogenization prior to hook-up to STU, it shall be adhered. Below are two recommended methods but if there are vendor recommended methods it shall be followed.

a) **Slurry Drum Tumbler (Shaker) Station** – Preferred Method

- i. Drum tumbler or shaker station shall be provided to make slurry in the drum homogeneous. Time duration for the operation of the tumbler mixers and the rpm shall be user configurable so that based on slurry properties, the required time and rpm can be configured.
- ii. Tumbler station shall hold the drums vertically and be able to rotate 360 degrees so that if there are any sediments at the bottom of the drum, the sediments shall be brought back into the liquid phase and be mixed well to have a homogeneous raw slurry. It can be chosen to operate either pneumatically or electrically depending on the RPM requirement which needs to be determined during the design stage.

- iii. For safety reason, the vertical drum tumbler shall have a protective barrier or cage on all four sides with leak detection (tape type) in place tied to SCADA. The enclosure shall have a safety interlock such that the tumbler automatically stops when the enclosure door is opened and shall not start if the enclosure door is not closed properly. Figure 1 shows a typical example of a Drum Tumbler with enclosure.
- iv. Appropriate drum lifting equipment shall be provided for ease of loading to and unloading from the drum tumbler.

Figure 1 - Typical example of a Drum Tumbler with enclosure



b) Mechanical Agitation - Mixer (Stirrer)

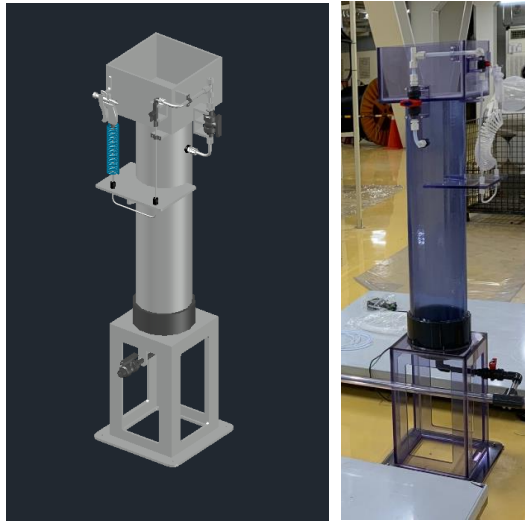
If mechanical agitation is recommended by slurry supplier, care should be taken to avoid conditions where splashing or air entrainment can occur. The recommended stirrer blade/propeller type and size shall be used. Normal variable speed motors shall be used for RPM control. Stirrer can be fixed on hydraulic rescissible fixtures so that stirrers can be moved in and out of the drum. Drum stirrer operator panel shall be either provided separately or integrated. The stirrer shall have a user configurable RPM and timer-based setting for periodic intermittent or continuous agitation at a specified speed. A recommended stirrer is CEMCO Mixer or otherwise recommended by slurry supplier.

6.2.3.2 Slurry /Additives Drum staging Drum Cabinet

- i. Staging cabinet for Drum shall be located beside each Slurry Transfer Unit (STU) where possible.
- ii. Slurry Drums to Slurry Transfer Units (SLTU) can be docked in drum enclosure or open docking staging depending on location of the drum cabinets as explained below:
 - Open drum cabinet located in Slurry Transfer Room. Each drum docking station shall be designed to hold two 200L containers, within an epoxy coated Unistrut frame, and a removable bottom pan for the containment volume of 1.1 x 200L of leaks.
 - Drum cabinet with enclosure if located outside a room or sub-fab. Same leak containment volume as above applies.
- iii. Bar code reader tie-in to STU controller for verification of correct container, expiration date, batch number, etc. There shall be an interlock control in place to prevent slurry/additives transfer in the event of wrong drum hook-up and/or validation failure. Refer to Chemical Error Proofing of Wrong Chemical Hook-up Standard link.
- iv. Barcode shall have high level password protection in place to bypass barcode scanning due to faulty barcode. The barcode high level password shall be restricted and accessible to only site appointed personnel and shall have the option to change password. Barcode bypass alarm shall be reflected in SCADA/HMI to alert unauthorized bypass and shall not be able to reset until barcode bypass had been turned off.
- v. Keyed slurry/additive specific drum quick connections (Entegris® QCIII or equivalent) with umbilical tubing in containment pipe with support as required. Double contained drum re-circulation tubing. Supports for the interconnecting tubes and cables shall be provided.
- vi. A slurry compatible roller conveyor shall be provided for ease of loading and unloading slurry drums. The level of the roller conveyor shall be adjustable to eliminate minor differences between roller conveyer and the drum cart rollers or any other staging stations.
- vii. A slurry drum stirrer as recommended by slurry supplier or based on site's experience using the particular slurry for agitation shall be utilized for the required applicable slurry as determined during the design stage. There shall be sufficient head space for

removal of stirrer during drum change. Based on requirement, stirrers can be fixed on hydraulic rescissible fixtures so that stirrers can be moved in and out of the drum. Drum stirrer operator panel shall be either provided separately or integrated so as to be operated from HMI. The stirrer shall have a user configurable RPM and timer-based setting for periodic intermittent or continuous agitation at a specified speed. There shall also be load cell for drum empty detection as well drum level monitoring which will interlocked to slow down stirrer rpm and stop drum stirrer (if utilized) to prevent raw slurry vortex or foam generation.

- viii. A sensor (proximity sensor) shall be installed on transfer pump suction line to detect drum/tote empty. A user configurable timer with 0.1 sec to 999sec shall be provided.
- ix. There shall be a leak sensor in the staging cabinet containment pan and tied to STU or STDU to stop drum transfer and recirculation operation in the event of leak detection.
- x. The staging cabinet shall have a minimum of two clean holder with conical bottom for drum quick connects placement and shall have a drainpipe connection at the bottom of the holder.
- xi. There shall be a design for drum diptube and dispense head cleaning system installed with UPW spray gun and drain connection located between the two drums for periodic cleaning of drum dispense head/diptube. A typical example of such a cleaning module is shown in **Figure 2**.
- xii. Remote EMO button shall be installed if the staging cabinet is more than 10 ft away from its associated Slurry Transfer Unit.
- xiii. The enclosed drum staging cabinet shall be exhausted with magnehelic gauge & damper. Fresh air intake grills with filters shall be installed as required to ensure adequate air changes in the cabinet. Intake grills installation location shall be at bottom of cabinet door.

Figure 2 – Slurry Dip tube and Dispense Head Cleaning Module

6.2.3.3 Slurry Transfer Unit (STU)

- i. The Slurry Transfer Unit cabinet shall be constructed of polypropylene or PVC (if installed in sub-fab, FM4910 PP or FM rated PVC shall be used), with access doors or panels containing clear PVC view ports as needed for maintenance and operations, and adjustable leveling pads to compensate for uneven flooring.
- ii. All access doors for wetted areas shall be equipped with interlock sensors to ensure a safe operating condition. The bottom of the cabinet acts has a 100 Liters of containment for internal leaks and shall be equipped with dual leak sensor and gravity drain port connection with manual isolation valve.
- iii. Utility connections for ICA, UN2, and DI Water supply/return shall be provided for system operation and maintenance, with humidified N2(>95%) blanketing of the supply source container to ensure slurry quality. A DI Water spray gun (dead leg free) shall be installed for wash-down of the wetted areas and a UN2 spray gun shall be installed for blowing dry those areas.
- iv. In general, the transferring of slurries shall be accomplished by redundant transfer pumps. The pumps shall be air operated diaphragm or bellows type with corrosion resistant bodies and PFA or PTFE wetted surfaces. Required Pump types can be ARO Pumps or equivalent. Tubing, valves, and fittings of PFA shall be used for slurry transfer. Pumps, valves and tubing, shall be sized to provide the flow rates as specified. STU shall be equipped with a drum

recirculation line via a filtration step. The transfer to the SBDU shall be accomplished by a loop return line back to the online drum via a 3-2-way air operated valve to prevent any slurry stagnation. During SBDU slurry request, the 3-2-way valve shall supply slurry to the SBDU blending tank without any return back to the online drum. When there is no request from SBDU, there shall be constant loop recirculation back to the online drum.

- v. There shall be a user configurable feature to have local recirculation of the offline full drum with dedicated flow path and pump without interrupting the online drum recirculation.
- vi. One 20" filter bank with bypass feature shall be available for raw slurry filtration.
- vii. Regulation of ICA and UN2 for system operation shall be provided and located inside the cabinet.
- viii. The Slurry Transfer Unit controls compartment (either internal or external) shall contain a dedicated PLC responsible for all operations within the module, and for sending and receiving network data that is required for operation of the Slurry Blend & Dispense Unit. A 15" color human interface panel (HMI) shall be included along with necessary solenoid valves and terminal blocks.
- ix. The controls compartment shall be purged with ICA or UN2 to keep corrosive slurry fumes away from electrical components, with no slurry wetted parts inside.
- x. All electrical components shall be easily accessible for maintenance and trouble-shooting purposes. The following alarms (at a minimum) shall be provided at the HMI and SCADA terminal:
 - a. Cabinet leak
 - b. System up/down
 - c. Cabinet door open for enclosed drum cabinet
 - d. EMO activated
 - e. Low exhaust flow
 - f. Maintenance bypass timeout alarm
 - g. Low ICA or UN2 pressure
 - h. Supply drum/tote empty
 - i. Start/End transfer to Blend Module
 - j. System using manual pump actuation
 - k. Drum failed by mechanical key code reader
 - l. Liquid in drain sensor

- m. In addition to the specifications noted above, each STU shall be provided with the following features and accessories:
- n. Magnehelic gauge on the exhaust connection
- o. Exhaust pressure (flow) switch with low alarm
- p. Emergency shut-off button (EMO) and power switch on operator panel
- q. Alarm buzzer and three-color light tower for system status indication
- r. A mechanical vertical float type leak switches
- s. Programmable time feature on door safety interlocks
- t. Maintenance bypass switch
- u. Touch screen display on operator panel (HMI)
- v. Sample valves to enable sampling after drum hook-up
- w. Fail safe pneumatic controlled valves with on/off indication
- x. PFA Tubing with PP header for drain piping
- y. N2 humidifier shall be constructed with transparent acrylic material. Proper mist separator has to be in place to avoid any water droplets been carried away into humidified N2 stream.
- z. Each pump shall have isolation purging ports for flushing upstream and downstream of the pump. Each pump shall have pressure transmitter with isolation valve downstream.
- aa. Each pump with dedicated post sampling point.
- bb. Filters with post sampling points.
- cc. Drum sampling option shall be present.
- dd. Fresh air intake grills with filters shall be installed as required to ensure adequate air changes in the cabinet. Intake grills installation location shall be at bottom cabinet.
- ee. All timers shall be user configurable.
- ff. All pressure transmitters shall have dual setpoint, prewarning and fault pressure settings.
- gg. Refer to **Table 4** below for a list of STU main components & features.

Table 4 – List of Main Components & Features for STU

S/N	System Main Components & Features
1	Dual redundant DC power supplies
2	Min 12" Color Touchscreen HMI
3	Dual 1" ARO AODD pumps or equivalent
4	Pulsation Dampeners on the discharge
5	Automatic flush/purge function
6	Pressure monitoring of dispensing and standby drum circuits
7	Generates humidified N2 for drum blanketing
8	Dual cabinet leak detection
9	Door switch interlocks with maintenance bypass
10	Gravity drains
11	GEMU or Entegris Integra Plus WS valves on slurry flow path
12	Automated drum switching based on drum scale empty detection.
13	Mechanical Mixer with interlock feature with drum scale to slow down Mixer RPM at user configurable low setting.
14	1 x 20" filter housing with filter bypass feature
15	All pneumatic tubing must be PFA/FEP material
16	Barcode
17	All solenoids to have "hot swappable" isolation valve manifolds

6.2.3.4 Slurry Blend and Dispense Units (SBDU)

- i. If SBDUs are to be designed with two engine/stations, blend/dispense tanks shall have two Levitronix dispense pumps (in parallel), and approval shall be obtained from Micron upon which design shall be appropriately done considering all the single point of failures.
- ii. Three separate engine/station system is preferred for avoiding single point of failures, dedicated tanks, pumps, valves, online instruments, metrology, filters, etc. shall be in place for each engine until the final dispense valve to the loop. If separate standalone filter station is utilized, there shall be redundant filter banks with filter bypass feature available.
- iii. SBDU shall either be designed with Single or Dual Global loop for the purpose of global loop flushing without taking down the entire SBDU. Dual loop design shall have the feature to offline one engine/station and carry out global loop flushing and maintenance with the required cleaning chemical fed to the offline engine/station and UPW as required and shall not affect the operation of the other two engines/stations supplying slurry to tools.

iv. Blended slurry requires thermal process management where temperature shall be maintained below 27°C or as specified based on slurry requirement. Process Cooling water supply/return assemblies consisting of isolation valves, pressure regulator, temperature sensors, flow meter, flow regulating valve, etc. shall be kept outside the SBDU cabinet. Pipes shall be converted to PFA tubing before the entry of the SBDU cabinet. Two different acceptable methods for thermal management are indicated below:

- **Method 1:** Heat Exchangers shall be placed at the loop return and at local blend recirculation in each blend station with appropriate capacity to maintain the desired temperature.
- **Method 2:** Blend/Dispense Tank with cooling coils and a heat exchanger at local blend recirculation in each blend station with appropriate capacity to maintain the desired temperature.

The heat exchanger controller shall be located within the SBDU electrical controls compartment. If system supplier has other methods for temperature control/management, Micron approval is needed.

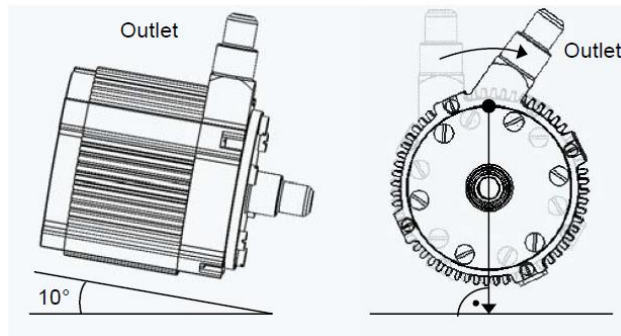
- v. The methods for two-part blending using concentrated slurry with UPW shall be primarily by weight (using scales or load cells), and by volumetric (using flow meters) if high dilution is required. A combination of both methods can be utilized depending on the slurry characteristics and dilution ratio. The quality of the blend shall be verified by dual conductivity probes before the mix tank content is qualified for dispensing. The two conductivity probes in each engine/station shall have a user configurable deviation alarm to highlight drift/failure of one of the probes. Other specified metrology shall be utilized if conductivity method is not suitable for a specific slurry. All systems shall be plumbed and ready for H₂O₂ dosing (3-part blending ready) for future use if needed.
- vi. The methods for three-part blending using concentrated slurry with UPW and Hydrogen Peroxide (H₂O₂) shall be by weight (using scales or load cells), and by volumetric (flow meters) for H₂O₂, or a combination of both depending on the slurry characteristics and dilution ratio. A user configurable auto replenish feature for H₂O₂ shall be provided to maintain batch quality. There shall be two sets of conductivity probes in each engine/station for verifying the mixture in the blend tank is within specification and qualified. The two conductivity probes shall have a user configurable
-

deviation alarm to highlight any drift/failure of one of the probes. For H₂O₂ assay qualification, it shall be done by either a titrator - via a slip stream pulling samples rotationally between each engine/station or by an inline concentration sensor (one number of K-Patents) in each engine/station which shall be selected based on slurry properties and concentration sensor measurement limitations. There shall be a second concentration sensor installed the global loop return line for verification if the dispensed slurry is still within specification. The quality of the blend shall be verified for conductivity, H₂O₂ assay, or other specific metrology before blend tank content becomes ready for dispensing.

- vii. The Slurry Blend/Dispense Unit cabinets if located in sub-fab shall be constructed of FM4910 listed polypropylene (or FM4910 listed PVC). If located at non-cleanroom areas such as dirty sub-fab Level 1 and slurry rooms, PP or PVC shall be used. All access doors or panels shall have clear PVC view ports as needed for maintenance and operations, and adjustable leveling pads to compensate for uneven flooring.
- viii. All access doors for wetted areas shall be equipped with interlock sensors to ensure a safe operating condition with maintenance bypass feature. The bottom of the cabinet acts as containment for internal leaks and gravity drain port connection with manual isolation valve. Each separated engine/station shall have its own set of two leak sensors to prevent shutdown of a complete system but instead should only shutdown the affected engine in the event of leak.
- ix. Utility connections for ICA, UN₂, and UPW supply/return, shall be provided for system operation and maintenance, with humidified N₂ blanketing of the blend/dispense tanks to ensure slurry quality. A UPW spray gun (dead leg free) shall be installed for wash-down of the wetted areas.
- x. The blending and dispensing of slurries shall be accomplished by magnetic flow - centrifugal pumps manufactured by Levitronix. The pumps shall be equipped with corrosion resistant impellers (two layers of PFA coating) as well as PFA or PTFE wetted surfaces. The pumps shall be primed (filled with liquid) before they can be operated. To prevent potential risk of entrapped air in the suction line, i.e. fittings, step junctions, valves, high spots etc. shall be minimized. Levitronix pump (if used) has a pump orientation instruction to minimize bubble entrapment and shall be applied. Entrained air in the pump could also result in

insufficient priming and shall be considered in the initial design. If the pump is evacuated through the outlet during the priming process the optimum orientation is a forward incline of about 10 degrees (Figure 3 left). An additional improvement can be achieved by turning the pump till the outlet of the pump housing is exactly plumb (Figure 3 right). The outlet hole in the pump housing is then positioned directly above the inlet. If, however, there are sufficient mitigation measures such as air removal devices and location of pump below and close proximity to the tank, the pump orientation described above may not be needed.

Figure 3: Forward incline (left) / Turned Pump (right)



- xi. Levitronix pumps shall be outfitted with an air-cooling unit. Each Levitronix or Iwaki pump shall have its own ELCB /RCCB/circuit breaker such that in the event of pump trip, power is only cut-off to the affected pump. All alarms from the Levitronix pump controller to be fully integrated such that any critical alarms from the controller will be reflected on the local HMI and SCADA (no need for an external laptop to plug into the controller to retrieve the alarms). All engine/station shall have temperature monitoring devices at the pump outlet. Design of the system shall be such that cavitation problem shall never occur. If cavitation is identified, the cause shall be determined and appropriate changes in tubing size (incorrect use of reducers), length, NPSH, etc. shall be implemented. Care should be taken to avoid this problem, i.e. minimize number of different sizes of tubing within cabinets. Levitronix pumphead bolts shall be Teflon coated stainless steel as PVDF bolts are not allowed.
- xii. Tubing, valves, and fittings shall be the same for all slurry distribution systems in order to optimize spare parts management. Sweep elbows shall be used instead

of 90° elbows to prevent any potential for dead legs. All slurry wetted parts shall be of PFA/PTFE. Pumps and tubing shall be sized to provide the desired flow rates as specified at the design stage. All dosing / blending lines to blending tanks shall have sufficient gradient to ensure that the contents are completely emptied to prevent any residual slurry causing stagnation.

- xiii. Each Slurry Blend/Dispense tank cabinet shall be sized as per design requirement. It shall have automatic user configurable flushing of the Slurry Blend/Dispense tank with UPW between batches. The UPW flushing shall be accomplished via 360° PTFE rotating nozzles with spray head that can be easily removed from the body for inspection and maintenance. The tank overflow pipe shall be connected to a p-trap that always contains water (due to residual water from 360° flushing) and which is then connected to the gravity drain.
- xiv. Blend/Dispense tanks shall be covered, and the headspace shall be blanketed with Nitrogen. Nitrogen shall be humidified to prevent evaporation of water and thus causing abrasive build up. It is recommended that the N₂ in the headspace of the tank is kept at >95% relative humidity. This reduces the possibility of slurry drying on the walls of the tank and subsequently falling into the solution. Nitrogen is put through a DI water bubbler in order to generate the humidified N₂. Each engine/station shall have its own humidifier depending on the design requirement or a minimum of N+1 redundancy.
- xv. All blend tanks should be designed so that they can be visually inspected periodically and cleaned out manually if needed via an opening with a clear cover. This should be located as close to the center of the top as possible to facilitate ease of cleaning. Each blend tank shall be constructed of HDPE or natural PP and equipped with level detection sensors.
- xvi. Bottom draw piping is required to add and remove the material from the tank. The tube for addition should come down from the top of the blend tank and extend at least several inches below the minimum liquid level in the tank to avoid air entraining in the slurry and possibly aggravating the drying of the slurry if any bubbles are formed. The withdrawal of slurry from the tank should be done from the bottom of the tank. This will aid in ensuring that adequate mixing of the slurry in the tank takes place.
- xvii. Mixing of the tank content shall be through a proper recirculation mechanism. Pump recirculation with

eductor is the recommended method for proper mixing. It is advisable to avoid any motorized stirring devices. However, upon approval from Micron, motorized stirrers can be used based on the appropriateness of the slurry properties or as per requirement of slurry supplier. If a motorized tank stirring device is incorporated into the design, the motor shall be air operated or electric, with stirring shaft and impeller that is non-contaminating, easily maintained, and will not impact the tank N2 humidified circuit. The sub-system shall be able to detect the rotation of the stirrer shaft. Stirrers will be able to be removed easily to be serviced periodically. RPM of the stirrer (stirring device) shall be user configurable.

- xviii. The three engine SBDU, shall have a dedicated pumps in series with identical flow path and components such that the system performance shall not experience any pressure degradation during a controlled engine/station swap. There shall be a user configurable option to have either an engine/station swap in the event of one of two pump in series had faulted or do not do an engine/station swap but ramp up the other pump in series to attain the required pressure/flow. This option shall be tested during SAT and set based on consumption and loop pressure requirements for specific slurry/additives.
- xix. Redundant segregated dispense engines/stations shall have its own individual solenoid block and supplied with individual CDA/N2 pressure regulators and manual isolation valves. Pneumatic solenoid blocks shall have Hot Swap function for replacing individual solenoid valves.
- xx. Each Pump with dedicated post sampling point.
- xxi. Filters with post sampling points.
- xxii. Each pump shall have post pressure transmitter. All pressure transmitters shall have dual setpoint, pre-warning and fault pressure settings.
- xxiii. All timers shall be user configurable.
- xxiv. Communication cards, I/O cards, terminal blocks, etc. shall be dedicated for each engine to ensure robustness. Thus, operation of PLC (i.e. the system) shall be assured even during replacement of any of the I/O cards. The PLC shall have dual supply cards with dual incoming source.
- xxv. Refer to **Table 5** below for a list SBDU main components & features.

Table 5 – List of Main Components & Features for SBDU

S/N	System Main Components & Features	Remarks
1	HDPE or Natural PP conical bottom blend tank with cu Jets (Eductor)	Size to be determined based consumption (500L Max)
2	In tank Cooling Coils or Return loop heat exchange system with dedicated Heat Exchanger for each Blend/Tank during local recirculation	
3	Tank level indication	
4	High accuracy scale metering of DI Water and/or Slurry	Preferred is Hardy
5	In-line Dual Conductivity measurement in each station	Preferred is E&H sensor
6	In-line Density measurement	Optional
7	Series (2) Levitronix BPS-2000 Centrifugal Dispense Pumps	
8	DI spray gun	
9	Local station recirculation	
10	Local station flush purge	
11	Sampling Points after pumps / filters / Global loop return	
12	Automated Back Pressure Regulation for Global Loop return to each tank	
13	All pneumatic tubing must be PFA/FEP material	
14	Single or Dual Global Loop switch	
15	Single or Dual global loop filtration with shared redundant filter bank, Two 20" filters housings in parallel per Bank and automated filter degassing function and automatic filter banks flush/purge including filter back charge to purge all air from filters. Automatic Filter vent needle valve flushing.	Single Global loop – 2 x filter banks with Bypass feature. Dual Global loop – 3 x filter banks with Bypass feature.
16	Chemical flush capability via chemical tote replacement or KOH Flush cart supply to provided connections.	
17	Exhaust monitoring of cabinets	
18	Dual cabinet leak detection	
19	Door Switches with interlocks and maintenance bypass	
20	Dual redundant DC power supplies	
21	Dual AC power feed with the load (pumps and power supplies, etc)	
22	15" Color Touchscreen HMI	

S/N	System Main Components & Features	Remarks
23	Gemu or Entegris Integra Plus WS valves on slurry flow path	
24	Electrical cabinet to operate <30°C in normal operation	
25	All auto valves have open/close indication	
26	ATRS Ready	Preferred: SemiChem
27	Chemical Addition for H2O2 with high accuracy course and fine fill flow meters	
28	Global loop Conductivity or K-patents Ready on loop returns. For dual loop, single sensor shared by both loops.	
29	All solenoids to have "hot swappable" isolation valve manifolds with dedicated to each engine/station	
30	Fine DIW addition to Blend tanks	
31	Manual isolation per filter bank	
32	Manual isolation valves for Global Loop switching	
33	Global Loop Degas Tees	
34	Sweep Elbow	
35	N2 Humidifier for each engine/station	
36	Extract Levitronix controller alarms from the controller via RS232 communication into the PLC for alarming through the HMI.	
37	The SBDU system utilizes a combination of Pump RPM and Back Pressure Control with multiple inputs to Maintain Global Loop Pressure within a tight tolerance. There shall be mid loop pressure transmitter for monitoring.	

6.2.3.5 Slurry Transfer and Blend Unit (STBU)

- i. Additives which are usually stored in slurry rooms are usually blended in this concept, and is generally coupled with Slurry Dispense Unit (SDUs) which has a day tank(s) to cater for high consumption and dispense to the CMP tools, however, depending on requirement, space and lower consumption, SBDU can also be utilized for this purpose and shall be determined during the design stage.
- ii. Additive transfer and blend takes place in the same unit or in separate units. The method for blending concentrated additives with ultra-pure water (and or a chemical) can be either by weight (using scales or load cells), or by volumetric (flow meters or flow controllers), or a combination of both depending on the additive's characteristics. There shall be redundant blending engine/station with dedicated flow path, redundant pumps/filter and metrology. The quality of the blend shall be verified before it is transferred to the day tank for dispensing. Blended additive is generally pumped to a day tank to meet high consumption.
- iii. The Slurry Transfer Unit (STU) can be built within STBU or built separately to convey the raw additive to the STBU. Day tanks can be equipped with eductors (preferred) or a stirrer (depending on chemical supplier recommendation) with a user configurable rotation per minute to ensure homogenization of the blend if mandated by chemical supplier. Blending & transfer of additive to day tank will follow the general guideline as explained under STU & SBDU description.

6.2.3.6 Slurry Dispense Unit (SDU)

- i. When additive blending is done in a separate unit such as STBU, a separate Dispense unit is required to supply the additive to all the manufacturing tools. SDUs shall have one standby engine for each loop designed. However, if SDUs are to be designed to have a common stand by pump for more than one loops, approval shall be obtained from Micron upon design is appropriately done considering all the single point of failures.
- ii. The primary purpose of the SDU is to draw additive from a nearby day tank(s) using an electrically powered centrifugal pump. The unit passes the chemical through filter(s), monitors its pressure and flow rate, sends it to the distribution pipeline, and allows proper amount of return flow from the distribution pipeline back into the Day tank via the

SDM. The SDU has duplicate pump (Levitronix pump in series per engine) and filter compartments (also called "engines") that allow flow to the distribution loop even while servicing the off-line pumps or filters. The SDU sends a day tank low electrical signal to the STU or STBU when its associated day tank has reached its Low sensor, which will fill the tank until Hi level sensor is seen.

- iii. During normal operation, the SDU dispenses the additive from the operator chosen primary engine while the secondary engine sits idle or performs tank recirculation functions based on user configurations via the HMI. Upon a fault condition, or when instructed to by an operator, the secondary engine will ramp its pump up to speed, open previously dispensing engine to Offline.
- iv. The two engine SDM, shall have a dedicated pump/filter with identical flow path and components such that the system performance will experience no degradation after an engine swap. The system design includes a two engine-bay cabinet with each engine in an isolated, dedicated area which allows for complete safe servicing of an offline engine while the other engine remain in fab distribution. A Dedicated flow path from each engine with AOV & manual isolation valve to fab supply - no common(shared) critical AOV (SDM last valve) – mitigation against SPOF (single point of failure).
- v. Each separated engine shall have its own set of two leak sensors to prevent shutdown of a complete system but instead should only shutdown the affected engine in the event of leak.
- vi. Redundant segregated dispense engines shall have its own individual solenoid block and supplied with individual CDA/N2 pressure regulators and manual isolation valves. Pneumatic solenoid blocks shall have Hot Swap function for replacing individual solenoid valves.
- vii. Communication cards, I/O card, terminal blocks, etc. shall be dedicated for each engine to ensure robustness. Thus, operation of PLC (i.e. the system) shall be assured even during replacement of any of the I/O cards. The PLC shall have dual supply cards with dual incoming source.
- viii. The pumps in series per engine can be configured to switch from primary to secondary mode based on a user-configurable timer and automatically switch in the event of a failure in the primary engine. In addition to

dispensing chemical, the two engine SDM shall make use of Stabilized Filtration via fab-wide or local recirculation by pumping a small volume of chemical across the online filter bank at all times to prevent filter shock associated with transitioning from zero flow to a high-flow condition. Stabilized filtration provides the best possible particle performance from a slurry distribution system.

- ix. Pumps in series configuration shall have isolation purging ports for flushing upstream and downstream of the pumps.
- x. Each Pump set and filter with dedicated post sampling points.
- xi. Each pump shall have post pressure transmitter with isolation valve downstream for ease of replacement. The post pump pressure transmitter shall be interlocked with the post-filter pressure transmitter to validate pressure loss prior to switching to secondary pump. The post pump and post-filter pressure transmitters shall act as a back-up if one of the pressure transmitters is faulty and there shall be an option to enable/disable any one of the pressure transmitters for replacement. All pressure transmitters shall have dual setpoint – pre- warning and fault pressure settings.
- xii. All timers shall be user configurable.
- xiii. The SDM shall be designed to remove any single point failures in the flow path including all supply and distribution valves, filters, pumps, and the CDA supply. In addition to component redundancy, each pump circuit includes its own cabinet area and isolation valves allowing for complete maintenance on the unit, protection from any pressurized chemical hazard, without interrupting flow to the fab. The SDU Module includes a keyed maintenance bypass switch for each engine.
- xiv. Loop return line shall be equipped with a stand-alone automatic back pressure control valve. There shall be a manual valve bypass system with sampling point in place to isolate and replace the faulty back pressure valve. All return lines shall have a non-intrusive flow meter for consumption monitoring & trending at SCADA.
- xv. The centrifugal pump system shall be able to operate in either a pressure-controlled mode or in speed mode (RPM control). The pressure mode shall be operated based on end loop pressure transducer and in the event of failure or out spec range, will automatically

switch to operate based on return pressure transducer and the third back-up is based upon dispense pressure transducer.

- xvi. The centrifugal Pump's rpm shall be user configurable and shall have a pre-warning setpoint for lower and upper rpm limit. For safety reason, the centrifugal pump maximum rpm shall be hardcoded based on the maximum allowable operating pressure of distribution system/component.
- xvii. The local recirculation shall have user configurable rpm or pressure setpoint option.
- xviii. All access doors for wetted areas will be equipped with interlock sensors to ensure a safe operating condition. The bottom of the cabinet acts as containment for internal leaks and gravity drain port connection with manual isolation valve.
- xix. Utility connections for ICA, UN2, and DI Water supply/return, shall be provided for system operation and maintenance. A DI Water spray gun (dead leg free) shall be installed for wash-down of the wetted areas and a UN2 spray gun shall be installed for blowing dry those areas.
- xx. The dispensing of additives shall be accomplished by magnetic flow - centrifugal pumps manufactured by Levitronix (BPS Series). The pumps shall be equipped with corrosion resistant impellers (two layers of PFA coating) as well as PFA or PTFE wetted surfaces.
- xxi. Tubing, valves, and fittings, of PFA /PTFE shall be same as slurry STU or STBU in order to optimize the spare part management.

6.2.3.7 Filter Design Configuration

- i. Each engine/station or a standalone filter cabinet, shall be equipped with filter housings, constructed of PFA 440 HP.
- ii. Single Global Loop design - A 20" filter in parallel for each engine/station or two redundant filter banks with filter bypass line.
- iii. Dual Global loop design - A 20" filter in parallel for each engine/station or three redundant filter banks with a filter bypass line.
- iv. A pressure transducer is installed upstream and downstream of the filters to indicate pressure loss due to clogging or de-wetting. Differential pressure setpoint shall be user configurable with two level, High & High-High setting and both alarms shall be warning only.
- v. The SBDU shall always be designed to provide filter stabilization to allow a minimum flow of slurry through the filters to avoid filter shock.
- vi. Filter shall have both automated and manual filter swap feature. If automated filter swap feature is enabled based on high differential pressure or runtime exceeded, it shall first perform standby filter slurry charging step and when ready shall flip to online filter status without any pressure bumps and the prior online filter thereafter shall flip to offline. The automated filter sequence shall continue to flush & purge the offline filter bank and provide alert to operator to replace filter.
- vii. Isolation valves upstream and downstream of filters and purging/flushing points, for safe maintenance and change-out.
- viii. The system shall have a feature for operator enabled pressure test function after filter change-out to ensure filter had been installed properly.
- ix. Take-off points to sampling station to allow sampling before and after filters.
- x. The filter venting line shall have an air-operated valve with sensor to detect

6.2.3.8 Other Requirements and Features

- i. All plastic wetted part connections/fittings shall be pure bond and/or PrimeLock® /Super-300 Type Pillar Fitting® or Flaretek® fittings
- ii. The bottom of STU, SBDU, STBU, SDM cabinet acts as containment (minimum 100L) for internal leaks and shall be equipped with a dual leak sensors and gravity drain port connection with manual isolation valve. Cabinet modules shall be constructed with sufficient slope directed to the discharge point or may be aspirated to allow for the complete removal of slurry/additives and water that eventually collects inside. A scrubbed exhaust connection shall be provided to capture fumes emitted during operations or maintenance functions. Fresh air intake grills with filters shall be installed as required to ensure adequate SEMI recommended air changes in the cabinet. Intake grills installation locations shall be at bottom of slurry Modules. In addition, utility connections for ICA, UN2, and UPW supply/return, shall be provided for system operation and maintenance. A flow thru UPW spray gun shall be installed for wash-down of the wetted areas.
- iii. Regulation of ICA/CDA and UN2/GN2 for system operation shall be provided inside the control cabinet. In addition, design accommodations shall be included to provide a N2 blanket on slurry drums/IBC, day tanks, and bulk storage tanks. Sufficient fluid velocity, controls, filtration, and instrumentation arrangements shall be considered to prevent back diffusion of chemical vapor into the ICA/CDA and/or N2 systems.
- iv. A 15" minimum Color Touch screen color display on operators' panel (HMI). HMI Panels where possible shall use DC power source. If HMI panel with AC power source were to be used, appropriate isolation shall be in place to replace it safely upon failure. All HMI panels shall come with external memory card with sufficient memory space to contain all programming / firmware / trends / alarms for replacement of faulty HMI panel replacement without re-programming.
- v. The main PLC for Slurry Distribution Systems shall be selected based on having dual power supply cards and fed with redundant power source to mitigate against power failure to PLC rack and electrical system. The PLC shall have an external memory card with sufficient memory space to contain all programming / alarms / trends such that a faulty PLC can be replaced by inserting the existing external memory without the

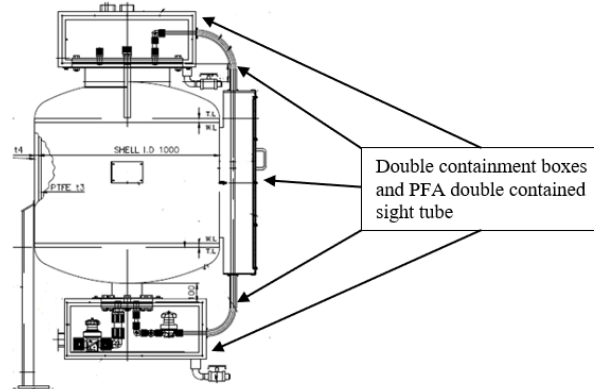
- need for re-programing. Alarm buzzer and three-color light tower for system status indication.
- vi. The controls compartment shall be purged with CDA to keep corrosive chemical fumes away from electrical components with no wetted parts inside.
 - vii. The exhaust for all STU, SBDU, STBU, SDM shall be monitored via a Photohelic gauge (combination of pressure switch and gauge) A N2 pressure regulator shall be provided to step down the house N2 pressure to 35-40psi or as required. All components downstream of the regulator, which are not directly vented to atmosphere, shall be designed to sustain the full house N2 pressure in the event of regulator failure. A pressure switch, downstream of the supply N2 regulator shall be used to provide a system warning/shutdown.
 - viii. All leak sensors shall be fitted with two switch points. One for pre-alarm (warning) and the second for alarm (shutdown). The shutdown alarm configuration shall be "AND" function whereby both warning and shutdown leaks shall be active to trigger a shutdown. Mechanical vertical float sensors shall be used for cabinet leak detection.
 - ix. All pneumatic tubing material shall be either PFA or FEP. There shall be no detachable electrical wire/cable connection in the wetted compartment. All pig tails from sensors etc. shall be long enough to make the connection in the electrical compartment of the associated slurry system. The wire/cable jackets/sheets shall be made out of FEP, PTFE, PFA or any approved chemical resistant material.
 - x. Refer to [Global Facilities TMAH Standard](#) for additional design requirement for safety for TMAH >1%.
 - xi. A summary of cabinet, piping and internal component material by media category for slurry / additive systems are shown in **Table 6** below.
 - xii. Refer to **Appendix 8** - Desired Design Requirements for Different Slurry System Type

Table 6 - Summary of Cabinet, Piping & Internal Component Material by Media Category

Media Category	Cabinet Material	Piping/Internal Components in Cabinet
Slurry / Additive (located in Slurry Room)	PP/PVC with clear PVC viewing panel	- Piping: PFA 451HP - Components: Wetted parts shall be PTFE /PFA / PTFM
Slurry / Additive (located in Sub-Fab)	FM4910 PP or Flame Retardant PVC with clear PVC viewing panel	- Piping: PFA 451HP - Components: Wetted parts shall be PTFE /PFA / PTFM

6.2.4 Day Tank Design and Configuration for SDU

- a) Chemical tanks shall be conical bottom, vertical cylindrical, designed to operate at or near atmospheric pressure and shall be both leak, fume, and vapor tight.
- b) As it is presumed there is no reaction taking place in the tank, day tank shall be designed to operate at atmospheric pressure (i.e., less than 2.5 psig). Based on the additive's properties, the maximum pressured exerted on any part of the tank (usually at bottom of the tank) can be calculated. However, the tank should be able to withstand the forces due to mixing, recirculation and vibration.
- c) The day tank shall have a bottom outlet port of 1inch (25.4 mm) diameter or greater with proper coupling to the pump suction manifold and manual isolation valve. It is not recommended to use PFA/PTFE lined stainless steel pipes or lined manual isolation valves at tank bottom but instead, the preference is to use PFA tubing with outer containment and Teflon valves.
- d) Tank shall have at least two spare flanges with blind flange on tank top according to the largest supply line. Tanks must be provided with one spare flange connection according to the largest discharge line at the bottom of the tank apart from the tank bottom drain line. Manual isolation valves shall be provided for all spare tank connections below media level. Tanks shall be designed with ANSI standard nozzles. Nozzle configuration shall consider layout of plumbing, location of tank, and required equipment.
- e) Depending of additives properties and compatibility, the tank material choices shall be HDPE, Natural PP, PFA Rotomolded tanks or equivalent.
- f) Tanks shall have top and bottom containment boxes with dual leak sensors (warning/shutdown with "AND" Logic) for detecting chemical leaks at the connection ports, discrete redundant level sensors as a primary method of fluid level detection, and 4-20mA radar sensors for analog level monitoring of the tanks. The sight tube for level monitoring shall be double contained. A typical tank drawing showing top, bottom and sight tube containment boxes is shown below in **Figure 4**.

Figure 4 – Typical Tank drawing with containment boxes

g) The ideal Day tank and tank discrete sensor arrangement shall be as follows:

- i. Overfill – Shutdown alarm: Supervised Circuit
- ii. High – Warning alarm / Transfer stop
- iii. Full – Normal operating range
- iv. Fill – Normal operating range (Medium)
- v. Low – Warning alarm
- vi. Empty / Low-Low – Shutdown alarm: Supervised Circuit

System programming should be able to tell if there is a sensor error and send a warning alarm that the sensors are out of sequence. For instance, if the Empty level sensor fails and the Low and Fill sensors are active then the program should be able to generate a "sensor error" maintenance alarm instead of shutting down the system.

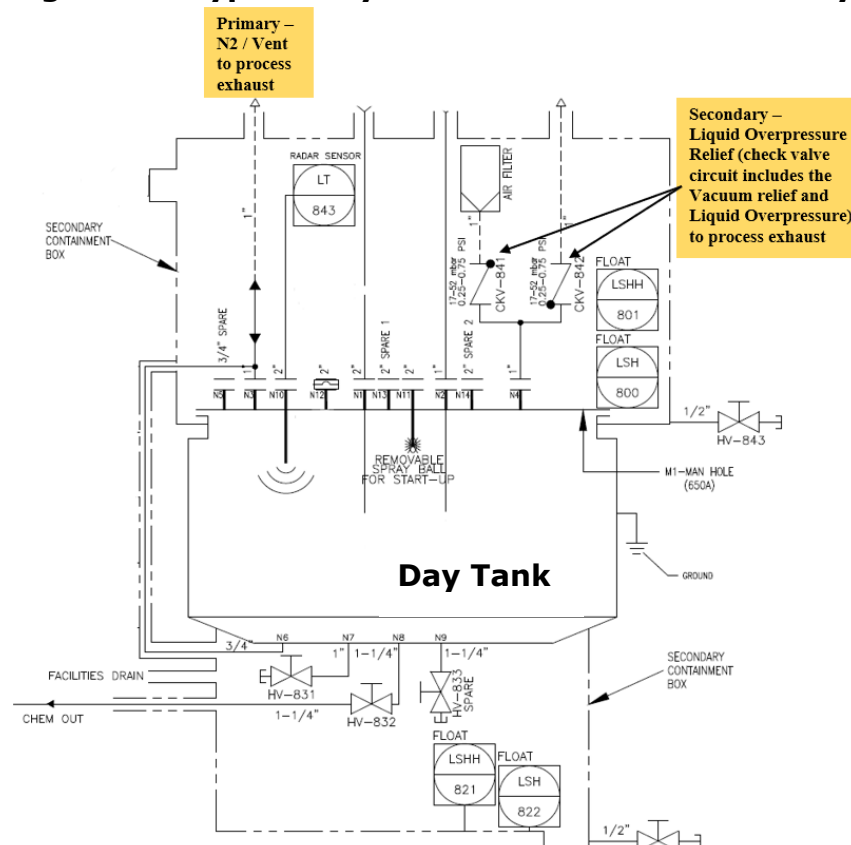
The sensors shall have a force-on bit option (PLC input page in HMI) to allow sensor replacement without stopping the system and access to the page shall be password protected. HMI / SCADA warning alarm shall be generated till the force-on bit is removed.

- h) Chemical compatible seals and gaskets shall be used.
- i) All tanks to be electrically grounded for static charges on tank surface.
- j) For tank top fill / return line, dip tube shall extend to within 6 inches the tank bottom and have a 45-degree cut end for slight mixing effect and prevent splash filling, creating turbulence and static generation. The fill/return dip tube shall have anti-siphon weep holes

(3mm or as per tank manufacturer recommendation) to prevent backflow.

- k) Based on requirements, tank liquid shall be blanketed with humidified nitrogen through blanket/vent connection in the SDU.
- l) Tanks shall be protected from over-pressurization and/or vacuum by approved methods such as conservation vents, flame arrestors, and shall be installed based on local code and safety requirements for chemicals utilized within the tank. Pressure-protection system shall be sized lower than recommended safe tank burst pressure. Typically, protection features are: Primary – N₂ blanket circuit and Secondary- Liquid Overpressure Relief (check valve circuit – includes the Vacuum relief and Liquid Overpressure). See **Figure 5** for Day Tank pressure protection system concept.
- m) Day tanks to be designed to retain 1.2x the 12-hour average daily consumption as a minimum and if space is available, then the ideal design factor shall be 1.2x the 24-hour average daily consumption.

Figure 5 – Typical Day Tank Pressure Protection System Concept



6.2.5 Distribution Piping Network and Valve Boxes

Polishing tools are fed with slurry/additive from the slurry distribution system that consists of single or dual global recirculation loops – one in use and other one is kept as auxiliary for usage during loop flushing. Dual global loops shall only be used for three engine/station SBDU where one engine/station can be taken down for loop flushing without affecting supply operation the other two engine/station. The reason for using the global recirculation loop is to keep the slurry moving and eliminate any possibility of the slurry settling anywhere in the system. The PFA tube in the loop and the pump shall be sized to obtain a minimum velocity of the slurry of 2 foot per second. The contractor as part of the tender for each slurry sub-system shall maintain the required slurry sub-system flow rates based on each slurry/additive's unique requirements.

6.2.5.1 Additives Distribution Piping System

- i. Each Additive chemical distribution network piping shall utilize containment boxes with interconnecting piping to house all primary tubing, valves, and fittings. Distribution pipelines will be routed from Slurry Dispense Area to distribution Valve Boxes (MHB, MHEB, JB etc.) located in the overhead utility racks just below the manufacturing area (at sub-fab zone) via specified piping hierarchies to a free standing Valve Manifold Box (VMB) on a rack on the sub-fab floor to supply the tools as required. The pipe routing shall follow the space management requirements. Vendor shall provide design, fabrication, mounting and installation details, delivery of all materials required, functional test and certification of the entire distribution loop. Pipe routing and installation shall be accomplished in a way such that dead space is kept to a minimum and having no trapped pockets. Based on requirement for different chemistries, future expansion valves for the supply and return shall be included in Main Header Expansion Boxes (MHEB).
- ii. The chemical Valve Manifold Boxes (VMB) will provide uninterrupted supply of process chemicals to the process equipment. The VMB shall contain all primary PFA tubing, valves, and fittings. The VMB shall be provided with an isolation valve, a branch valve, drain valve, a leak tight support structure for the base, and a leak and spray tight clear PVC window. Each chemical valve manifold box will have from four to eight valves sticks to which hook-up piping to the process equipment will be connected.

Each chemical valve stick shall be provided with a sample port valve, an air operated valve, and a manually operated valve. The sample valve shall be located between the manual and air-operated valve (AOV) in each point of use (POU) stick.

- iii. All VMB housings shall be constructed of chemical resistant material, FM4910 approved PP/PVC and will have a transparent and removable cover. All housings shall have a drain valve and dual leak sensors. All MHB/MHEB/JB which are normally installed at a height of ~3-4m, shall be constructed of chemical resistant material, FM4910 approved clear PVC for visibility of internal components from ground and with the exception of TMAH, which shall be PP. All valve boxes shall come from the same PP/PVC supplier to assure a consistent shade of color.
- iv. There is a remote solenoid valve panel (SVP) that contains the solenoid valves that actuate the VMB AOV valves. The remote solenoid valve panel shall be installed below the VMB. The contractor is also allowed to combine a VMB and SVP into a single housing with separate compartments. Refer to **Figure 6** for a typical VMB design.
- v. Hook-up pipelines with processing tools is not included in this scope. The distribution PFA tubing shall be designed with no dead space. The pipelines are to be installed such that complete emptying of the pipeline is possible (no low point, which cannot be drained). PFA tubing between MHB and VMBs shall be in one piece and pull/union box with dual leak sensor to be considered for main supply line if required.
- vi. Each VMB, MHB, MHEB, JB, shall have two leak sensors. In the event of low-level leak, a warning alarm will be activated and upon the high-level sensor activation ("And" logic), shall stop supply from SDM or SBDU. The manual shutdown of an individual VMB shall not impact operation of other VMBs connected to same distribution pipeline.

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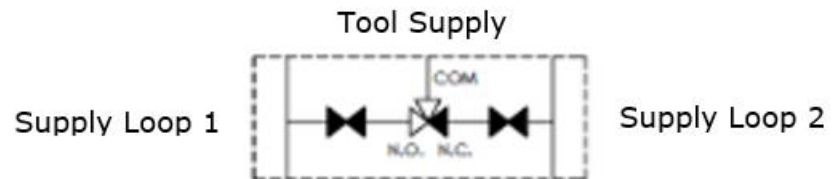
- viii. The following minimum alarms/sensor status shall be provided at the SCADA terminal:
 - a) Leak sensor (Low and High).
 - b) Auto valve (open/close) status.
 - c) Auto valve cycle count.
 - d) Auto valve opening and closing duration with time stamp.
 - e) Door open warning alarm
 - f) VMB Maintenance Bypass alarm
- ix. In addition to the specifications noted above, each VMB or Containment Box shall be supplied with the following features and accessories:
 - a) Provide Levelling feet for the VMB's.
 - b) Provide necessary mounts and brackets to support internal valve (manifold) installations. Design valve boxes to have a leak-tight window for observation. Each box should be factory tested for leaks before installation.
 - c) Provide individual solenoid valves, hot-swappable type for each AOV, located at the respective VMB and mounted on a panel in a solenoid box below the VMB or as a separated compartment in the VMB. The manual CDA/ICA valve shall be located inside the solenoid box or compartment.
 - d) All the control penetrations are to be located on the side of the solenoid box and shall be sealed and liquid tight.
 - e) Each air-operated valve shall be provided with an on/off indicator whose position is clearly visible through the valve box cover. Valves shall also be clearly marked with the manufacturers name and data. Manual valve handles shall have directional or visual indicators to indicate open and closed positions.
 - f) Keyed "maintenance bypass" switch in all valve boxes or software enabled maintenance bypass feature in SCADA with high level password protection shall be available to safely change-out faulty leak sensors. The respective valve box maintenance bypass alarm shall appear in SCADA until the bypass is turned-off.

- g) Valves shall be laid-out to support the installation of COHE(Lockout/Tagout) devices.
- h) Valve connections shall be of the same size as the tube/pipe in which they are installed unless stipulated otherwise.
- i) The valve box shall be designed in such a way that a leak shall flow to the location where the drain valve is located.
- j) The wetted compartment door shall have both door warning sensor and lockable latches instead of multiple screws as to allow operator quick access.
- k) Standardization of valve manufacturer/type and fitting shall be required.

6.2.5.2 Slurry Distribution Piping system

- i. Each slurry distribution piping loop shall utilize containment boxes with interconnecting piping to house all primary tubing, valves, and fittings. Distribution pipelines will be routed from Slurry Dispense Area to Slurry Distribution Valve boxes located below the raised floor located at CMP tool functional area in the fab. The pipe routing has to follow the space management requirements. Vendor shall provide design, fabrication, mounting and installation details, delivery of all materials required, functional test and certification of the entire distribution loop. Pipe routing and installation shall be accomplished in a way such that dead space is kept to a minimum and having no trapped pockets.
- ii. Certain slurry loops need to be cleaned as the particles agglomerate over a period of time and can cause some wafer issues. Therefore, in order to ensure continuous slurry supply to the process tools while loop flushing is being carried out, two loops can be installed instead of one loop. Both loops shall go through all VMBs and all VMB sticks shall be connected through valves. By charging and operating the relevant loop with using the appropriate valve configuring in VMB, continuous supply of slurry becomes possible without any interruption.
- iii. The dual loop with POU tool supply/return utilizes a block valve to switch between loop 1 or loop 2 as shown below. A data cutsheet of a typical Swagelok block valve is shown in **Appendix 7**. The block valve

comprises of a 3-2way valve used for the purpose of switching between supply loops and an additional air-operated valve for each supply loop which acts as a double isolation valve. Similar arrangement shall exist for the return loop. Refer to **figure 7** for detail. Replacement of the isolation valve can be achieved by switching loop.



- iv. At any one time, only one loop will be active (in operation) and other loop shall remain idle & dry and shall only switch loop upon operator intervention via SBDU HMI.
- v. Depending on slurry properties and site's preference, a tool supply and return hook-up line might be needed where the tool has the provision to cater for supply/return configuration and shall be decided during design stage. Refer to **figure 7, 8 & 9** for single loop with VMB tool supply/return and Dual loop with VMB tool supply/return design concepts. If at design stage, it had been decided to proceed with single loop and single POU tool install line without return, the design shall generally follow as per **figure 9** but without tool return line. For both VMB design concepts, the minimum number POU shall be 4 to a maximum of 6.
- vi. To cater for different type of tool install supply/return connections within the CMP tool based on tool Make & Type, **figure 10** shows an example of a Dual loop POU supply/return VMB design configuration connected to different types of CMP tool. If there are different type of tool internal configurations not stated in **figure 10**, design shall into the consideration the necessary changes as needed.

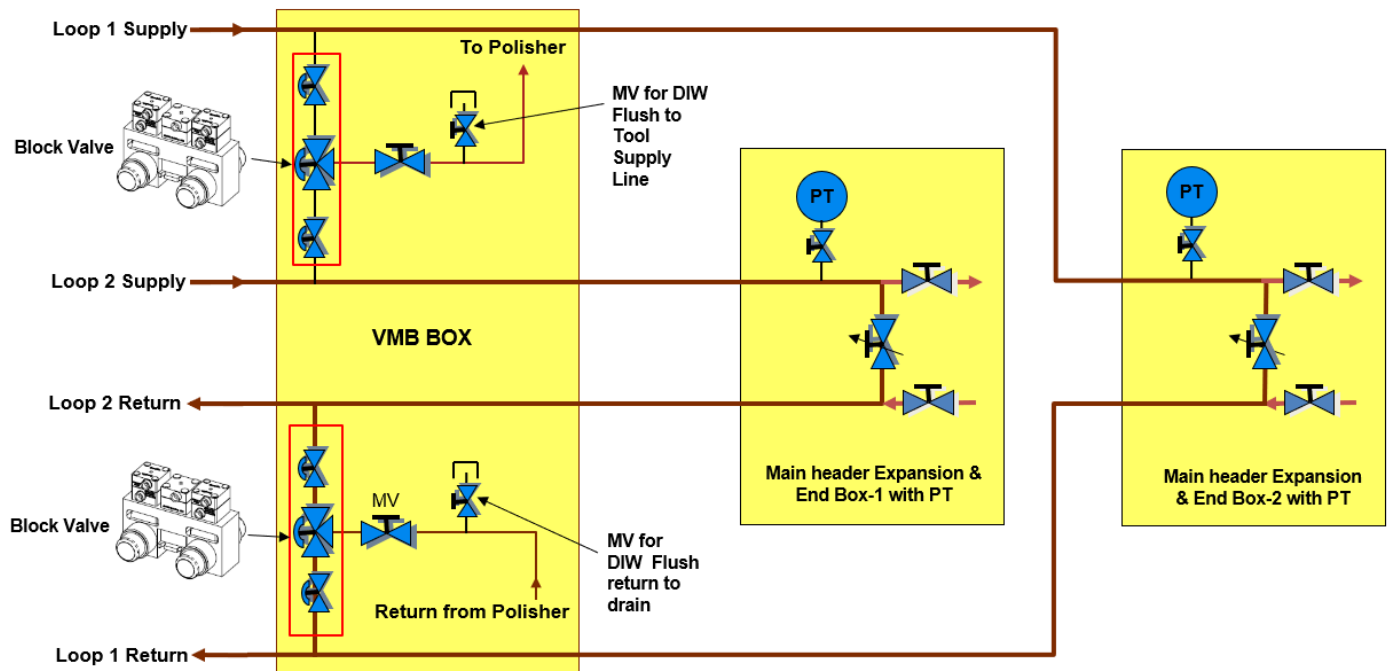
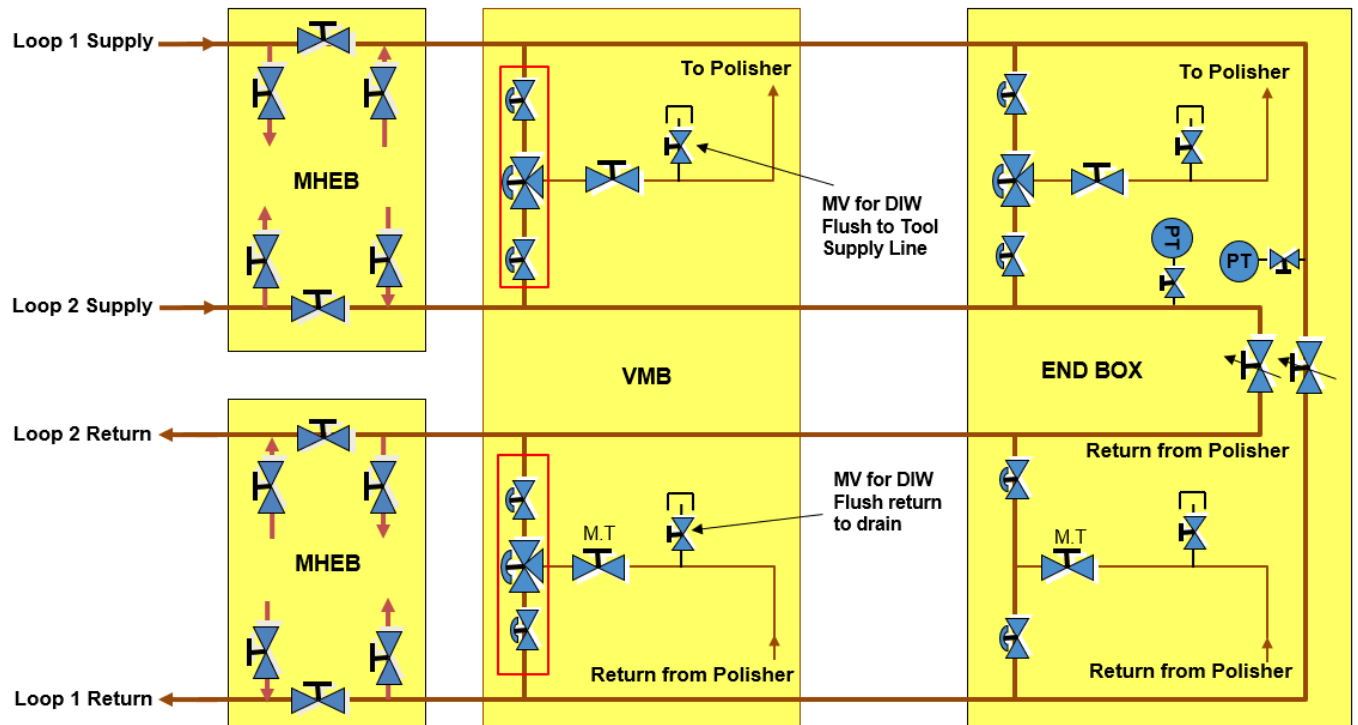
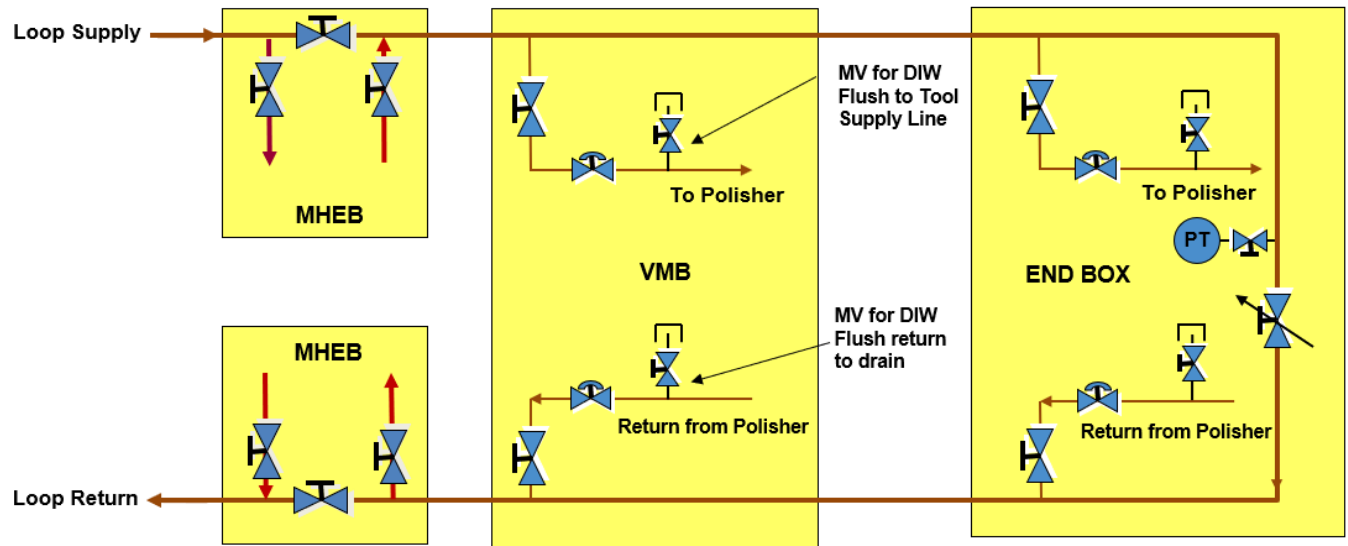
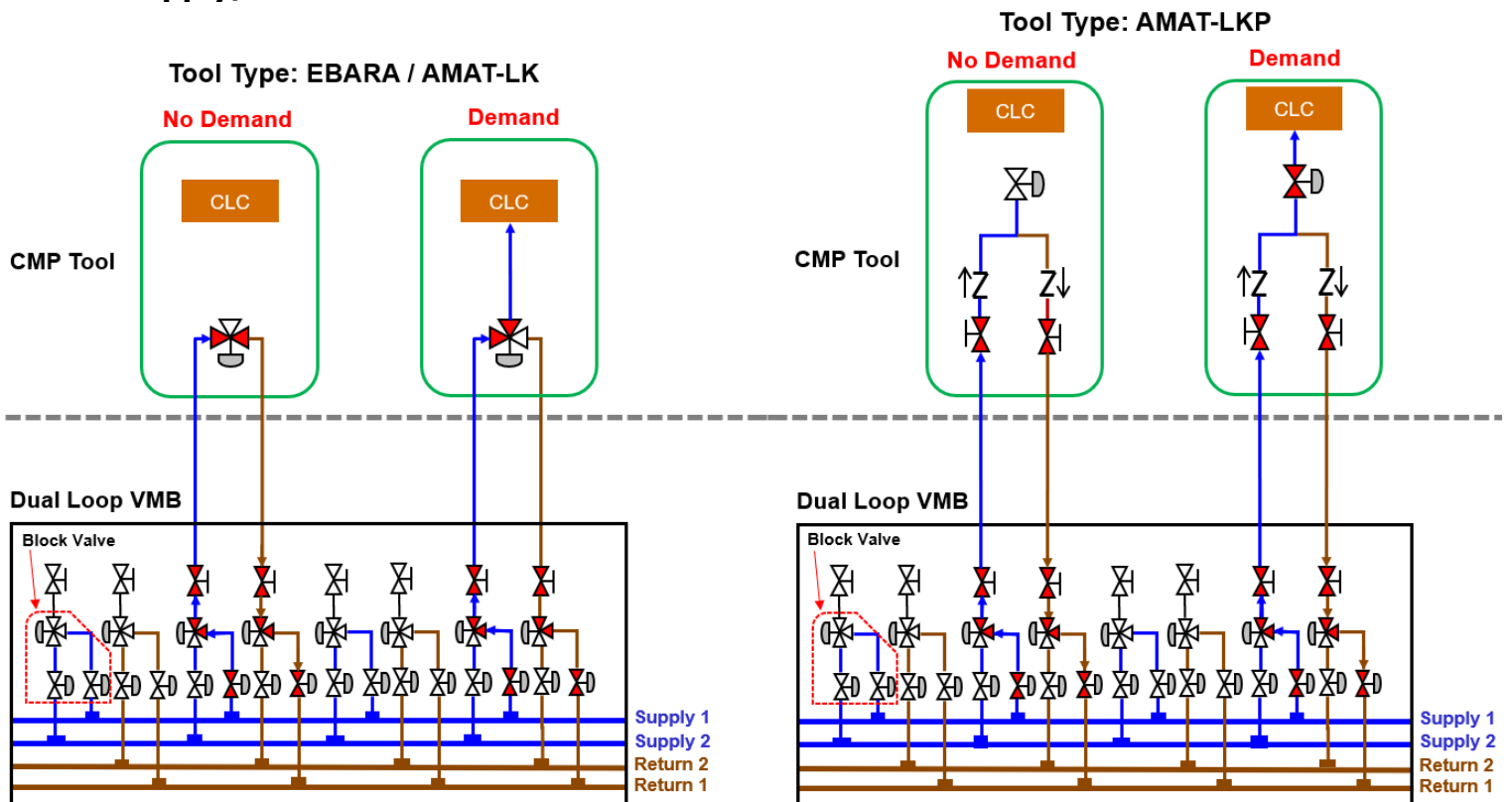
Figure 7 – A Typical Dual Loop VMB with POU Supply/Return and combined MHEB & EB Configuration**Figure 8 – A Typical Dual Loop VMB with POU Supply/Return with separate MHEB & EB Configuration**

Figure 9 – A Typical Single Loop VMB with POU Supply/Return Configuration**Figure 10 – Typical VMB to CMP Tool configuration for Dual VMB with POU Supply/Return line**

6.2.5.3 Distribution Network Design Concepts

- a) Main Header Distribution (MHD) Concept - Application: Additive Chemical
- Additive loop passes through MHB and the tee-off line is connected to a respective a VMB or a maximum of two VMBs. MHB shall have dual leak sensors and isolation manual valve to isolate the VMB from loop. The VMB configuration shall follow the typical VMB design in **Figure 11**.
 - The VMB design provides additional flexibility to expand to another one VMB instead of expanding from MHB if flow/pressure drop calculations allow. This shall be the **Preferred** loop design concept for Global loop re-circulation. This design provides ease of VMB relocation without shutting down the entire loop and provides better loop pressure management as there is no significant dynamic pressure drop. The typical MHD concept design and different configurations of MHB and MHEB are shown in **Figures 11, 12 and 13**.

Figure 11 – Typical Main Header Distribution Concept

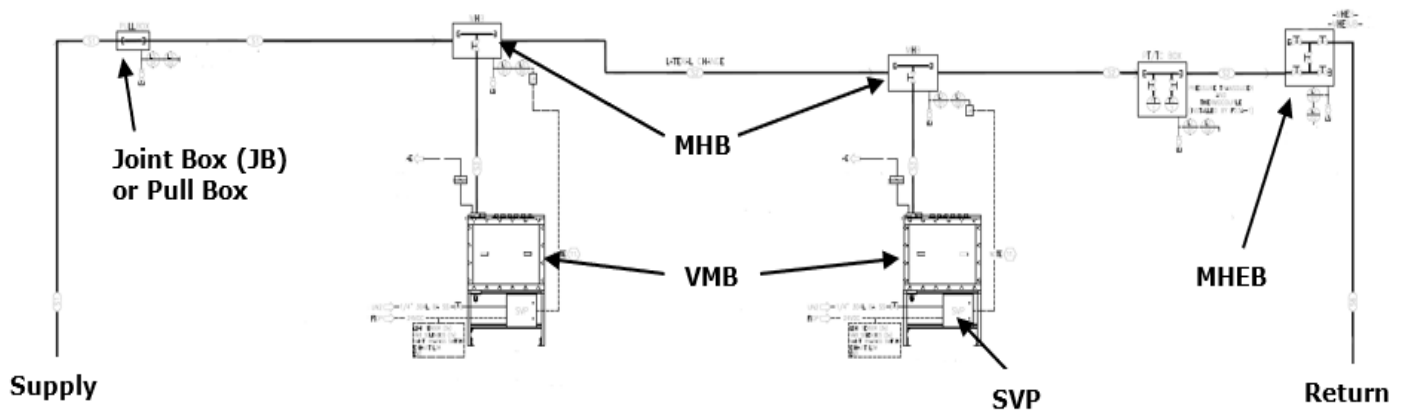
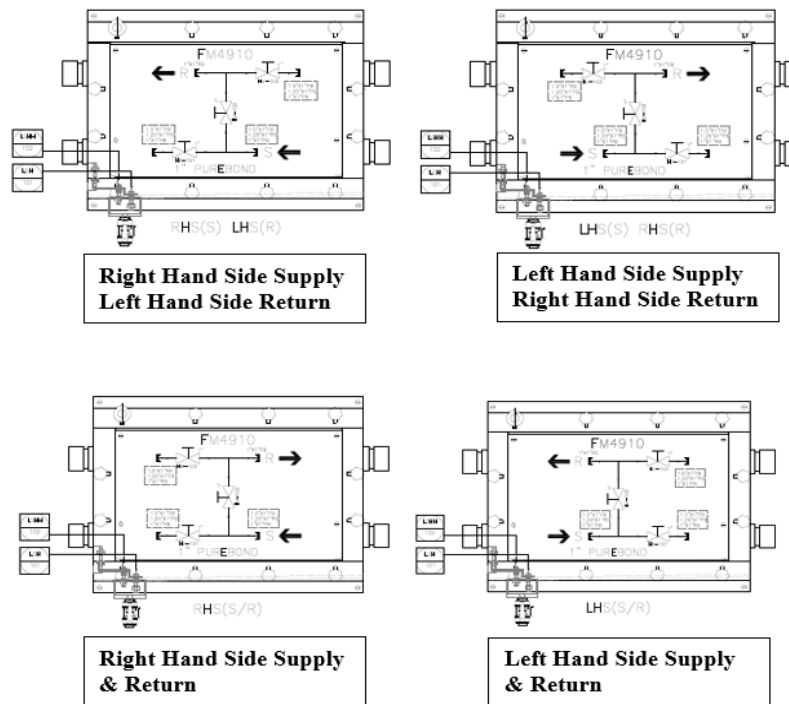
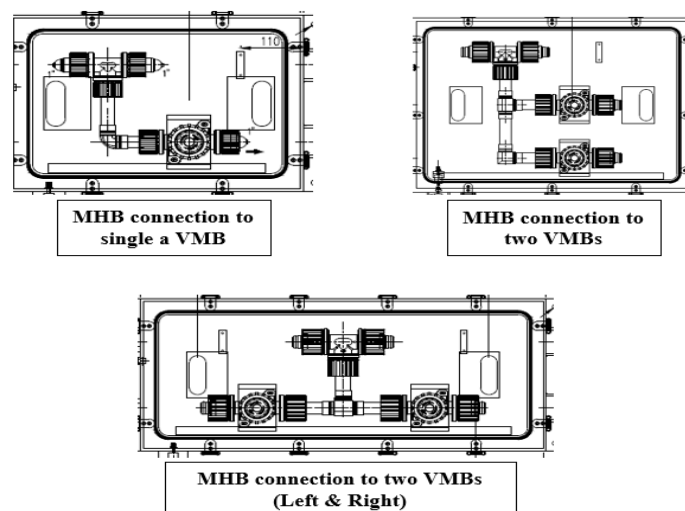
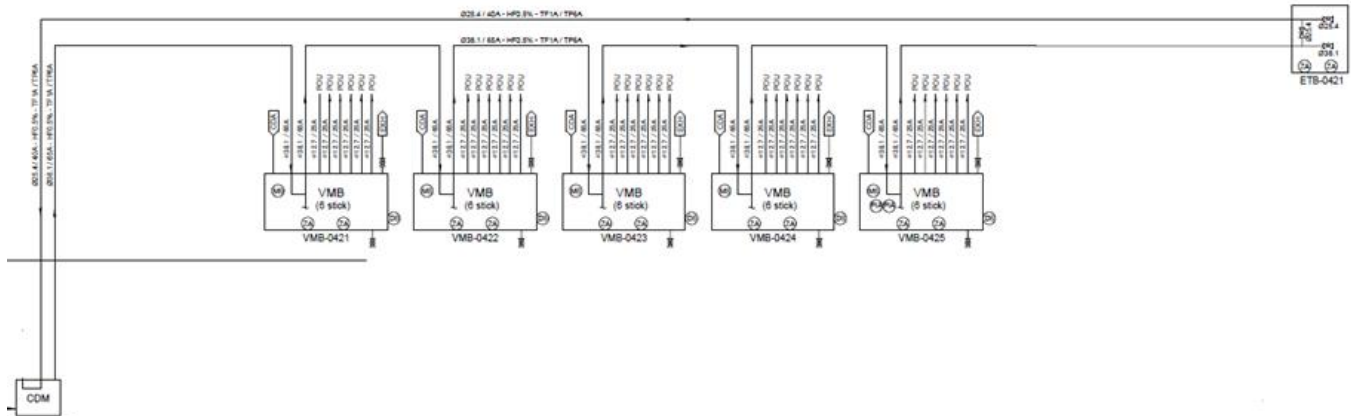


Figure 12 – Different MHEB Typical Configurations**Figure 13 – Different MHB Configurations**

b) Daisy Chain Distribution (DCD) Concept – Application: Additive Chemicals

- i. In this concept, MHB shall be incorporated into the VMB itself. However, the loop will have additional bends in order to fulfill the space requirement of the VMBs. When Tee-Off is taken from loop in the VMB, there shall be an isolation Tee-Valve so that if the particular VMB has any leak within the VMB, the required VMB can be isolated without having any impact to the loop. Typical Daisy Chain Distribution design is shown in **Figure 14** below.
- ii. This is **NOT** a preferred loop design concept as VMB relocation requires shutting down of the entire loop and there will be dynamic pressure drop as well. This design shall **ONLY** be applied if there is space / budget constraint and shall be decided during design review. The VMB design shall follow as per **Figure 6**.

Figure 14 – Typical Distribution Loop with Daisy Chain VMB Concept



c) Flow Through Distribution (FTD) Concept - Application: Abrasive Slurry and Additives:

- i. The slurry/additives loop passes through VMBs in such a way that elbows, and tees are avoided as much as possible. In this arrangement, the Slurry / Additives supply shall go through the VMB manifold and exit after last POU stick and returns back to the SBDU/SDU. Tee-manual or block valves shall be used at each point of use stick to prevent stagnation of slurry.
- ii. In order to have close proximity of the VMB to the CMP tool, VMBs shall be installed under the raised floor in the cleanroom at the location of the CMP tools. The ideal space allocation below the raised floor is along the front of the CMP tools (below the walk path) where there is minimal services conflicts. Where possible, the raised floor panels, shall be clear visible type. Refer to **figure 15 & 16** for

typical Flow Through VMB distribution layout and installation concept and **figure 17** for different configuration of main header expansion box (MHEB) and End box (EB).

- iii. Depending on slurry properties and needs, Point of Use (POU) supply and return from tool shall be incorporated in the VMB design and shall be determined during the design stage.
- d) Minimum requirements and the relevant components for distribution network are given below in **Table 7**.
- e) Materials for distribution network by media category is shown in **Table 8**.

Figure 15 – Flow Through VMB Installation guidelines below Raised Floor in cleanroom

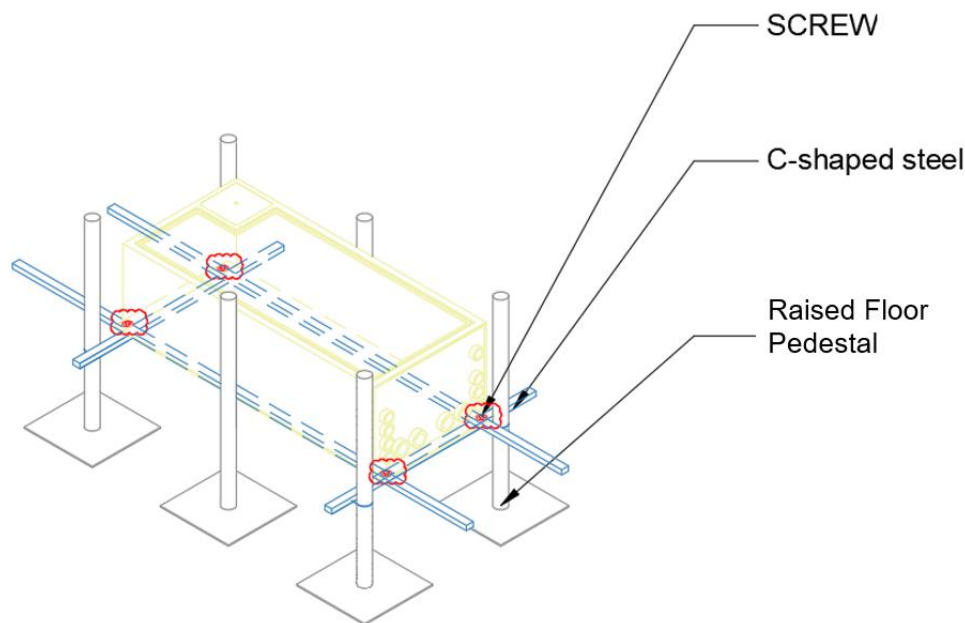


Figure 16 – Flow Through VMB Distribution layout below Raised Floor

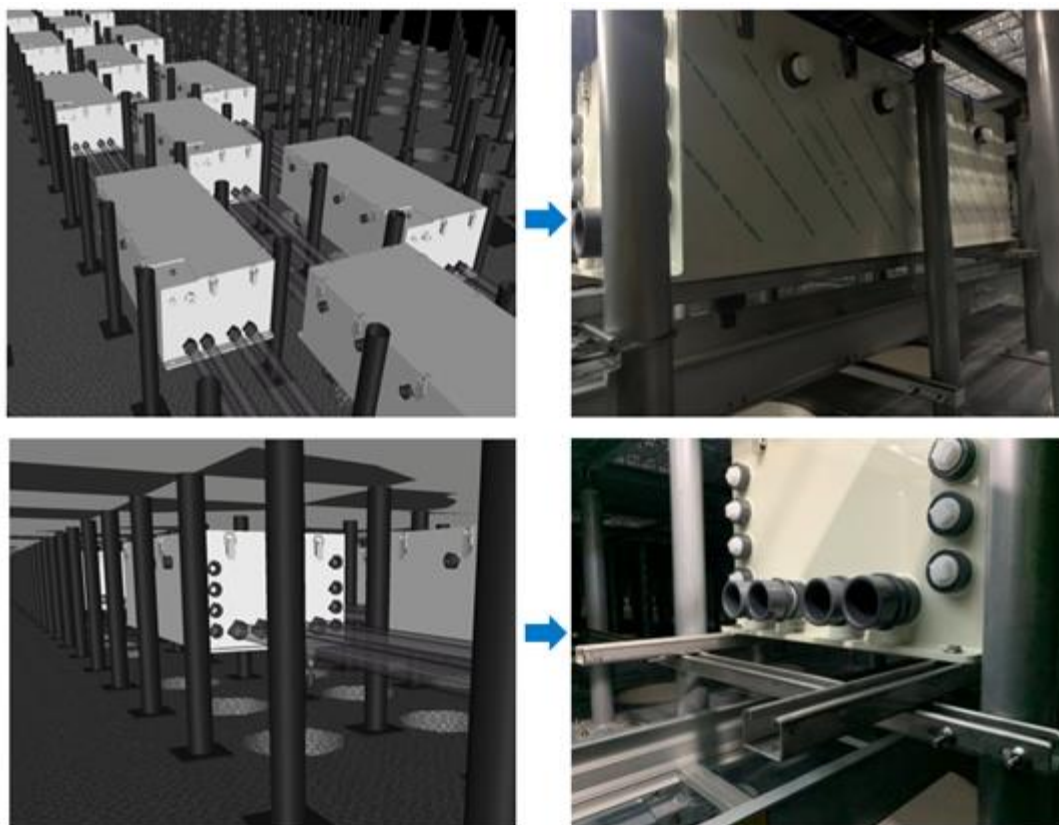


Figure 17 - Different configuration of MHEB and EB

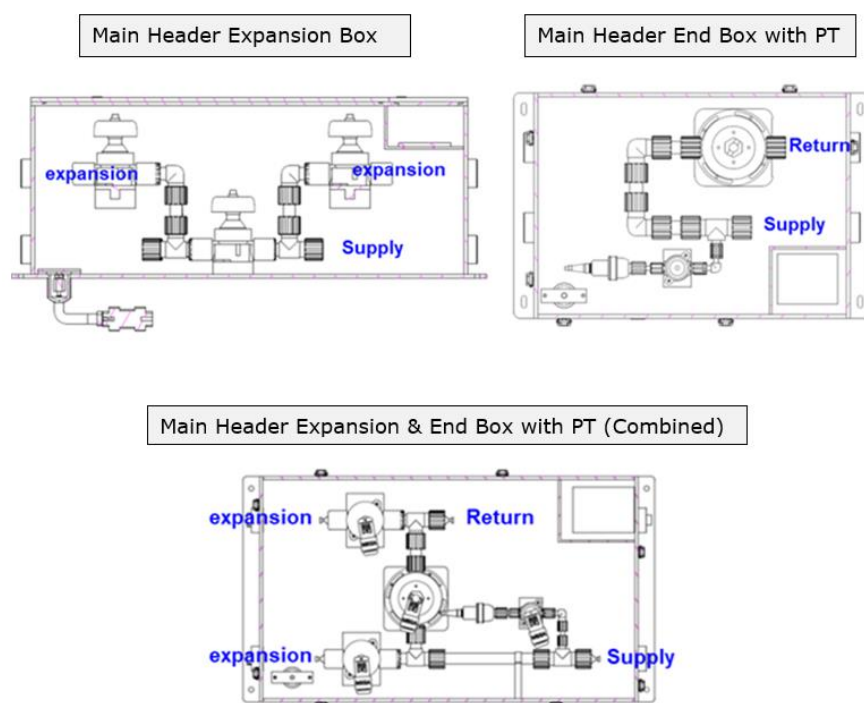


Table 7 – Minimum Requirements and Relevant Components

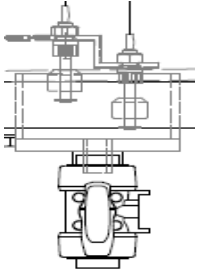
Description	Minimum Requirements	Remarks
Distribution Supply Piping Size	Minimum size of 1" for Global loop supply and ¾" for Global loop return.	Actual pipe size shall be based on capacity requirements, pressure drop calculations, and minimum loop velocity based on slurry requirements.
Dual Leak Sensor (Warning/Shutdown)	Dual Leak sensor configuration shall be "AND" Logic. Mechanical vertical float sensors for all valve boxes. The configuration is as shown on the right, figure 18. VMB leak sensor cable shall be seal tight with chemical resistant cable.	 Figure 18
Connection / Fittings	Pure bond and/or PrimeLock® /Super-300 Type Pillar Fitting® or Flaretek® connections	Space saver fittings are not preferred for Flaretek® connections
Each POU at VMB	Flush & purge manual Tee-valve (To purge/flush or perform DIW flush of tool install line).	MHEB & EB does not require flush & purge manual Tee-valve.
All Valve boxes	Door warning sensor and lockable latches. Door panel O-ring shall be Viton or other materials based on chemical compatibility. All PT with manual isolation valves for ease of replacement.	
Solenoid Valve Panel (SVP)	Hot swappable solenoid valves. Main Incoming CDA/ICA manual valve location inside SVP. Pneumatic Tubing for AOV actuation shall be PFA or FEP.	

Table 8 – Materials for Distribution Network by Media Category

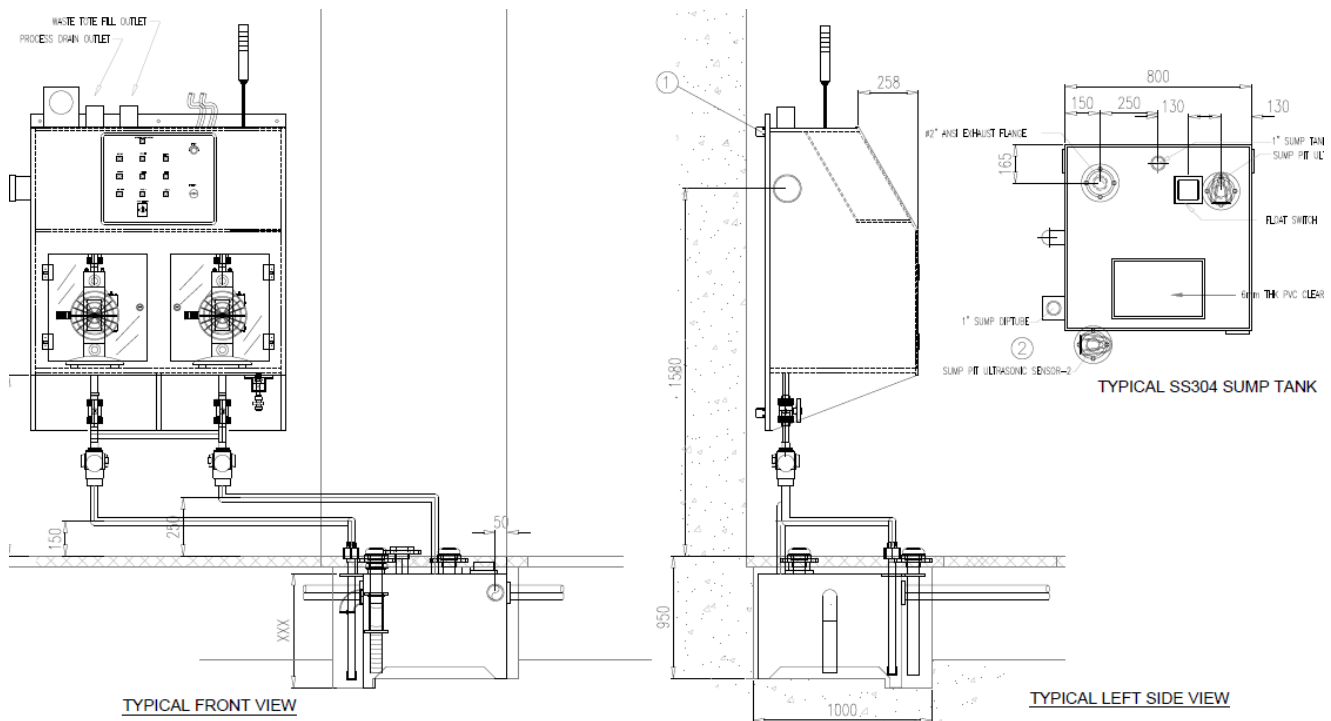
Media Category	VMB / MHB / MHEB / JB Construction Material	Piping Material in VMB / MHEB	Distribution Piping Material
Slurry / Additive	VMB: FM4910 White PP/ Flame retardant PVC MHB/MHEB/JB connecting to free standing VMBs in sub-fab: Flame retardant Clear PVC Slurry MHEB / EB (below raised floor): FM4910 White PP/ Flame retardant PVC	PFA 451HP	Primary: PFA 451HP Secondary: Schedule 40 Clear PVC

6.2.6 Slurry Room Sump System

- a) Each Slurry room is equipped with containment trenches and sump pits to contain any potential chemical spill inside the room. All waste from the slurry system cabinets shall be either gravity drained or have pressurized drain lift functionality to the appropriate exhausted sump tank in the respective rooms. Drain slopes in the slurry room shall be a minimum of 1%. Each room sump pit is equipped with a sump tank, sump tank level sensor, sump pit leak sensor and sump pump. Where an exothermic reaction can occur a tank temperature sensor shall be provided with N+1 redundancy.
- b) Each sump system shall also have a local control panel responsible for all operations and indicate status of leak sensors. The control panel shall be purged with ICA. Audible alarm shall be activated in the event of leak in the sump box or sump pit. The sump pump system should have ability to operate both in manual and auto mode. The following minimum alarms/sensor status shall be provided at the SCADA.
 - i. Sump tank level sensor (Low, high and High-high).
 - ii. Sump pit level sensor.
- c) Sump stations shall not be wall mounted (where possible) but mounted on a stand or a standard cabinet with leveling legs similar to SDS equipment cabinet. The sump systems are dual pump, PLC driven modules that pump liquid waste to bulk waste tanks for further treatment or disposal.

- d) Each sump system includes two pumps, designated Pump I or Pump II. A selector switch on the sump system allows the operator to select which pump will be used to evacuate the sump tank or sump pit. The sump tank collects waste from the various systems in each slurry room via a gravity drain system. The sump pit accumulates the liquid from the trenches and overflow from the sump tank in the event of abnormal system operation. If the operator does not select a primary pump, the system will be placed in an idle mode. Sump tank and sump pit level is monitored by a pair of analog ultrasonic level sensors. A float sensor is located in the sump tank to indicate High-High sump tank level. The sump tank and sump pit are mounted below the main floor level to allow for gravity fed fluid to drain into the sump tank. Contractor to assure the sump tank is anchored for tank buoyancy up to 2 meters. The pumps, PLC, indicating lights, and audible alarm are contained in an enclosed housing module. In addition to monitoring the level of the sump tanks, leak detection is provided in the pump and control box, which will shut down the system if a fluid leak is detected. A typical PFD of a sump pump system is shown in **figure 19**.

Figure 19 – Typical Dual Sump Pump Lift Station & Pit Tank



6.2.7 Power Distribution Design Criteria:

The Design of Power Distribution System for slurry systems is very important for continuous operation of the slurry systems and therefore, shall be designed to provide a continuous power supply to slurry distribution equipment and the respective controls systems. Power Distribution System shall mainly consist of Main Power Distribution Panels, Room power panels and all required cabling and connections to each piece of equipment. The slurry distribution system contractor shall supply the Main Power Distribution Panels for termination of the feeder cables. The room power panels can be placed generally outside the slurry rooms unless any other location is specified due to space constraints.

The AC Main Power Distribution Boards are the main receivers of incoming power for most of the Electrical and Control systems. Incoming normal utility power shall be continuously monitored in the Panel. Any anomaly such as voltage sags will cause the internal circuitry to command to transfer to the standby source while additional switch circuitry determines the source has the proper voltage and frequency. When normal utility power returns or no anomalies have occurred, the operator can then manually transfer back to utility power when it is deemed appropriate. When one bus is under maintenance, other bus which supplies power through dual power supply maintains power in the panel and thus to slurry systems.

Following points shall be generally considered in the power distribution design for the slurry systems.

- a) Power distribution design to address electrical redundancy and single-point-of-failure mitigation.
- b) Any AC CDU or AC stirrer directly supplied from the Dual Feed AC Main Power Board requires dual feed power supplies.
- c) Points of connection (By Others) for Normal Power and UPS/CPS (Uninterrupted/Continuous Power System) power will be provided in the area designated for the process slurry Systems and terminated at fusible disconnect switches in the Electrical Room – slurry system contractor shall liaise with electrical contractor to ensure the feeder cables are terminated in the slurry electrical panels as per agreed design.
- d) Connect critical equipment (i.e. cannot withstand prolonged power interruption) to UPS/CPS power. There will be no interruption to UPS/CPS in the event of a momentary or prolonged Normal Power outage. However, the contractor shall have to make sure how critical equipment will be in operation in the event of power supply lag. Equipment connected to Normal Power will experience a power loss in a

Normal Power outage and remain offline until Normal Power is restored.

- e) As power source supply from grid is 400 VAC, 3 phase, 50 or 60 Hz (depending on region/country), provide transformers and panel boards to derive 380V, 200-240V/100-120V, 24VDC or other 3-phase and single-phase sources.
- f) Furnish and install motor control centers, VFD's, instrument power transformers, electrical enclosures, equipment power disconnects, conduit, and cabling for proper operation of the System
- g) If Solvents are used, electrical for solvent collection and transfer operations shall be Code compliant.
- h) Provide protective devices within the slurry Electrical equipment to prevent the propagation of electrical faults generated within the slurry dispensing rooms or within slurry dispensing or distribution equipment
- i) Label electrical equipment (panels, transformers, motor control centers), conduit and wire per the Equipment Coding specifications in the Electrical Specification Design Standard.
- j) Contractor to follow project Ferrule requirements as specified in the Electrical Specification Design Standard.
- k) Contractor to follow project electrical panel (MCC's) spare requirements as specified in the Electrical Specification Design Standard.
- l) Future Circuits: 20% spare or as specified in the single line diagram, whichever is greater.
- m) The Electrical distribution will be modeled in a Power Systems Study Program by others. Prior to equipment installation. Contractor to provide loading data for panels and equipment within the slurry transfer / dispensing rooms for incorporation into the database. Power Systems administrator will review submitted load data to ensure panels are not overloaded.
- n) Do not install liquid piping immediately over or within a 3-foot plan view clearance of any electrical panel, motor starter, or control panel. Where piping must be located within these zones, either install piping inside a PVC conduit or shield the electrical device to prevent direct liquid access to electrical equipment. Locate piping such that bottom of pipe (or pipe covering/shield, if pipe is covered/shielded) is not less than a clear height of 6 feet, above the highest part of electrical equipment located directly below the pipe.
- o) Ingress Protection requirements for Main Power Distribution Panels, Room power panels, electrical panels for individual

slurry systems or other related equipment, motors, etc. shall be according to the zonal specifications stipulated in Electrical, Control, and Monitoring Specifications.

- p) Following electrical design configurations and parameters shall be observed in the electrical design with respect to slurry systems:
- i. Slurry systems are typically connected to direct dispense polishing tools and as such shall be equipped with centrifugal pumps. The pumps controls shall be powered from two separate power sources to provide maintainability (A/B philosophy): a normal power source and a UPS source.
Note: Power Distribution Panels (PDPs – 24VDC) or Power Distribution Units (PDUs – 24VDC) are not recommended for power to slurry systems. If they are utilized, they shall have redundant power supplies that can automatically switch over upon a failure and the incoming power supply shall have SAG mitigation or a UPS feed.
 - ii. For protection from power disturbances, the slurry dispense system controller for a direct dispense system and all systems where continuous control is required shall be powered with a UPS source.
 - iii. Power requirements such as UPS, SAG mitigation, and normal power for slurry systems and components shall be applied based on system and tool needs.
- q) Refer to Electrical Specification Design Standard for additional required scope of work. Link: [Global Facilities-Electrical Specification Design Standard](#)
- r) If there is any discrepancy between the above electrical requirements and specifications mentioned in Electrical Specification Design Standard, specifications shall be followed after obtaining approval from Micron.

6.2.8 Instrumentation, Control and Monitoring

The slurry Instrumentation, Control and Monitoring System shall be designed to control, monitor and report leak sensor activity for the distribution piping networks, slurry system activity (events) and alarms, room sumps, and as well as process tool requests based on the requirements. The main components shall consist of process instruments, slurry system PLCs, Data Concentrator(s), a SCADA terminal, Field I/O Panels (RIO), Master and Room EMO panels, sump station controllers, an Ethernet cable network for communication between the various control system components, and a network for communication with various operation stations. Brand & Model of PLCs,

acceptable brand for I/O & Other Cards, Power supply unit and OS Platform and version of SCADA shall be advised by Micron. SCADA shall be located in the TGCM Room. The Field I/O Panels shall be located as required by VMB requirements and tool requirements. Following points shall be observed for design and design configuration of instrumentation, control and monitoring:

- a) Slurry dispense systems shall be equipped with a Programmable Logic Controller (PLC) to execute all sequences of operations, and a Human Machine Interface (HMI) to allow interfacing with all set points, alarm thresholds, and loop tuning parameters.
- b) Data Concentrators used for control & communication between CDU, distribution network and tools shall be equipped with primary and secondary PLCs with hot back-up feature for redundancy.
- c) A "System Available/Not available" status signal shall be available for all tools.
- d) A "Chemical demand" signal shall be included in the PLC input as feedback for the slurry system based on tool demand.
- e) All analog signals and alarms shall be monitored and historized by some type Facilities Monitoring Control System (FMCS) or System Control And Data Acquisition (SCADA) system. The following information shall be provided as a minimum on the SCADA screen:
 - i. Individual system status.
 - ii. Network and PLC communication status.
 - iii. All system alarms and EMO.
 - iv. Individual equipment P & ID's.
 - v. VMB/MHEB and other distribution box leak sensor status.
 - vi. Valve Box and Tool activity
 - vii. Monitoring & Control of slurry/additive systems
 - viii. Chemical consumption monitoring and reporting.
 - ix. Preventive maintenance monitoring and reporting, such as valve cycles, filter hours, pump stroke counts, analyzer and instrument calibration intervals.
 - x. Room sump leak detection (status).
 - xi. System performance trending.
- f) All instrumentation and control equipment shall be utilized within the manufacturer's recommended operating parameters.
- g) Leak detection shall be provided within the Slurry Distribution system and response strategies shall be

identified based on site-specific safety requirements, chemical properties, and local/national codes and regulations.

- h) PLC and instrumentation cabinets/panels shall have exhaust and Z- purge incorporated into the system design to prevent and control corrosion of electronic components.
- i) Metrology shall be monitored with the SCADA system, and related data returned for trending and troubleshooting.
- j) Temperature transducers or thermocouples that are compatible with corrosive chemical or slurry and environments shall be used when temperature control is identified during the design phase.
- k) To prevent filters from clogging, the Maximum Delta Pressure (dP) through the filter assembly shall be monitored via the pre and post filter pressure transducers. The PLC will have the ability to calculate dP and shall be settable within local PLC and SCADA alarming protocols. The dP set point shall be identified based on slurry properties and consumption, tool process, and design criteria.
- l) Pumps are switched over based on settable timers built into the PLC program or other means. For centrifugal pumps, a pump start sequence shall be implemented with a dedicated sequence of operation to prevent abnormal pressure fluctuation.

6.2.8.1 Tool Signal Interface – SBDU/SDU to Process Tool

- i. The slurry Control System requires certain inputs from the POU's (Points of Use, typically FAB production tools) to function properly.

Each process tool communicates with the Slurry Control System through the hard-wired tool interface. The tool interface consists of dry contact signals between the tool and a FIOP. The FIOP are responsible for monitoring leak sensors, activating the distribution system valves, and providing all signal interfaces to the manufacturing tools. FIOP shall be configured as required to provide the necessary inputs and outputs for the location in which they will be installed.

- Chemical Demand Signal

For each chemistry stream utilized by a process tool, a Chemical Demand signal is required; if a tool uses the same chemical from two sources (i.e., 2 valves within the same or different valve manifold boxes), a separate Chemical Demand signal is required for each stream. A Chemical

Demand signal is the result of a contact closure at the POU. The signal that passes through this contact closure originates at the FIOP or chemical dispense unit depending on the specific system. Facilities provides IO dry contact and 24vdc received from Tool.

- Chemical Ready Signal

The Slurry Control System generates Chemical Ready signals to the process tool for each chemistry stream programmed; all of these relays are installed even though it is optional for the tool to pick up these signals. If not used, a jumper will be installed at the RIO panel. Similar to the Chem Demand signal, if a process tool uses the same slurry from two sources, a separate Chem Ready signal is sent for each stream.

The Chemical Ready signal to the process tool is an accumulation of the distribution system status (SBDU and valve boxes) up to the Air Operated Valve (AOV) within a VMB. A Chemical Ready signal is the result of a contact closure in the slurry dispense unit. The signal that passes through this contact originates at the FIOP or POU depending on the specific system. Facilities provides IO dry contact and 24vdc received from Tool.

- VMB – Demand Duration & Demand Interval Timers on SCADA

For each POU, the control system has the ability to enable or disable alarms for the monitoring of the duration of each Chem Demand signal, and the frequency of those demands. These alarms are known as "Demand Too Long" and "Demand Too Soon". Enabling this feature at SCADA/FMCS enables both alarms.

For each POU, from SCADA, Maximum Demand Duration and Minimum Demand Interval Duration values can be set. These set points are values from 0 to 999 minutes. If enabled, and if a POU's Chem Demand signal stays ON beyond the Maximum Demand Duration set point, then a Demand Too Long alarm is issued. If enabled, and if a POU's Chem Demand signal goes ON prior to the Minimum Demand Interval Duration set point, then a Demand Too Soon alarm is issued.

In either of these cases, the POU's AOV within the VMB remains open. These alarms are indicative of problems within the tool, such as the slurry flowing directly to drain.

- SBDU / SDU Signal Interface

The SBDU/SDU in the distribution system are running under Constant Pressure operation. Under Constant Pressure, the SBDU/SDU's tunnel valve is always open, and the SBDU/SDU is always delivering slurry/additive to the fab, regardless of process tool requests. The operating status condition of a dispense module shall be visible at the local SCADA and FMCS screens.

6.2.8.2 Control Panel and Enclosure Fabrication Guidelines

- Fabricate control enclosures larger than 48 inches by 36 inches as freestanding panels fitted with removable lifting lugs and legs minimum 12 inches high. Provide steel nuts and bolts with gaskets to fill holes after removal of lifting lugs.
- Fabricate control enclosures 48 inches by 36 inches or smaller suitable for wall or a strut rack mounting.
- Standard size prefabricated enclosures manufactured by Hoffman or Rittal or approved equal.
- Enclosures will be NEMA 4X for wet, damp and corrosive locations; NEMA 12 for all other locations.
- Oil-tight latching mechanisms with keyed-alike locking handle. Tape keys to inside of doors.
- Gasketed overlapping front access doors, with heavy gauge hinges and three-point latch rods with rollers.
- Internal LED panel light with safety cover, end cap, and convenient on/off switch. Provide one fixture for each 48 inches of panel width.
- 9 inches by 12 inches by 1-inch print pocket to store arrangement, schematic, and loop drawings.
- Minimum 14-gauge steel.
- Electrical power sockets that are isolated from the control wiring.
- Cooling package (e.g. fan exhaust, enclosure air conditioner coolers) sized for published heat rejection information for powered components. Incorporate integral filters and sealing gaskets into fans and exhaust vents without compromising the panel environmental rating.

- xii. Install enclosure heaters on enclosures that will not be located within climate-controlled spaces to prevent condensation and resulting corrosion.
- xiii. All field wiring shall be bottom entry.
- xiv. Provide wiring between terminal blocks and panel equipment.
- xv. Fasten components (e.g. terminal blocks) to interior surfaces of panels. Mount operator interface components on front access door/panel.
- xvi. Install copper grounding bus bar and studs and provide 10% (minimum of 3) spare taps and screws for future and field grounding use.
- xvii. Provide ground cable connection over hinge between sub-panel and door.
- xviii. Provide separate wireways for AC and DC signals. Separate AC wiring from DC wiring. Maintain a 4-inch minimum spacing between parallel runs of power and control wiring and electronic instrument wiring.
- xix. Size wireways so that total cross-sectional area of installed wire will not exceed 20% of cross-sectional area of wireway. Locate wireways to eliminate interference with access to panel-mounted instruments or devices. Do not encroach on space allotted for future instruments and do not install terminal strips and splices in wireways.
- xx. Group wires to panel-mounted instruments and switches through wireways or cable ties.
- xxi. Provide finger-safe terminals and components per IP-20.
- xxii. Cut, punch, or drill panel cutouts and smoothly finished with rounded edges.
- xxiii. Provide one-piece back panels.
- xxiv. Mount components to the back panel using machine screws or bolts with tapped holes according to manufacturer's recommendations. Provide lock washers with bolts and screws.
- xxv. Mount components such that:
 - Adjustments can be made without the removal of instruments or interruption of service to adjacent components.
 - Their installation will not be affected by, or interfere with, the removal of panel instruments.
 - Components can be removed from the front without removal of the back panel.

- xxvi. Support door-mounted equipment such that no appreciable or noticeable sag occurs and no excessive wear on the hinges or wiring occurs.
- xxvii. Provide lockout/tagout capable main power switch with circuit breaker or fused disconnected and terminal block for termination to each AC power (120 or 240V) feeder circuits. Provide a power terminal block for termination of the wires.
- xxviii. Provide power distribution component layout and wiring in accordance with the Drawings. Provide individually protected Vac and 24Vdc circuits for instruments as required plus 20% for future.
- xxix. Provide a circuit breaker panel with circuit breakers for each specific power source to accommodate panel lighting, power supplies, instruments, I/O modules, and processors.
- xxx. Provide sufficient space to allow AC power cables to run directly from the power block to the circuit breakers.
- xxxi. Terminate internal power wiring so that an interruption at any one point will not disrupt power to remaining components.
- xxxii. Wiring for Front of Panel Devices:
 - Provide hinge wiring wrapped with plastic cable stays. Secure hinged wiring at each end so that any bending or twisting will be around the longitudinal axis of the wire. Protect the band area with a sleeve.
 - Attached wiring to interior side of panel door to house wires for panel front-mounted devices.
- xxxiii. Terminate interconnecting wires, including spares, between panel-mounted equipment and external equipment at numbered terminal blocks. Multi-conductor cable is not allowed except for I/O module wiring harnesses.
- xxxiv. Arrange wiring neatly, cut to proper length, and remove surplus wire. Provide abrasion protection for wire bundles, which pass through holes or across edges of sheet metal. Spliced or taped wiring, wire nuts, and butt splices are not acceptable.
- xxxv. Wire Labels:
 - Preprinted, slide-on type or heat shrink cover Brady type, which may be wrapped around wire.
 - Affix wire labels at both ends of wire.
 - Cloth or adhesive tape is not acceptable.

- xxxvi. Install wiring in such a way to provide protection from inadvertent grounding and cross connection.
- xxxvii. Service Legends and Nameplates: Engraved, rigid, laminated plastic type with adhesive back; White with black letters unless otherwise noted.
- xxxviii. Firmly affix interior front of panel nameplates to the panel without compromise enclosure rating.
- xxxix. Firmly affix interior nameplates next to the device with industrial grade adhesive.
- xl. Comply with grounding requirement as per NFPA 70 or equivalent.

6.2.9 Slurry EMO System

A Master slurry EMO shall be located in the TGCM Room and shall be configured to provide individual shut down of each slurry sub-system in case of an emergency. In addition, the Master Slurry EMO can shut down all site Process Slurry Systems.

In addition, EMO remote pushbuttons shall be located at the entrance to each slurry room and shall be configured to provide shut down of all slurry systems in the associated room. All signals to or within the system shall be fail safe. Fail-safe is defined as, if power or pneumatic supply is lost, or if the PLC malfunctions. In such a state, all valves will fail close, the process flow stops, and the system sounds an alarm. Manual system reset is required.

7.0 Labeling Specification

Safety Signages and Labelling of slurry/additives lines, slurry systems and components are very important because they contain information on them about usage, direction, application and storage amongst other equally important information.

7.1 Labels – Equipment / Components

Printed labels shall be used on all required systems and components. Safety warning label shall be provided by equipment suppliers with hazards and the safety precautions identified to ensure safe operation and maintenance of the equipment. These requirements shall fulfill relevant section in EN61010 – Electrical warning labels, relevant sections in Semi S2 and S1 – Hazard Warning Labels & Hazards Labels, and IEC 61310 Safety of Machinery – Indication, Marking and Actuation. Additional labeling guidelines for TMAH >1% shall be adhered as per [Global Facilities TMAH Standard](#).

Required color code shall be used and there shall not be any deviation that can cause any misleading. As a minimum, the following shall be labeled:

- a) Slurry Distribution Systems.
- b) Tanks – While the standard requirements shall be met to ensure the compliance to hazardous material, tanks shall also have the name plates for the tanks.
- c) System Doors and Removable Panels.
- d) Any other cabinets.
- e) Labelling for electrical cabinets and cable trays shall follow electrical requirements/code/standards.

There shall be required tags for the following components:

- a) All Pumps, blenders, mixers, Valves, Regulators, etc.
- b) Pressure gauges, pressure sensor/transmitter, flow sensor/ transmitter, flowmeters, etc.
- c) Quality measuring instruments such as conductivity, density, concentration, pH, etc.
- d) Heat exchangers, dampeners, humidifiers, etc.
- e) Tagging for electrical components and cables shall follow electrical requirements.

7.2 Labels – Piping and tubing:

- a) Self-Adhesive Pipe Labels: Printed plastic with contact-type, permanent-adhesive backing.
- b) Pipe Label Contents: Include identification of piping service using the Sub-System Identifier and an arrow indicating flow direction.
- c) Direction of Flow: Where applicable, arrows will indicate the direction of fluid flow within the pipe or system. Arrows in both directions shall be used to indicate systems having bi-directional flow.
- d) Labels shall be placed such that they are visible from whatever angle(s) or positions a person might be expected to be looking at the pipe/system.
- e) Labels shall be placed within close proximity of valves and branch line intersections.
- f) Ensure all materials and adhesives are compatible with the environment in which they are to be installed (e.g. corrosive atmosphere).
- g) Label Spacing: Every 20 feet for open areas, every 10 feet in congested areas. Both sides of floor and wall penetrations and changes in direction (within 2 feet of direction change).
- h) Lettering Size: As required by piping size, as indicated in **Table 9**.
- i) Label color code: As per media category, as indicated in **Table 10**. If there are specific local code requirement, sites follow as per local code. If existing, site label color code does not match **Table 10**, for expansion within the same fab, sites may follow existing color code to prevent confusion. Sites shall follow below requirement for new greenfield fab.

Table 9 – Lettering Size

Pipe Size (OD)	Text Height	Label Height
1/4"	1/4"	1/2"
1/2" to 3/4"	3/8"	1/2"
1" to 2-1/2"	7/8"	1"
3" to 5-1/2"	1-7/8"	2"

Table 10 – Label Color Code

Media Category	Label Color code	
	Label Background	Label lettering
HPM Chemicals	Yellow	Black
Non-HPM Chemicals	White	Black
TMAH > 1%	Orange	Black

7.3 Slurry/Additives Labeling Convention

As slurry/additives changes frequently over time, whenever slurry/additive is replaced with a new type, all labeling such as system names, line labels, alarm tag names and SCADA pages will need to be changed costing time and money. An example of how two types of slurry/additive (D9228 & PL6501) shall be labeled is indicated in **Table 11** below.

Table 11 – Example of Slurry Labelling Convention

Labelling Location	Naming Convention	
Drum Cabinet / station	Slurry01(D9228)	Slurry02(PL6501)
All Main Slurry and Sub-systems	Slurry01(D9228)	Slurry02(PL6501)
Global Loop & Inter-connect piping	Slurry01	Slurry02
All Valve Boxes	Slurry01(D9228)	Slurry02(PL6501)
HMI Alarm Tag Name	Slurry01	Slurry02
SCADA Pages (actual Slurry name at every page shall be editable)	Slurry01(D9228)	Slurry02(PL6501)
SCADA Alarm Tag name	Slurry01	Slurry02

8.0 Design Review

Minimum three Design Reviews shall be carried out at the project site and/or at Micron office, at partial completion milestones. Acceptance and written approval of the partially completed design is required as a basis for subsequent design work. Minimum expectation requirements for each design review include:

- Process Development including Process Flow Diagrams and Pressure Flow calculations along with Material and Energy Consumption and Production. Preliminary equipment plans (Layout drawings), FMEA review and Safety in Design with HAZOP score according to Micron requirement.
- Process Design including Piping and Instrumentation Diagrams along with equipment sizing, piping sizing, component selection, and General Equipment and Piping Arrangements (Layout drawings).
- Detailed Design including Detailed Equipment, Piping and Support Drawings (Detailed Layout Drawings); equipment, materials, and component selection; and utility and accessory design. Final review of FMEA and Safety in Design.
- At different stages of BIM development.

Though the following information is related to design review, contractor shall inform Micron within the given time frame as this information is related to other disciplines.

- The Contractor shall indicate not later than 3 weeks after award of contract the necessity of any foundations or structural supports needed to move in equipment and tanks. Contractor shall review BIM or 3D Drawings to evaluate the move in equipment path.
- Provide space management design within each of the slurry rooms and areas. Meet and plan with other contractors to ensure a coordinated design including overhead lighting. Prepare drawings for Micron approval.
- Review the make-up air and room exhaust system design, including distribution ducting, inside the designated areas.
- Contractor shall study the civil work drawings, wall and floor penetration drawings, and space management drawings carefully to evaluate free

height, necessary openings, access ways, common supports, etc. that are available. If there is any additional opening or supports required, contractor should include the cost for all additional requirements. All pipe routing and cable tray drawings shall be submitted within 45 days of contract award.

9.0 Approved Manufacturers

To minimize spare parts inventory, Micron may standardize equipment/component make such as pumps, motors, valves, instrumentations and etc.

Equipment/component which requires routine maintenance and replacement parts shall be "standard manufactured" items with replacement parts readily available. In addition, preference is given to equipment/component make which has local service and support.

Acceptable equipment/component make shall be as indicated in **Table 12**.

Table 12 – List Approved Manufacturers

Component Type	Approved Manufacturers
SDS Equipment	Air Liquide, Diversified Fluid Systems (DFS), MEGA Fluid Systems, Sumitomo & Asia ICMP – any other manufacturers requires Micron approval.
High Purity PFA Tubing	Entegris, GEMU, Nichias, Parker, Nippon Pillar, or equivalent
Clear PVC (Schedule 40) Piping	George Fisher, Hershey, Spears, Landmark Plastics or equivalent
High Purity PFA Fittings	Entegris, GEMU, Nippon Pillar, or equivalent
Ultra-High Purity Valves	Entegris (Integra Plus WS), GEMU (Weir Type), Swagelok, or equivalent
Ultra-High Purity SS Valves	GEMU, FUJIKIN
Ultra-High Purity Back Pressure Valve	Parker or equivalent
Solenoid Valves	Festo, SMC or equivalent
Stainless Steel EP Pipe	Kaze, Valex

Component Type	Approved Manufacturers
Ultra-High Purity Pressure Regulator	Parker, SMC or Equivalent
Dispense Pump	Levitronix (Centrifugal pump),
Transfer Pump	ARO (high purity – Air operated), Nippon Pillar / Iwaki (Bellows – Air operated) or equivalent
Low Purity Air Operated Sump Pumps	ARO, Yamada, Wilden or equivalent
Tank and Vessel	Applied Supreme, Edlon, Entegris, NGC, Plasticon, Sun Fluoro, Valqua, Fluoro Tech, Yodagawa or equivalent
Conductivity Sensor/Transmitter	E&H, Horiba, Burkert, Hach or equivalent with assurance for application.
Flow meter /Controller/Transmitter	Entegris or equivalent with assurance for application
Pressure Sensor/Transmitter	Entegris or equivalent with assurance for application
Concentration Meter	K-PATENTS or Equivalent
Other Instruments	Micron Approved ones only or Prior approval to be obtained.
Load Cells/Transmitter	Hardy
Titration	Semi-Chem
Programmable Logic Controllers (PLC)	Allen Bradley, GE Fanuc, Mitsubishi, Omron, Siemens, or equivalent

10.0 Testing, Commissioning and Acceptance

During the system fabrication, shipping, installation, and start-up phases, each Slurry distribution equipment, associated components and sub-assemblies are subjected to stringent quality checks and testing as defined in the above Section. Further the assemblies of systems and components shall be in a clean manner to maintain cleanliness during the assembly and packaging processes.

As needed all units shall be qualified both at factory and at field. Thus, the continuity shall be maintained to the extent that observations and non-conformance noted during factory test can be validated during field test. Required information with respect to FAT and SAT (or Site Testing and Commissioning) are provided in the relevant sections.

10.1 Commissioning Plan

Start-up activities are critical to the overall success of any installation. System Supplier needs to have most qualified commissioning team/field services with chemical, mechanical, control and electrical engineering capabilities. The Defect Liability Period ("DLP") of the system shall start upon the acceptance of the system.

Upon award of the contract, system supplier needs to prepare a Commissioning Plan including Certification and Validation if applicable. The Commissioning Plan contains all relevant information: test dates, type of test/inspection, detailed test procedure description, and specification, checklists, systems and components involved in the test, block diagrams, piping network diagrams, test equipment and calibration data, test results form and factory test reports, certificates included. Prior to start of commissioning works, system supplier shall provide commissioning plan to Micron and only when the commissioning plan is accepted Micron, commissioning activities can be started.

The System Supplier commissioning team shall consist of a Project Manager and Commissioning Engineer(s). The team will verify the installation of the equipment, interconnecting process piping, utility piping, loop piping, instrumentation, electrical and control wiring supervision of the cleaning, start-up and qualification leading to customer acceptance of the system. Upon the End of Installation, System Supplier can start the pre-commissioning activities. During this pre-commissioning period, System Supplier may stop all or part of the equipment or run it in various modes, with the aim of making the necessary settings and ensure it is working properly. However, if there is any damage to the slurry system or associated equipment or piping or associated systems/utilities, system supplier shall bear the sole responsibility without any claim from the Customer for the damage.

During this pre-commissioning period, all preliminary trials can be performed, concerning every equipment or step of treatment. These trials will enable to check the correct functioning of the mechanical, electric, electronic, thermal and hydraulic devices, according to the corresponding standards, and the planned conditions of functioning.

As minimum requirement, Commissioning Plan Shall Include the following information:

Activity	Date for the Activity	Required Documents	Micron's Responsibility	Remarks
Factory Acceptance Test	Date to be provided By Vendor	Refer to FAT Section	Approval of FAT Procedures and Check Lists	Documents to be submitted to Micron Minimum 3 months prior to the activity.
Installation Inspection	Date to be provided By Vendor	Refer to Installation Inspection Section	Approval of Installation Inspection Procedures and Check List	Documents to be submitted to Micron Minimum 3 months prior to the activity.
Site Acceptance Test	Date to be provided By Vendor	Refer to SAT Section	Approval of SAT Procedures and Check Lists	Documents to be submitted to Micron Minimum 3 months prior to the activity.
Site Testing, Start-up and Commissioning	Date to be provided By Vendor	Refer to Site Testing and Commissioning Section	Approval of Testing, Start-up and commissioning Procedures and Check Lists	Documents to be submitted to Micron Minimum 3 months prior to the activity.

10.2 Factory Acceptance Test

Factory Acceptance Test (FAT) is to be carried out in order to demonstrate operational and functional testing of the system. Though FAT is carried out at the factory, detailed testing of the I/O list, I/O Loop test, verification of assigned tagging both on component and on HMI, I/O firing test, field calibration of necessary instruments, instrument scaling both at HMI and SCADA, at least 2 point random check to verify the scaling of instruments, etc. shall be performed.

All functional testing with all possible scenarios shall be carried out to confirm the sequence of operation and theory of operation. Functional check list shall be submitted before starting the performance test. Soft functional testing shall be performed in software simulation (or with simulator panel) and Actual functional checking shall be thoroughly performed at site.

All the alarms, conditions for the alarms and scenarios arising from the alarms shall be validated. Alarm priorities based on the Micron standards shall be in compliance with Micron Requirements. Alarm list comprising all alarms with different categories, priorities and follow up action to be taken upon the occurrence of the alarm shall be submitted before the starting any performance tests.

FAT is not just limited to the below statements as the tests can vary based on the chemistries used. Generally, FAT has to be performed in the presence of Micron staff and based on the circumstances, this particular FAT requirement may be changed.

- a) Contractor shall provide FAT procedure which shall include Checklist, all the tests carried out at factory prior to FAT, all the tests to be carried out during FAT, how all the tests are carried out, handling of change control forms with respect to FAT, etc.
 - b) What have been approved in P&IDs shall not be compromised unless deviations granted.
 - c) Component Tagging list, Equipment list (Components, Instruments and Hardware/Software information), I/O list, pressure test reports, tank lining integrity test report, factory calibration certificates, DI water grade used for flushing the system, DI Water sample analysis if available, etc. shall be provided prior to FAT.
 - d) During FAT, at least the followings will be tested at the Factory.
 - i. Vendor's QA/QC procedures and QA/QC data of the system.
 - ii. System size, tank sizes, mechanical integrity, accessibility to components, etc. shall be checked. System has to match the 3D modelling provided by the contractor during design stage.
 - iii. Mechanical, Pneumatic and Electrical segregation for individual engines to be checked.
 - iv. Leak test for tanks
 - v. Leak test for cabinet whichever portion is necessary – cabinet bottom, sample compartments, other compartments through which slurry or UPW water or any other chemical or wastewater is passing.
 - vi. Leak test for all wet tubing/piping. Perform the same for dry tubing/piping too.
 - vii. System start up test
 - viii. EMO test
 - ix. I/O Loop test
 - x. Valve firing test
 - xi. Instrumentation scaling
 - xii. Functional Tests. If wet test cannot be performed, all functions shall be simulated either by a simulator or by a software simulation.
 - xiii. Communication with remote units/stations
 - xiv. If tank lining Integrity Test can be carried out, it shall be performed.
-

- e) Upon completion of FAT, Micron shall approve if the system can be sent to site.
- f) All rectification works shall be completed by contractor and shall be either retested at factory in the presence of Micron staff or at site based on the severity of the rectification work.
- g) **Appendix 1** shows a guideline for FAT.

10.3 Installation Inspection

Installation Inspection is to be carried out in order to demonstrate readiness of the system installation so that relevant Testing, Start-up and Commissioning work can be started. Thus, Site Inspection for Start-up shall precede testing and commissioning activities. The following points shall be observed prior to site inspection works. During Design of installations, some conflict can occur with installation of other disciplines according to BIM and therefore these conflicts should have been resolved before installations.

- a) Contractor shall provide Installation Inspection procedure which shall include Check List, all the tests carried out prior to/During/after Installation, how all the tests are carried out, handling of change control forms with respect to FAT, etc.
- b) All installations shall comply all relevant codes or standards.
- c) Once the Installation of Tanks, Systems, Piping work, VMBs, Power Panels, Control Panels, other field service devices, etc., all relevant reports with respect to Installation have to be check and approved by Micron.
- d) All mechanical and electrical terminations are to be checked and thus, there shall not be any open-ended pipes, tubes, ducts, cables, etc.
- e) When there are many installations proceed in parallel, relevant installation inspection can proceed one after the other based on installation completion.
- f) When there are many systems are installed, inspections and relevant documents are to be submitted for individual systems.
- g) During Installation Inspection, at least the followings checks shall be carried out.
 - i. Vendor's QA/QC procedures and QA/QC data with respect to installation.
 - ii. System size, tank sizes, mechanical integrity, Electrical Installations, etc. shall be checked.
 - iii. All piping and their integrity shall be checked
 - iv. All electrical and instrumentation cables shall be checked.
- h) Upon completion of Installation Inspection, Micron shall approve if the installations shall be used for its purpose.
- i) A guideline for Installation Inspection is provided in **Appendix 2**.

10.4 Site Acceptance Test

The Site Acceptance Test will be carried out after the completion of installation of the system at Micron facility. The purpose of the SAT is to verify that the system is supplied and installed for the system to operate according to the designed specification. At the completion of installation of the system, as stated under QA/QC, field service engineer from the contractor shall be in-charge for the SAT. Field Service engineer shall certify that the system is installed as per the requirements.

Contractor shall provide SAT procedure which shall include Checklist, all the tests to be carried out at Site prior to start-up, how all the tests are carried out, handling of change control forms with respect to SAT, etc.

If FAT was not carried out with the agreement of Micron due to some unavoidable circumstances, a thorough FAT shall be performed during SAT as per FAT guidelines provided under FAT. If FAT was done properly, some critical checks shall be performed to confirm the integrity of the system as below by a trained personal:

- a) Mechanical and electrical Integrity of the installed system.
- b) Ensuring that all the loads of the load cell-based blend tank is taken up by load cells. For this purpose, load cell calibration and linearity check shall be performed with calibrated weights. Following points shall be noted for calibration:
 - i. Based on the designed blend weight, 10 standard weights each with 1/10th of the design blend weight (approximately) shall be prepared.
 - ii. Standard weight shall be gradually loaded one by one and using the respective readings at HMI. HMI Reading vs Standard Weight shall be established for loading.
 - iii. Similarly, after the full designed blend weight is loaded, weights shall be reduced one by one and using the respective readings at HMI, HMI Reading vs Standard Weight shall be established for unloading.
 - iv. All the data points captured in the SCADA or HMI shall be attached to calibration curve for reference.
 - v. If there is any other method is established by the system supplier, it shall be given in writing and get the necessary approval.
 - vi. If the weight of any components of the blend composition (mixture) is lower than the 1/10th of the designed blend weight, that particular weight shall be taken as incremental/decremental weight for calibration of the load cell.
 - vii. When error is calculated between HMI and actual weight, reference shall be the smallest weight. If the smallest weight is 500g and HMI shows 490g, the error % is 2%.

- viii. When calibration is done, the highest error value during loading or unloading has to be recorded and used for error calculation.
 - ix. R^2 (coefficient of determination) for the correlation (calibration curve) shall be very close to 1.0.
- c) Ensure integrity of the load cells, flow controllers, flow and pressure regulators, temperature transmitters, etc.
- d) Ensure the completion of all the rectification works to be performed based on FAT.
- e) Electrical and Control Panel Verification Testing: At the completion of panel installation, each Electrical and Control (Remote I/O Panel -RIO) panel to be tested by a factory-trained and certified field service engineer according to Electrical and Instrumentation & Control specification.
- f) Confirm input voltage (120Vac or 240Vac or appropriate 3 phase supply) is within appropriate tolerance range, and that the panel indicates normal operating condition upon the 120 or 240 Vac power application.
- g) Confirm the operation of EMOs – System, Remote panels, Room EMO and Remote EMO in TGCM room
- h) Ensure the integrity of the PLC/HMI/SCADA programs
- i) Ensure the availability of all the modes of operation and operability of the system different modes.
- j) Test the Day-tank level settings and comparison with analog reading to confirm the instrumentation scaling
- k) Ensure the system is ready for start-up.

10.5 Testing, Start-Up and Commissioning

As stated under commissioning plan sections, Prior to any testing activities, Contractor has to submit the Test Plan (TP) or inspection Test Plan (ITP) for approval of Micron.

10.5.1 Testing and Start-Up

- a) Before starting any testing activities, Contractor has to inform the Owner and Purchaser (based on purchase requirement) the beginning of the tests at least one week in advance.
- b) Contractor shall produce the list for the testing and as minimum, Contractor shall perform following tests as indicated below: Based on the SAT requirement, some of the test supposed to be performed before SAT and it is the responsibility of the contractor to ensure the completion of those required tests before SAT.

Equipment	Activity	QC Point	Quality/ Test Record	Responsibility of System Supplier
Tank	Pressure Test	Off-site and On-site	Pressure Test Report	Preparation for test followed by execution of the test
	Spark Test or Any other Integrity Tests for Lining	On-site	Spark Test or Any other Integrity Test report for Lining Test Report	
	Lining Thickness Test	Off-site	Test Report	
	Harness Test	Off-site	Test Report	
	Leak Test	Off-site and On-site	Test Report	
Utilities readiness before hook-up to system - UPW	Pressure Test, Metal, Particle	On-site	Pressure Test, Vacuum Test, Particle and other analytical test reports	Preparation for test followed by execution of the test and third-party analysis
Utilities readiness before hook-up to system - PN2, GN2	Pressure, Vacuum (He Leak test), Particle,	On-site		
System	FAT	Off-site	FAT Check List	Preparation of Check list and other preparations for FAT
System and all other associated equipment	Installation Inspection	On-site	Installation Inspection Check List	Preparation of Check list and other preparations for Installation Inspection
System	SAT	On-site	SAT Check List	Preparation of Check list and other preparations for SAT

Equipment	Activity	QC Point	Quality/ Test Record	Responsibility of System Supplier
Inter-connections	Pressure Test	On-site	Pressure Test Report with sections of piping	All preparation work for pressure test and execution of the test
Distribution Loop	Pressure Test	On-site	Pressure Test Report with sections of piping	All preparation work for pressure test and execution of the test
Tank	Flush with UPW followed by Metal and particle analysis	On-site	Full Metal analysis report and Particle count report	All preparation work for flushing & sampling, execution and third-party analysis
System – Through all wetted parts and take samples from - Final Discharge point to Loop - Local Recirculation before tank	Flush with UPW followed by 5 key Metal and particle analysis	On-site	Metal analysis report and Particle count report	All preparation work for flushing & sampling, execution and third-party analysis
Loop Return before tank	Flush with UPW followed by 5 key Metal and particle analysis	On-site	Metal analysis report and Particle count report	All preparation work for flushing & sampling, execution and third-party analysis
VMB	Leak Test	Off-site for all On-site for some	Leak test Report	All preparation and execution
MHEB / EB	Leak Test	Off-site for all On-site for some	Leak test Report	All preparation and execution

Equipment	Activity	QC Point	Quality/ Test Record	Responsibility of System Supplier
Any Enclosures	Leak Test	Off-site for all On-site for some	Leak test Report	All preparation and execution
Electrical/ Control / Instrumentation Test	EMO Test for all Panels	On-site	EMO Test Reports	All preparation and execution of work
	Cabling Continuity Test	On-site	Cabling Continuity Test Reports	
	Loop Test	On-site	Loop Test Reports	
	Earthing continuity check	On-site	Test Reports	
	Panels earth connections if required	On-site	Test Reports	
	Turning on Electricity to system by LEW	On-site	LEW's Power Turn-on Report	
	Any Functional test that is not performed during SAT	On-site	Functional Test Reports	
Load Cell	Calibration with Std weight Vs. HMI Readings	On-site	Calibration report with calibration curve (Upwards and Downwards)	All preparation including volume calculations for blend tanks and execution of work
Conductivity Meter/ Transmitter	Calibration	On-site	Calibration Certificate	Calibration based on relevant procedure

Equipment	Activity	QC Point	Quality/ Test Record	Responsibility of System Supplier
Density Meter/ Transmitter	Calibration/ Verification	On-site	Calibration/ Verification Cert	Calibration/ verification based on relevant procedure
Concentration Meter/ Transmitter	Calibration/ Verification	On-site	Calibration/ Verification Cert	Calibration/ verification based on relevant procedure
Flow Controller	Calibration for the chemical UPW or any other blending component to be established based on density & viscosity	On-site	Calibration Cert	Calibration based on relevant procedure
Radar Sensor/ Transmitter for Day tank	Mapping and Verification	On-site	Tank Mapping information	Mapping using relevant software
Digital discrete Level sensor settings for Day tanks	Verification of Levels and matching with Design basis	On-site	Digital Level Sensor Settings	Set the digital level sensors at correct location and fine tune the sensor sensitivity
Slurry / Additive Quality	Based on slurry / additives properties, Solid Content, Density, Conductivity, Concentration of H2O2	On-site	Third Party Analytical Results	Sampling and third- party analysis
Verification of online / Inline metrology	Based on slurry / additives properties, Density, Conductivity, Concentration of H2O2	On-site	Analytical Data and corresponding Field instrument values and SCADA Data	Verification of field SCADA values with lab analysis

Equipment	Activity	QC Point	Quality/ Test Record	Responsibility of System Supplier
Sump pump Enclosure	Leak test	On-site	Test Report	Preparation and execution
Sump	Leak test	On-site	Test Report	Preparation and execution

Note:

- i. As for all the test, Micron shall witness, approve and accept. Without witness, approval and acceptance of Micron, commissioning data shall not be accepted.
 - ii. Some of the line items are common for many sections or equipment and therefore, contractor's checklist has to be with individual ones so that testing will be done individually, and reports can be generated accordingly.
 - iii. Some of the above tests are to be performed after start-up and therefore, contractor shall perform those tests after start-up.
 - iv. Load cell calibration and linearity check methodology is explained in previous section.
 - v. When conductivity cell calibration is performed, conductivity cell shall not be removed from the measurement location. Liquid (blended sample also can be used) with Known conductivity value shall be filled in the required section of the instrument line and calibration shall be performed. Zero calibration can be performed without any content in the required line. Zero Calibration shall be performed first.
- c) Once all the required test for start-up is completed, system will be ready for start-up and start-up shall be performed based on start-up procedure submitted by the contractor. Though start-up procedure is available, a start-up check list to be submitted by contractor prior to start-up to confirm and system shall be checked based on start-up checklist. As minimum, start-up check list shall include the following points:
- i. All the required safety checks.
 - ii. Approval of the start-up and shutdown procedures by Micron.
 - iii. System installation/construction acceptance from Micron.

- iv. Start-up can be initially performed with UPW only. All the wet tests to be performed based on the chemical /slurry properties and shall be listed.
- v. Availability of all required medias (slurries, chemicals, etc.).
- vi. Availability of final draft of Operation and maintenance Manual.
- d) Based on the checking performed in accordance to start-up check list, the readiness for start-up will be confirmed.
- e) Based on the chemistry criteria, system start-up shall be performed using UPW. If UPW cannot be used, correct chemistry shall be used for start-up.
- f) Once start-up is performed according to start-up procedure, system shut-down shall be checked in accordance to shut-down procedure. Thus, start-up and shutdown is completed.

10.5.2 Commissioning

- a) Based on the given schedule, Contractor shall inform Micron the beginning of the commissioning at least two weeks in advance.
- b) Commissioning of all components and the total system shall be witnessed by the Micron.
- c) If FAT or SAT was performed thoroughly, selected critical components such as pumps shall be commissioned based on equipment commissioning guideline provided by the equipment manufacturer. Commissioning checklist for the critical components shall be provided by the contractor.
- d) Contractor shall also provide the commissioning checklist for the system.
- e) All the system parameter settings set during commissioning shall be approved by Micron. Until final agreement is achieved between contractor and Micron, contractor is liable for the parameter settings set on the system HMI/SCADA.
- f) Instrument and equipment accuracies shall be confirmed during commissioning activities
- g) When conductivity or any other parameter is used to represent slurry/additive concentration, calibration curve of Conductivity (or other relevant parameter) versus concentration shall be established for the desired concentration range. This curve shall be established using the actual slurry/additives(s) used for blending.
- h) If conductivity (or any other parameter) is selected for representing concentration, as conductivity is a temperature dependent parameter, temperature compensation

coefficient shall be found for the required blend. This temperature compensation coefficient shall be used for temperature compensation coefficient and shall be used appropriately in the PLC programming. It shall be noted that cell constant is different from temperature compensation coefficient.

- i) All the functional checks shall be carried out during commissioning
- j) As minimum, following activities as shown below shall be performed during commissioning.
- k) Qualification procedure needs to be submitted by contractor and to be approved by Micron. Sampling points, how many samples to be taken for accuracy, repeatability and reproducibility. Measurement detection limits and percentage of error, etc. shall be specified in qualification procedure.

Equipment	Activity	QC Point	Quality/Test Record	Responsibility of System Supplier	Responsibility of Micron
Pumps and other critical equipment	Commissioning based on equipment manufacturer's guideline.	On-site	Commissioning report	Performing commissioning for the respective equipment	Witness, approval and acceptance.
Instruments	Scaling between field and Remote (HMI and SCADA)	On-site	Instrument Scaling Reports	Preparation and execution	Witness, approval and acceptance.
Metrology Instruments	Accuracy of the instruments for the particular chemistry	On-site	Verification Reports	Performing verifications	Witness, approval and acceptance.
System	Functional Test comprising all possible scenarios	On-site	Functional Test Reports	Preparation of Functional Test checklist and performing functional test	Witness, approval and acceptance.
Sump Pump	Pumping Rates	On-site	Pump Capacity Test Report	Preparation and execution	Witness, approval and acceptance.

Equipment	Activity	QC Point	Quality/Test Record	Responsibility of System Supplier	Responsibility of Micron
Sump Pump	Functional Checks	On-site	Test Reports	Preparation and execution	Witness, approval and acceptance.
System	Verification of Alarms – This also can be covered under functional test	On-site	System alarm and SCADA alarm reports	Preparation and execution	Witness, approval and acceptance.
Blended Slurry/Additive	Qualification accuracy, repeatability, and reproducibility	On-site	Qualification Test Reports	Preparation of blends in automatic mode, sampling and analysis	Witness, approval and acceptance.
System	Leach test based on chemistry	On-Site	Leach Test Report	Preparation and execution	Acceptance
System	Simulation for Power Failure test	On-site	Power failure simulation test report	Preparation and execution	Coordination, witness, approval, and acceptance
System	Simulation of power failure from one source to ensure seamless take over.	On-site	Power failure simulation test report	Preparation and execution	Coordination, witness, approval, and acceptance
SCADA	Pages, Process Flow Diagrams, Trends, SPC if included,	On-site	Printed pages, PFDs, Trends, SPC information	Preparation and work with SCADA Vendor depending on scope	Coordination, witness, approval, and acceptance
SCADA	Alarm Testing, Event Tracking, Scaling, Prioritization, etc.	On-site	Generated Alarm list, Event list, Scaling information, priority setting, etc.	Preparation and work with SCADA Vendor depending on scope	Coordination, witness, approval, and acceptance

- l) Startup and validate System in compliance with requirements of the contract documents. Provide third party analytical testing as necessary to verify proper performance for System qualification and acceptance. The Slurry Distribution Systems shall continue demonstrating adequate performance under full load conditions.
- m) Contractor shall submit requirement of Slurry/ Chemical volumes for commissioning. Slurry and chemical logistics, including slurry /additives required for startup are the responsibility of MICRON
- n) Simulate power failure and Confirm no loss of functionality occurs. Test and confirm the supply from UPS or battery for specific components.
- o) Simulate power failure from one source and ensure the operation of the system without any impact. For both sources this test to be performed.
- p) Commissioning is deemed complete with following document to be proven.
- q) System Commissioning Completion Certificate signed-off by Micron.
- r) During the commissioning the training of the future operation team shall be conducted.
- s) If there is any additional requirement specified in other disciplines for commissioning of the respective disciplines such as SCADA, Electrical and Instrumentation, refer the respective discipline document.

10.6 Performance Test

Performance test of the system shall be performed before the final acceptance of the system. This includes running of the complete system – transfer, blending and dispense based on the design concept used – at the required flow without any interruption or any major changes in the system installation or system control units for at least continuously 14 days (guarantee parameters in specification).

The test is considered failed if:

- a) Failure by a self-cause and/or substantial intervention at the system, to ensure function.
- b) Specification parameter not achieved.
- c) Safety devices not responding on failure.

Upon failure, re-start of the tests is expected, and the costs shall be fully borne by the Contractor.

Performance test can only be performed only after the following actions

- a) Acceptance of qualification reports.
- b) Acceptance of all other commissioning reports.
- c) Settings of all agreed parameters on HMI/SCADA.

When the performance test is in progress, Micron shall request to pull samples and analyze to verify the blending accuracy, repeatability and reproducibility of the system. Contractor also shall submit the sampling plan during the performance test period.

Performance test will be accepted upon submission of performance test report in addition to all required site recordings. Performance test report shall comprise the followings:

- a) Blender application summary – This is a table with all basic recipe calculations, flow rates, consumption rates, etc.
- b) Blender performance trend (conductivity or concentration trend during qualification or dispense for individual batches.
- c) If conductivity is used as the reference parameter, Conductivity Vs temperature for a blend closer to the required recipe. This shall be conducted in controlled environment at different temperatures such as 19°C, 21°C, 23°C, 25°C and 27°C. The equation obtained from the graph shall be used for temperature compensation.
- d) Temperature trend during local recirculation and dispense.
- e) Table comprising online conductivity, lab conductivity and lab concentration for target recipes. Develop useful graphs to explain the correlations.
- f) Once target conductivity or recipe versus concentration is developed, target conductivities for the required upper, lower and middle specifications (or control limits based on what is required) shall be predicted for the performance of the system.
- g) Actual Performance Trends and tables be based on the predicted Targets.
 - i. System performance trends.
 - a. Trends to prove the qualification period.
 - b. Trend of reference parameter (such as conductivity) with specified bandwidth of the target value.
 - c. After qualification, total time out of control limit shall be calculated and shown in the performance trend.
 - d. Requirement for load cell or flow controller taring or zeroing after each blending cycle.
 - Requirement for flushing after each blending cycle.
 - ii. Sample analysis results
 - a. Samples shall be taken for accuracy, repeatability and reproducibility as shown below. S1-9 refers to **Sample 1 – 9**.

Sample	Lower Control or Lower Spec. Limit	Control Limit	Upper Control or Upper Spec. Limit
For Accuracy	Fresh sample after stabilization – S4	Fresh sample after stabilization – S1	Fresh sample after stabilization – S7
For Repeatability	Fresh sample for new blend prepared – S5	Fresh sample for new blend prepared – S2	Fresh sample for new blend prepared – S8
For Reproducibility	Fresh sample for new blend after shut-down and start-up the system – S6	Fresh sample for new blend after shut-down and start-up the system – S3	Fresh sample for new blend after shut-down and start-up the system – S9

Note:

- For each tank 9 samples shall be taken as prescribed above
- If there are 3 blend tanks, 27 samples to be taken for qualification. However, if the number of samples can be reduced based on the approvals granted by Micron system engineer.

10.7 Acceptance and Certification

- a) Equipment vendors performance report shall be accepted by Micron which is a prerequisite for acceptance and certification.
- b) Certification shall be done by the System Supplier together with the Micron or appointed TGCM contractor
- c) The purchaser may request a third party to be involved in the certification phase. This will be at no additional cost to the system supplier.
- d) The start of the certification measurements and tests shall be announced at least two weeks in advance. An introductory meeting and a presentation of the certification set-up shall be organized by the System Supplier for all involved parties.
- e) For the certification and acceptance criteria for the system shall have the followings as Minimum:
 - i. Meet all requirements on EHS
 - ii. Meet all installation requirements
 - iii. Meet all required technical specifications
 - iv. Completion of all set points with logical reasons – System Parameter Settings with approval from system engineer
 - v. No critical alarms for 14 days
 - vi. Submission of all relevant documentation as stated under section submission of documents.

- f) Acceptance is deemed to be the complete specified and fully functional delivery of the works (System) to the Owner/Purchaser after testing, complete with service requirements e.g.
 - i. Completed Quality Assurance Program (QAP) report.
 - ii. Completion of the as-built drawings of the complete installation.
 - iii. Documentation of operational and functional acceptance tests, and of third-party laboratory analysis.
 - iv. System has been validated and performance satisfactorily demonstrated.
 - v. Completion of the operation and maintenance manuals
 - vi. Completion of the warrantee certificate(s)
 - vii. Completion of punch list items.
 - viii. Completion of training programs.
 - ix. Any other points stated in the contract. An official report (System Final Acceptance Certificate) of this successful acceptance shall be prepared by the Contractor and signed off by Micron.
- g) If during operation of the system, it is determined that the systems/sub-systems do not function properly or do not comply with the specified requirements of the contract, a renewed testing and acceptance procedure cycle shall take place.
- h) Material, components, equipment, assemblies, and subsystems will be considered defective and replaced if unable to pass tests and inspections in conformance with contract requirements.
- i) With respect to the rework, all associated costs shall be borne entirely by the Contractor, including the fees and transportation / accommodation costs of the Owner/ Purchaser's representative(s).

11.0 Quality Management

11.1 Quality Control Requirements

QA/QC is an indispensable part of high purity system manufacturing, installation, testing and commissioning. The following points with respect to QA/QC shall be observed during the project.

- a) Provide skilled installation personnel who are certified in their trade.
 - b) Provide job training and instruction of pipe joining techniques for personnel assigned to such duties. Welders certificate to be submitted as part of the submittal process. Clear PVC bonders need to be certified onsite.
 - c) Include training and certification by pipe manufacturer for joining technique selected.
 - d) It is the scope of the contractor to co-ordinate with Micron to schedule and conduct source inspections at any manufacturing facility or assembly plant prior to shipping any equipment designated for this project.
-

- e) Performing appropriate inspection and testing is necessary for all equipment and provide inspection report prior to source inspection by Micron.
- f) Maintaining records is very important and Micron will audit the documents when necessary.
- g) In order to control and assure quality of all works related to the slurry system, establish a quality assurance program (QAP) which shall be maintained throughout the project by Contractor. Clearly document QAP in the Bid. QAP is subject to monitoring during the course of project execution. QAP may include:
 - i. Material handling
 - ii. Material acceptance or rejection criteria
 - iii. Replacement of material when it is rejected.
 - iv. Incoming Quality Checks – material and equipment inspections.
 - v. Skill level of Thermoplastic (non-clear PVC / PFA) Pipe fitter / Welder and trainings certificates.
 - vi. Skill level of clear PVC & PFA Pipe fitter / Bonder and training certificates.
 - vii. Skill level of Stainless-Steel Pipe fitter / Welder and training certificates.
 - viii. Filed performance and credentials of these skilled workers
 - ix. Fabrication procedures and clean room protocols
 - x. Thermoplastic and Stainless-Steel high purity pipe welding procedures
 - xi. Design compliance checks.
 - xii. Monitoring of submittals provided, and incorporation of comments received.
 - xiii. Development of the detailed System startup and validation plan.
 - xiv. Coordination between Contractor and Micron.
 - xv. Coordination between the Contractor and site installation activities. Coordination of multiple disciplines within the Contractor's organization (i.e. electrical, I&C, shop fabrication, mechanical).
 - xvi. Compliance on all necessary tests that are performed on-site and off-site
 - xvii. Verification the test reports are generated in accordance with the project requirements, including appropriate signatures.
 - xviii. Verification of Licenses used for PLC programming and HMI/SCADA development
 - xix. Certifications and credentials of PLC programmer and SCADA developer

- h) Micron reserves the right to review facilities or fabrications in process at any time or finished equipment prior to shipping, and to witness shop testing.
- i) Contractor shall assign an individual who will be solely responsible for quality control functions, resident on job site. Quality control functions to include as minimum; cleanliness checks, component inspections, welding inspections per SEMI F81-1103, conformance to F78-0304 installation practices, bonding inspections, craftsmanship inspections.
- j) Workmen shall be thoroughly experienced in their trade and fully capable of manufacturing and installing the assigned works. All works shall be performed in accordance with the best practice of the trade, as recommended by the manufacturer and/or as directed and approved by the Micron. Any work that is determined to be of substandard or poor quality by the Micron shall be removed and reinstalled at no additional cost to the Contract. All costs resulting from any schedule impact generated by such actions shall be fully borne by the Contractor.
- k) An experienced supervisor shall be continuously in attendance at the jobsite during all phases of construction to co-ordinate and supervise the works.
- l) All workers shall follow the site safety rules, regulations and practices imposed by Micron.
- m) Based on the size of the project, Micron shall request a QA/QC engineer during site construction.
- n) All QA/QC reports shall be submitted weekly during construction period. The submittals shall be in accordance with latest Micron practice.
- o) Factory test reports from manufacturer prior material/equipment delivery are to be submitted to warrant that the delivered material/equipment meet the required specification.
- p) The full authority of the contractor management shall be given to a member of their full-time site staff. This employee shall have the authority to commit financial and technical decisions of a magnitude associated with the size of the contractor's works' scope.
- q) Based on site conditions and progress, Micron has the right to request for additional QC and/or certification services which can be provided partly or fully by a third party during the installation of the systems and in regular intervals.
- r) Engage a qualified testing agency to perform tests and inspections that is outside the core competency of the Contractor.
 - i. Perform tests and inspections in conformance with equipment manufacturer recommendations.
 - ii. Prepare test and inspection reports.
- s) It is contractor's responsibility to provide factory trained field service engineer or engineers to perform System checkout and startup. Engineers who performed Factory FAT/SAT test prior to shipment are preferred to be the field service engineer. Field service engineer needs to certify in writing that the supplied equipment has been installed correctly with proper piping and wiring connections.

11.2 Design Validation

Complete design including process flow diagrams, P&IDs, equipment & pipe specification, pipe and equipment plans and details, equipment skid general arrangements, storage vessel fabrication drawings, equipment support and restraint details, and pipe support and restraint details along with the referenced specifications, shall be validated prior to start of any other work in compliance with Micron requirements. All the source calculations are to be submitted to support the design review. HAZOP and FMEA shall be performed in detail such that safety and robustness of the system shall be maintained throughout the life cycle of the system.

3D modelling drawing validated to make sure of the requirements on system segregation, component accessibility, space management within the system and operation & maintenance. NPSH shall be validated through modelling such as Computational Fluid Dynamics (CFD) software. Hydraulic modeling to validate flow requirement shall be performed with AFT Fathom 7.0 or equivalent software. BIM shall be submitted in different phases in order to evaluate the piping and system layout and space management.

Sequence of operation (SOO) shall be submitted, reviewed and validated. Sequence of operation is the basis for safe, risk free and robust operation. Based on SOO, functional tests and alarm list shall be prepared and validated.

HAZOP and FMEA shall be submitted and reviewed. Based on HAZOP and FMEA observations, SOO followed by Functional test and alarm list shall be updated and validated.

In addition to the general design validation requirements, there are specific design validation requirements for Slurry Blend and Dispense systems (SBDU) and following areas of design for Slurry/Additive blending shall be validated:

- a. Methodology of Blending for the specific final product. Suitability of the blender for possible quality and quantity of the blend.
- b. Quality measuring (conductivity, concentration) and quality control (flow controller, load cell) instrumentation used in the SBDU / STBU.
 - i. Integrity of the instrument including material of construction.
 - ii. Measurement accuracy of the instruments.
 - iii. Measuring range and required range.
 - iv. Usual life span.
 - v. Required frequency for calibration.
 - vi. Past experience on drifting for such blend.
 - vii. Effective use of factors such as cell constant, temperature compensation factor.
- c. Selection of mixing methodology.
- d. Time required for stabilization of the blend.

- e. The required operational control and specification range for direct or indirect parameters (e.g. concentration can be direct parameter and conductivity can be indirect parameter) that are measured online.
- f. What is the acceptable frequency for out of the control and specification limits?
- g. Blend quality RE%: Decision on the control limits and specification limits and the appropriate usage of control limit and specification limit. It will be critical to ensure which one (control or specification limit) will take precedence for RE % calculation.

12.0 Training Program

Training Micron staff or the appointed TGCM contractor is very essential to ensure safe and consistent operation of the systems. The actual training dates shall be agreed between Micron and system supplier. System supplier shall submit a detail training program for approval 4 weeks before training commence

- a) Classroom and In-the-field training shall be provided in Micron 's premises
- b) As explained below the training shall be conducted in three categories.
- c) If special equipment or instruments used in the system, equipment / instrument manufacturer's personnel shall be brought in for instruction on that particular equipment/instrument.
- d) As operation teams are working on shift, training is required for 2 teams

12.1 Training for System Operation

As a minimum system operation training shall include:

- a) Basic P&ID.
- b) Basic theories of the system and sequence of operation.
- c) Properties of the slurry/chemicals.
- d) Operation of the system using Human Machine Interface.
- e) Modes of operation and when/how to be effected.
- f) Critical/warning/other alarms and appropriate measures to ensure proper operation of the system.
- g) Critical observations for the components of the system.
- h) Basic troubleshooting of the system.
- i) Slurry/Chemical preparation (if applicable).
- j) Sample collection.
- k) Explain how to guarantee accuracy of online & batch blending and dilution. Discuss the properties of the blended slurries (for example: conductivity curves) and the of role of the metrology analyzers.

12.2 Training for System and Equipment Maintenance

As the group of team members attending this training are from service team, this training shall be conducted to address the service issues. As a minimum, system and equipment maintenance training shall include:

- a) All major components of the system and their operation and maintenance
- b) Troubleshooting of major components such as pumps, VFDs, Instruments, etc.
- c) Control system and basic PLC training to troubleshoot control issues
- d) Electrical power system training to troubleshoot electrical related issues
- e) Instrument calibration/verification/validation to ensure the proper cooperation of the instruments.
- f) Explanation on Critical/warning/other alarms and appropriate measures to ensure proper operation of the system.

12.3 Training for System Engineers

As the group of team members attending this training are engineers, this training shall be conducted to ensure the operation and maintenance of the system without much support from system supplier. As a minimum, training for system engineers shall include all the points in detail requested in the above sections and detailed system troubleshooting methodologies.

13.0 Documentation

Documentation is very important to carry out and complete the project very effectively. Submission of the required documents at the right time moves the project as per the project plan. **Appendix 6** shows the summary of documents to be submitted at different phases of the project by the vendor. The following list below shows some critical document details that shall be submitted at the different phases of the project:

13.1 Calculation Documents

When components or process parameters in the systems are sized or rated, these have to be proved with calculations. Thus, all the relevant calculations shall be submitted to Micron for review and approval. System supplier shall bear all responsibilities for any error in the calculations and thus bear whole responsibility to ensure the system operation within the required specification. As Minimum, the following calculations shall be provided by system supplier.

- a) System Capacity
- b) Tank Sizing
- c) Pump Sizing
- d) Dampener Sizing
- e) Pipe Sizing for suction, discharge and distribution
- f) Pipe sizing for all Interfaces
- g) Pipe sizing for Drain line and Drain Headers
- h) Filter Sizing
- i) Heat Exchanger Sizing
- j) In-line Blender sizing
- k) Mixer Sizing
- l) Minimum Recirculation flowrate
- m) Time durations for different stages
- n) System and Loop pressure loss calculation
- o) Exhaust Calculations
- p) Instrument Air consumption calculations
- q) CDA consumption calculations if CDA is used.
- r) Any other required System or Equipment calculations

13.2 Sequence of Operation (SOO) Document

Sequence of operation of a system will have to explain in detail how the system operates to meet the required specifications of the system and how the operators and engineers shall intervene to ensure smooth operation of the system if the system goes into alarm mode. When sequence of operation is presented by System Supplier, the following points shall be considered:

- a) SOO can be presented in paragraph format or point format or flowchart format.
- b) If the complete system is supplied as two separate modules such as transfer unit and dispense unit, etc., while SOO for each unit is presented, all interfaces and impacts due to operation of separate modules to be clearly discussed in SOO.
- c) The sequence from Drum hook-up (or even Bulk Transfer) to dispense shall be discussed in SOO.
- d) All different stages of operation such as rinsing, qualification, stand-by, switching, etc. shall be discussed in detail.
- e) SOO for different modes of operation shall be discussed
- f) All scenarios that can occur during operation shall be considered in SOO
- g) All alarms that will be generated during operation of the system shall be linked to SOO
- h) The reasons for alarm status and required actions to be taken by operators and engineers are to be explained.
- i) Limitations of the automation and control available in the system shall be clearly discussed so that the operators and engineers would know the requirement of their actions for the smooth operation of the system.

13.3 Software Modeling Documents

Software models become inseparable from design of any system. Software designs shall be available not only for design of the system but also for space optimization and space management.

13.3.1 Building Information Modelling

BIM for Rooms, Sub-Fab Area, Common Corridors, Etc. BIM shall be used to get all intelligent engineering information about the facilities prior to construction.

13.3.2 3-Dimensional Modelling

3D Modelling for Systems to get all intelligent engineering information about the system prior to construction.

13.3.3 Computation Fluid Dynamics or Equivalent

CFD or Equivalent shall be used to identify the hot spots for slurry batch blender. While CFD can be used for tank levels and pump suctions, length of dip-tube, transfer of return bubbles into pump suctions, etc., it can be used for predicting homogeneity of the product of blenders. Time duration for blending (either stirrer or recirculation) can be properly predicted for batch blenders. Flow rate ranges and time for stabilization can also be properly predicted using CFD. All relevant hot spots shall be discovered so that the system shall be designed for better performance.

13.3.4 Fluid Dynamic Simulations for Network Pressure Drop and Flow Distributions

As the tool consumption is dynamic, system slurry/additive supply shall be equivalently dynamic to maintain system supply specifications. This kind of software shall be used to ensure system capabilities to deliver slurry/chemical within the system specifications at different scenarios. Software such as AFT Fathom or PIPEFLO can be used for this application.

13.4 Equipment List Document

Equipment List usually consists the list of major equipment and other major components of the systems along with the information such as Equipment Name, Manufacturer/Brand, Model, Part number, serial number, contact information for future purchasing (Principal and Local Agent), etc. Equipment list is to be used as an easier aid not only for future purchasing of equipment or spare parts but also to know the services rendered by the equipment supplier. Continuous improvements and issues related to the equipment and instruments can be known from the supplier.

13.5 Operation and Maintenance Manual

Operation and Maintenance manual is not just the compilation of different equipment and instrument manuals. O&M manual shall be very detail explaining about the design, configuration, operation, and control philosophy of the system that includes process, control network, and control and other components.

13.6 Summary List of Final Project Hand-over Documents

Below is the minimum acceptable requirement for final project hand-over documentation.

- a) Contract documents
 - b) Specifications: Mark and record each Product Section description of actual products installed with the following details:
 - Manufacturer's name and product model/number.
 - Product substitutions or alternates used.
 - Changes made by Addenda and Modifications.
 - c) Mechanical shop drawings:
 - Room layout drawings
 - VMB/MHB/MHEB Schematics (inclusive Equipment P&ID, Distribution P&ID, Utilities P&ID)
 - Piping layout drawings (slurry rooms, Sub-fab, CMP areas, Maintenance rooms)
 - Section drawings of pipe layout
 - Auxiliary equipment (such as dip tube change cabinets, drum cabinets, sampling station etc.)
 - Sump pump system schematics
 - Life support system (such as CO2 Fire suppression system etc.)
 - Bill of Material for components
-

d) Electrical Shop Drawings

- Slurry tool interface drawing
- Overall single line drawing
- Single line drawing for individual panel
- Panel drawings
- Panel location drawings
- Overall control architecture drawing
- Individual IO panel schematic and location drawing
- Database or I/O allocation matrix
- I/O schematic
- EMO schematic
- Electrical circuitry drawing for all equipment
- Bill of Material for components

e) Engineering Calculations

- Utilities flow and pressure calculations.
- Electrical load calculations
- System flow and pressure calculations
- R&R data from equipment manufacturer for slurry and blend systems
- Design matrix

f) Product Data

- Technical specifications of systems
- Cut sheets for all major components
- Spare parts list

g) Test Certificates

- Pressure decay test reports
- Liquid particle count test report
- PN2 particle count test report
- Metallic test report
- Assay R&R test report
- Controls Checklist
- Safety Checklist
- Functional check list of CDS
- Functional check list of SCADA
- Functional checklist of SCADA/CDS interface
- Functional checklist for all auxiliary equipment
- Field I/O and Solenoid Panels Test Report

h) Operation and Maintenance Manuals for all installed equipment.

i) Softcopy of all "As-built" drawings and documentations.

j) Softcopy of source code for all PLC programs (if contractually agreed).

k) Certificate of Warranty(s).

14.0 Reference Specifications:

14.1 Codes and Standards

Slurry distribution systems and sub-systems shall be designed according to IBC, IFC, NEC, NFPA and other applicable local codes and regulations and/or adopted equivalent standards. The following Guidelines and Codes in **Table 13** below shall be applied appropriately.

If the local code requirements are stricter than the any of the codes/standards mentioned in the table below, compliance to the requirements of the local code/standard shall be met by the contractor. Otherwise compliance to the international/semiconductor codes/standards shall be met as stated above.

Table 13 – Reference Specification: Codes & standards

S/N	Standard/Code		Description
1	FM	7-7/17-12	Semiconductor Fabrication Facilities
2	IBC	2018	International Building Code (IBC)
3	ICCEC	2006	ICC Electrical Code (ICCEC)
4	IFC	2015	International Fire Code (IFC)
5	IMC	2006	International Mechanical Code (IMC)
6	NEC		National Electric Code
7	NFPA	30	Flammable and Combustible Liquids Code
8	NFPA	55	Compressed Gases and Cryogenic Fluids Code
9	NFPA	70 – 2005	National Electrical Code (NEC)
10	NFPA	110 – 2002	Emergency and Standby Power Systems
11	NFPA	318 – 2012	Standard for Protection of Semiconductor Fabrication Facilities
12	NFPA	704 – 2001	Identification of the Hazards of Materials for Emergency Response
13	OSHA	29 CFR 1910	United States Occupational Safety and Health Administration (OSHA)

S/N	Standard/Code		Description
14	P&FM	2005	Import of Petroleum & Flammable Materials (P&FM: Fire Safety Regulations)
15	SEMI	E49.2	Guideline for the qualification of polymer assemblies used in ultra-pure water and liquid chemical systems in semiconductor process equipment
19	SEMI	E49.6	Guide for Subsystem Assembly and Testing Procedures – Stainless Steel Systems
20	SEMI	E49.7	Purity Guide for the Design and Manufacture of Ultrapure Water and Liquid Chemical Systems in Semiconductor Process Equipment
21	SEMI	F1	Specification for Leak Integrity of High-Purity Gas Piping Systems and Components
22	SEMI	F2	Specification for 316L Stainless Steel Tubing for General Purpose Semiconductor Manufacturing Applications
23	SEMI	F3	Guide for Welding Stainless Steel Tubing for Semiconductor Manufacturing Applications
25	SEMI	F34	Guide for Liquid Chemical Pipe Labeling
26	SEMI	F39	Guide for Guideline for Chemical Blending Systems.
28	SEMI	S2	Safety Guidelines for Semiconductor Manufacturing Equipment
29	SEMI	S6	Safety Guidelines for Ventilation
30	SEMI	S7	Safety Guidelines for Environmental Safety and Health (ESH) Evaluation of Semiconductor Manufacturing Equipment
31	SEMI	S8	Safety Guidelines for Ergonomics/Human Factors Engineering of Semiconductor Manufacturing Equipment
32	SEMI	S10	Safety Guidelines for Risk Assessment
33	SEMI	E33	Specification for Semiconductor Manufacturing Facility Electromagnetic Compatibility

S/N	Standard/Code		Description
34	SS	532: 2007	Code of Practice for the Storage of Flammable Liquids (Singapore Standard)
35	UFC	-	Uniform Fire Code (UFC) (especially articles 51,79 & 80)
36			All other references attached within the Micron Documents – Electrical, Controls, and Safety.

The table above specifies relevant codes and standards for different disciplines. Some specific ones related to chemical and solvent are listed below:

- a) Exhausting a Cabinet
 - i. Exhaust and Ventilation – Semi S2 (Section 22)
 - ii. Design Principles Test Methodologies for Equipment Exhaust – **Appendix 2** of Semi S2
- b) Labels, Warning Indications
 - i. Electrical warning labels – Relevant section in EN61010
 - ii. Hazard Warning Labels and Hazards Labels – relevant sections in Semi S2 and S1
 - iii. IEC 61310 Safety of Machinery – Indication, Marking and Actuation.
- c) Safety Interlocks
 - i. Semi S2 (Section 11, Safety Interlock Systems)
 - ii. Semi S2 (Section 12, Emergency Shutdown)
 - iii. Semi S2 (Section 17, Hazardous Energy Isolation)
- d) Seismic Protection – this is generally covered under I & C
 - i. Semi S2 (Section 19, Seismic Protection)
 - ii. Semi S2 (Related Information 5, Seismic Protection)
 - i. UL 698 Industrial Control Equipment for use in Hazardous Locations
 - ii. NFPA 30 – Flammable and Combustible Liquids Code

15.0 Document Control

15.1 Approval: GLOBAL_FAC_GFTT_APPR

15.2 Notification:

GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

15.3 Final Viewer Access

All Micron sites Facilities Team Members

15.4 Retention

Adherence to the requirements specified in the
[**Global Facilities - Document Control Procedures**](#)

15.5 Review

Biennially (two years)

16.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	04/16/21
1	6.0	6.2.2 item v was added	03/28/23

APPENDIX

Appendix 1 – Guideline for FAT

Name of Inspector:	FAT Date:
Name of Organization:	Person in Charge from Vendor:
Name of Fab:	Name of Vendor/Supplier:
Name of System:	Job/PO Number:

S/N	Items for Inspection	Verification	Satisfactory Remarks
1.0	Readiness of vendor with FAT checklist		
2.0	Availability of approved P&IDs		
3.0	Availability of single line diagrams		
4.0	Mechanical Checks		
4.1	Ensuring all required codes, standards and regulations are followed based on requirements		
4.2	Cabinet Material Cert		
4.3	Checking for appropriate fit of panels, doors, hinges, hood, adjustment of legs, pipe supports & required pipe sleeves. Tightness of flared fittings, and that no leaks are found.		
4.4	Availability of components and their tags as per P&ID (list along with the purpose of following components to be provided by Vendor) - Tanks - Pumps - Heat Exchangers - Filters - Manual Valves - Auto valves - Check/Non-Return Valves and their direction - Pressure Regulators - Back pressure valves - Pressure Relief valves - Flowmeters - rotameters - Pressure transducer/transmitter - Pressure switches - Pressure gauges - Temperature sensor/transmitter		

S/N	Items for Inspection	Verification	Satisfactory Remarks
	- Conductivity sensor/transmitter - Densitometer - Concentration meter - Sensors for door status, leak, drain, vent, empty, valve status, discrete level, etc. - UV/IR flame detectors - Load cells - Cooling fan for elect cabinet		
4.5	Torque (for tightness) check has to be performed for required components. Random check shall be performed if too many to check.		
4.6	Leak test for the cabinet containment area and other relevant sections in the cabinet such as sampling box, etc.		
5.0	Tanks		
5.1	Ensuring all required codes, standards and regulations are followed based on requirements		
5.2	Manufactures certificate		
5.3	Tank sizes		
5.4	Bottom shape and requirement		
5.5	Installation of dip-tubes		
5.6	Installation of spray balls based on drawings		
5.7	Leak test		
6.0	Pumps		
6.1	Check Pump Models to correspond P&ID		
6.2	Check Inlet and Outlet sizes		
6.3	Availability of surge suppression system		
6.4	Check Availability of Cooling line		
7.0	Interfaces – Size and Ratings		
7.1	Chemical		
7.1	Chemical Drain Lines		
7.2	Instrument Air Interface Line		
7.3	CDA Interface Line		
7.4	Process and Utility N2 interface Lines		
7.5	UPW Interface Lines		
7.6	Process Cooling Water		
7.7	Interconnection Lines		
7.8	Exhaust Line		

S/N	Items for Inspection	Verification	Satisfactory Remarks
8.0	Electrical and Controls		
8.1	Ensuring all required codes, standards and regulations are followed based on requirements		
8.2	Availability of list of Components in Electrical Cabinet (list of following components to be provided by Vendor) - EMO - Tower light - ELCB/RCB and other circuit breakers - Terminal blocks - VFDs - Fuse blocks		
8.3	Availability of list of PLC Components in Electrical Cabinet (list of following components to be provided by Vendor) - PLC with Removable Memory card - HMI with Removable Memory card - Communication cards - Digital and Analog cards - Segregation of cards based on segregation of engines if necessary - Communication links through DLR and check for dual link communication. Check if activation is needed and completed.		
8.4	Availability of list of Components in Pneumatic Compartment - Solenoid blocks - Segregation of solenoid blocks based on individual engines		
8.5	Do electrical panels meet required IP standard or other code of practices – prove documents to be included in manual		
8.6	Check cable conduits, din rails, solenoid valve back panels, breakers, etc. are securely mounted. Make sure no sharp edges are present.		
8.7	Check lugs and ferrules are tight and according to specification		
8.8	Panel earth connections if required		
8.9	Location of components as per approved drawings		
8.10	Check Tagging and labelling are correct		
8.11	Earthing continuity check		

S/N	Items for Inspection	Verification	Satisfactory Remarks
8.12	Wiring terminal check		
8.13	Check if all grounding is done correctly		
8.14	Continuity of Safety Earth Bar (Power Earth – PE) and interconnecting links		
8.15	Continuity of Instrument Earth Bar (IE) and interconnecting links		
8.16	Ensure integration is done as per architecture drawing		
8.17	Dry I/O Test – All continuity to be tested and proved.		
9.0	Instrumentation		
9.1	Ensuring all required codes, standards and regulations are followed based on requirements		
9.2	Door and panel switches – List along with functions shall be provided by vendor		
9.3	Leak Sensors – List along with functions shall be provided by vendor		
9.4	Empty Sensors – List along with functions shall be provided by vendor		
9.5	Level Sensors – List along with functions shall be provided by vendor		
9.6	Flow meters – List along with functions shall be provided by vendor		
9.7	Pressure switches – List along with functions shall be provided by vendor		
9.8	Pressure Transduces – List along with functions shall be provided by vendor		
9.9	Temperature Sensors – List along with functions shall be provided by vendor		
9.10	Conductivity Sensor/Transmitter – List along with functions shall be provided by vendor		
9.11	Density Meter – List along with functions shall be provided by vendor		
9.12	Load Cells – List along with functions shall be provided by vendor		
9.13	Pump leak sensors for pneumatic pumps – List along with functions shall be provided by vendor		
9.14	Z-purge instrument based on necessity – functions shall be provided by vendor		
9.15	RPM – Counters if any		

S/N	Items for Inspection	Verification	Satisfactory Remarks
9.16	Thermostats		
10.0	System Power Supply Check		
10.1	Ensuring all required codes, standards and regulations are followed based on requirements		
10.2	Verify all cabinet circuit breakers are open/off		
10.3	Verify all knife edge field terminals are open		
10.4	Connect the required power supply from field to incoming terminals		
10.5	Before switching on MCBs, ensure no voltage is present at the reference point		
10.6	Before switching on MCBs, ensure the system is ready for power supply. Any loose connections or any connections that can cause problems to be rectified.		
10.7	Operate Distribution MCBs within the cabinet and confirm correct voltage		
	Specified Voltage A/C: Actual Measure Voltage:		
	Specified Voltage D/C: Actual Measure Voltage:		
11.0	Wet I/O Testing		
11.1	All pneumatic operated valves shall be tested by manual activation at solenoid side and HMI side		
11.2	All door and panel switches shall be tested by vendor		
11.3	All leak Sensors with functions shall be tested by vendor		
11.4	Empty Sensors – List along with functions shall be tested		
11.5	Level Sensors – All functions (low, high, filling, stop, etc.) shall be tested		
11.6	Flow meters – All functions (low, high, etc.) shall be tested		
11.7	Pressure switches – All functions (low, high, etc.) shall be tested		
11.8	Pressure Transduces – All functions (low, high, etc.) shall be tested		
11.9	Temperature Sensors – All functions (low, high, etc.) shall be tested		

S/N	Items for Inspection	Verification	Satisfactory Remarks
11.10	Conductivity Sensor/Transmitter – All functions shall be tested		
11.11	Density Meter – All functions shall be tested		
11.12	Pump leak sensors for pneumatic pumps – All functions shall be tested		
11.13	Z-purge instrument based on necessity – All functions shall be tested		
11.14	RPM – Counters if any		
11.15	Thermostats and its functions to be tested		
11.16	Load Cells – All functions shall be tested		
11.17	Load cell taring and repeatability		
11.18	Test Related to Pumps		
11.19	VFD settings and ramp-up and ramp down to be tested through mA injection or some other means		
11.20	Check VFD digital signals and their appropriation		
11.21	Max RPM setting based on necessity		
11.22	Whatever field instruments cannot be tested due to unavailability of instrument, FAT testing shall be performed by injecting appropriate 24V DC or required mA.		
11.23	NPSH testing when max flow takes place – If system is hooked up to a loop only this test can be performed.		
12.0	Functional Test		
13.0	Alarm list and testing of alarm list – Vendor needs to provide the alarm list and needs to create such alarms		
14.0	Scenario Tests – Vendor needs to provide list of all possible different scenarios such as tank switch, preparation, qualification, engine failure, auto filter change, etc. and actual sequence of operation during these scenarios.		
15.0	Pressure control modes and appropriate tests		

Note:

1. The above table is only a guideline and actual full list shall be provided by the vendor.

Appendix 2 – Guideline for Installation Inspection

Name of Inspector:	FAT Date:
Name of Organization:	Person in Charge from Vendor:
Name of Fab:	Name of Vendor/Supplier:
Name of System:	Job/PO Number:

S/N	Items for Inspection	Verification	Satisfactory Remarks
1.0	Readiness of vendor with Installation Inspection checklist		
2.0	Availability of approved P&IDs and Lay Out Drawings		
3.0	Availability of single line diagrams and Electrical & I/O Panel Installation Lay Outs		
4.0	Mechanical Checks		
4.1	Ensuring all required codes, standards and regulations are followed based on requirements		
4.2	Ensuring the installations is as per approved lay out drawings and all the installations are fastened properly, securely and firmly on mounting supports.		
	- Day Tanks		
	- Slurry Transfer Units		
	- Slurry Transfer & Blending Units		
	- Slurry Dispense & Blend Unit / Slurry Dispense Unit		
	- VMBs		
	- Slurry Sump Pump Systems		
4.3	Ensure all weld tags according to Semi S2 Standard		
	Ensure the installation integrity of the major components such as pumps, filters, valves, regulators, pipes, etc. – Individual Checklists are to be submitted		
	Ensure the ratings of pumps and filters as per requirement – Individual Checklists are to be submitted		
	Ensure the mechanical installation integrity of the major instruments such as flowmeters, conductivity meters, concentration meters, pressure transmitters, etc. – Individual Checklists are to be submitted		
4.4	Ensure all Labelling and component Tags According to Semi S2 Standard		
4.5	Ensure all edges of Unistrut are filed and covered with approved end covers to avoid any injuries		
4.6	Ensure usage of appropriate chemical resistant anchor bolts and anchoring accessories		
4.7	Ensure Appropriate leveling legs used for the system		
4.8	Ensure Levelling of the system done appropriately such that the system does not exert any pulling or pushing effect to the associated pipes and supports		
4.9	Ensure appropriateness of Drain Line design		
4.10	Ensure appropriateness of drain lines connections		

S/N	Items for Inspection	Verification	Satisfactory Remarks
4.11	Ensure Drain ports at relevant locations and the use of appropriate valves and plugs.		
4.12	Ensure there are no conflicts with other installations.		
4.2	Pressure Ratings of Pipes and Tubes used in slurry Rooms		
	- PFA Tubing		
	- Outer Containment Tubes		
	- Drain Lines		
	- Instrument Air Interface Line		
	- CDA Interface Line		
	- Process and Utility N2 interface Lines		
	- UPW Interface Lines		
	- Process Cooling Water		
	- Interconnection Lines		
4.3	Pressure Ratings of Pipes, Tubes and other components used at the Field (VMBs or any other remote slurry related units)		
	- PFA Tubing		
	- Outer Containment Tubes		
	- Instrument Air Interface Line		
	- CDA Interface Line		
	- Process and Utility N2 interface Lines		
	- UPW Interface Lines		
	- Process Cooling Water		
	- Interconnection Lines		
4.4	Material and Brand of Pipes and Tubes used		
	- PFA Tubing		
	- Outer Containment Tubes		
	- Drain Lines		
	- Instrument Air Interface Line		
	- CDA Interface Line		
	- Process and Utility N2 interface Lines		
	- UPW Interface Lines		
	- Process Cooling Water		
	- Interconnection Lines		
	- Exhaust Line		
4.5	Availability of Pressure Test Reports		
	Slurry Loop		
	outer Containment		
	- Instrument Air Interface Line		
	- CDA Interface Line		
	- Process and Utility N2 interface Lines		
	- UPW Interface Lines		
	- Interconnection Lines between Day Tank and System		
	Interconnections between Transfer unit and Day Tank		
	Interconnections between Blender Unit and Day Tank		

S/N	Items for Inspection	Verification	Satisfactory Remarks
	Interconnections between Bulk Tank to other transfer/blending/dispense sections		
	Tanks		
	VMBs if it is required		
4.6	Material and Brand of other Installation Components		
	Exhaust Dust		
	Exhaust Damper		
	Exhaust Pressure indicator/transmitter		
	Solenoid block for VMB		
	VMB Valves		
	VBM Leak Sensors		
	Pipe Support Hangers		
	Pipe Support Clamps		
	Pipe Support Unistrut		
	Pipe Support		
4.7	Sump Pump System and its components		
	Pump Cabinet		
	Sump pit and sump tank checks for its integrity		
	Capacity of pump for the required usage		
	Sizes of Drain Line Connections to Sump Pit and Sump Tank		
	Capacity of Drain Line Pipes		
	Integrity of Piping system		
	Requirement of double containment		
5.0	- Electrical Checks		
5.1	- Follow Required Installation Inspection Checklist specified in Electrical Specification		
5.2	Ensuring all required codes, standards and regulations are followed based on requirements		
5.3	Ensuring the installations is as per approved lay out drawings and all the installations are fastened properly, securely and firmly on mounting supports.		
5.4	Ensure Control cables are not bundled with power cable to avoid any EMI related issues.		
5.5	The panel passes a visual inspection and complies with the following:		
	Ensure Outer enclosure of Electrical, I/O Panel, and Electrical & Control panel of the system are labeled appropriately with ID tag.		
	Ensure Devices and modules are installed per general assembly drawings and single line diagrams.		
	- Wiring, devices and modules are labeled within the panel in accordance with the Single Line and general assembly drawings.		
	- Wiring terminations (with lugs, ferrules, etc.) are adequate and firm, and wiring utilizes the color code established within the drawings.		

S/N	Items for Inspection	Verification	Satisfactory Remarks
	- Wiring cable shields are terminated at one end to power supply ground as appropriate.		
	- Wiring is gathered with tie-wraps or bundled within wire-mold channels.		
5.6	- Verify Ratings of all cabinet circuit breakers		
5.7	- Verify all incoming and outgoing connections to and from the panels. Other ends need to be verified.		
5.8	Ensure all the electrical and control connections to I/Os and controllers of Pumps, VFDs and associated controllers – Individual Checklists are to be submitted		
5.9	Ensure all the electrical and control connections to I/Os and controllers of instruments, transmitters, etc. – Individual Checklists are to be submitted		

Note:

1. As components in the system are checked during FAT and SAT, the verification of the system components is not necessary unless there is any change done after FAT or SAT.
2. If the components are more than one, the vendor shall prepare the list based on the actual installation data.
3. Installations related to Fire Sprinkler or other Safety Related Installations will have to be inspected by relevant parties.

Appendix 3 – Chemical Receiving Checklist

Chemical Receiving Checklist			
Chemical System Vendor		Date/Time of Delivery	
Chemical Supplier		Supervisor in Charge	
Area of Inspection		Inspected By	
S/N	Items	Yes/ No/ NA	Actions / Comments
1	Prior to Delivery		
1.1	Confirm chemical name, container type, and quantity to be delivered onsite.		
1.2	Is calcium gluconate onsite for hydrofluoric acid delivery?		
1.3	Confirm hazard type for chemical to be delivered and any specific safety requirements.		
1.4	Ensure access is prepared - area clear with no materials or vehicle obstructions		
1.5	Ensure all safety paths are cleared for access to safety shower and for evacuation		
1.6	Ensure safety showers are tested and functional		
1.7	Are all room life safety devices tested and functional - fire suppression, fire sprinklers, fire alarms, etc.		
1.8	Ensure room exhaust is active and adequate		
1.9	Ensure all PPE, spill kits, fire extinguishers, SDS, and first aid kits are present		
1.10	Ensure proper chemical signage, including room signage is displayed and the appropriate locations		
2	Onsite Delivery / Receiving of Chemicals		
2.1	Is the point of transferring being guarded and kept out of unauthorized personnel and work?		
2.2	Are proper signs erected to caution/warn unauthorized team members to keep out?		
2.3	Are other incompatible works being asked to cease during the transferring of chemical - like hot work, grinding, etc.		
2.4	Are incoming chemical containers in good condition		
2.5	Verify containers are labeled correctly and SDS is present		
2.6	Check that all delivery crews are wearing appropriate PPE		
3	Storage of Chemicals		
3.1	Is storage space sufficient for qty of chemical to be delivered		
3.2	Sufficient passageway and access are provided to store and deliver chemical for use		
3.3	Storage racks of of adequate strength with labels for storage weight limits and chemical types		
3.4	Chemical bottle/carboy labels face outward for ease of identification and retrieval		
3.5	Incompatible chemicals are isolated and stored separately		
3.6	Adequate ventilation and safety equipment are provided and readily available		
4	Handling and use of Chemicals		
4.1	Workers know the chemical name and safe handling procedures		
4.2	Workers have undergone hazard communication by their leader		
4.3	Workers have access to SDS at area of work at all times		
4.4	Workers are wearing the appropriate PPE		
4.5	Workers are aware of procedure for spill clean up and location of spill kits		
4.6	Appropriate waste collection bins are in place and labeled		
4.7	Workers know proper disposal procedures		
4.8	Workers are aware of location of nearest emergency shower/eyewash station in case of chemical exposure		
4.9	Workers observe personal hygiene after handling of chemicals		

Appendix 4 – Checklist for First Charge of Chemical / Slurry

Preparation Checklist for first Charge of Chemical into System & Loop Distribution			
Inspected By		Date/Time	
Area of Work		Chemical Name	
Equipment ID		Chemical Category	
S/N	Items	Yes/ No/ NA	Remarks
1	Have all piping and drain lines passed pressure testing?		
2	Have all chemical system leak sensors, interlocks, pressure relief devices and EMO been tested in AUTO mode? Has Micron verified?		
3	Are all valves in chemical and drain piping in closed and capped position?		
4	Are all utilities systems turned ON and in required spec with stable supply ? - Power - Exhaust (for chemical system, room emergency exhaust) - UPW - CDA - N2 - Instrument Air (if applicable) - City water (for Emergency shower)		
5	Are all containment boxes closed and sealed properly?		
6	Are all distribution valves manifold boxes closed and sealed properly?		
7	Have all chemical distribution VMB/MHB/MHEB leak detects been tested with the chemical system in AUTO and has Micron verified that the chemical system shuts down when those leak detects are activated?		
8	Has the sump been tested and in AUTO mode?		
9	Are all exhaust systems turned ON and balanced - chemical system, sump, and room?		
10	Is Solvent system (tank, piping & VMB) connected to appropriate grounding bar?		
11	Are the waste systems ready and available?		
12	Has the quantity and timing of chemical being flushed to WWTP and waste chemical being trucked out to outside vendor been verified and communicated to the WW Operations team?		
13	Is the SDS readily available and location known?		
14	Has ERT been informed of the charging of chemical and on standby for the event.		
15	Is the ERT contact information posted in the chemical room?		
16	Has the charging of chemical been communicated to the ENTIRE work force in the affected building?		
17	Are there cautionary signs in the area of chemical charging, including all areas of the sub-fab where the chemical piping is present?		
18	Are spill kits, first aid box, and PPE in the area of work?		
19	Are the emergency eye wash/shower stations tested and ready, including all areas of the sub-fab where the chemical piping is present?		
20	Are all life safety systems - fire sprinklers, fire alarms & fire suppression system (where applicable), are tested and ready in the chemical room, including all areas of the sub-fab where the chemical piping is present?		

Appendix 5 – Utilities Interfaces and Requirements

UTILITIES / CATEGORY		Slurry Room	Sump Pump System	Slurry System
Electrical		- Room EMO - Room Lighting and power sockets - Power supply to Room electrical panels - Earth Grounding bar	Power supply to sump system	Power supply to system
I&C		- Room EMO at door exits - Room door card access	Sump system alarms to SCADA / FMCS.	- Individual System EMO and Global EMO at FMCS or TGCM room. - All required system alarms to SCADA. - Required instrumentation information and interlocks.
Mechanical	IA or CDA (if IA is not available)	20% spare points	Supply of required amount of CDA for sump pumps.	Supply of required amount of IA or CDA for valves & other instrumentation.
	Emergency Exhaust	Yes	NA	NA
	HVAC	- Room Temperature shall be maintained 20–25°C. - Relative humidity < 65 % - Sufficient air exchange as per room hazard classification and individual system air exchange requirement.	NA	NA
	PCW	20% spare points - Based on requirement, Chilled water may be used for Heat Exchanger.	NA	Depending on system requirement
	Acid / Caustic / Exhaust	20% spare points	Required for Sump pump cabinet and sump tank.	- Meet code requirement if any. - Depending on the nature of the chemical.
	Door Air Curtain	Depending on high humidity regions, an automatic air curtain shall be installed on the outer door of slurry rooms where drum/tote delivery frequently happens.	NA	NA

UTILITIES / CATEGORY		Slurry Room	Sump Pump System	Slurry System
	Emergency Eye Shower	As per EHS requirement	NA	NA
	Fire Sprinkler	As per relevant Fire code requirement.	NA	Not required unless otherwise specified.
Gas	GN2	20% spare points	NA	1. Blanketing 2. Purging
	PN2	20% spare points	NA	Based on requirement GN2/ UN2/ PN2 will be used.
Water / Waste	UPW	1. 20% spare points 2. UPW Return Line Valve Manifold assembly	NA	1. Preparation / Recipe 2. Flushing & Rinsing 3. Humidification
	Drain / Waste sump	Sump tank / pit connection to appropriate sump system.	Double contained connection to appropriate drain header.	Drain connection to sump tank.
	Non-Portable /City Water	For Emergency Shower & Eye Wash units	Drain line to sump tank	NA
Civil		1. Room Doors with Push Bar Safety Panic Bar Exit handle 2. Sump Pit(s) 3. Equipment Move-in access 4. Trenches with sufficient gradient leading to sump pit. 5. Provisions for pipe supports 6. Loading bay and Drum transport passage. 7. Discharged Sprinkler water gravity drainpipe to designated collection tank. 8. Tank plinths and bund wall (if required during design phase) 9. Removable Wall for bulk tank access	1. Sump Pit 2. Trenches leading to sump pit	
EHS		LSS Alarms Room PPE Cabinets	LSS Alarms	

Appendix 6 – Documents to be submitted at different phases of the project by vendor

S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
1	Company Information and corporate overview with Local Organization Chart	Progressive report upon Changes	Progressive report upon Changes	Progressive report upon Changes	Progressive report upon Changes	
2	Past projects or References	NA	NA	NA	NA	NA
3	Standard Project Management and Project Control Flowchart	Project Management and Project Control Flowchart relevant to this project	Accepted Project Management and Project Control Flowchart	Progressive report upon Changes	Progressive report upon Changes	Applied Project Management and Project Control Flowchart
4	Standard Project Organization Chart	Project Org Chart with names	Progressive Project Org Chart with names upon changes made – Installation team needs to be included with details.	Progressive Project Org Chart with names upon changes made – commissioning team needs to be included with details.	NA	Final Project Org Chart and Contact List of key personals
5	Relevant approvals and certificates for manufacturing slurry/chemical systems and Tanks	Manufacturing site information and required certificates including Clean Room (Clean Environment) Certificates (or Information) for system manufacturing	Manufacturing site information and required certificates and acceptance from Micron	NA	NA	Relevant approvals and certificates for manufacturing slurry/chemical systems and Tanks
6	Scope work Matrix	Revised scope matrix without deviating the agreed scope as per PO	Progressive scope matrix upon agreed changes	Progressive scope matrix upon agreed changes	Progressive scope matrix upon agreed changes	Executed Scope Matrix and Gaps between agreement and execution.
7	Schedule with Critical Path	Schedule with Critical Path for approval with actual dates honoring	Progressive schedule with appropriate mitigation schedule.	Progressive schedule with appropriate mitigation schedule	Progressive schedule with agreed completion dates	

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S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation and Commissioning Phase	Project Completion
		the agreement as per PO				
8	Standard Process Flow Diagrams	Process Flow Diagrams	Reviewed PFDs	Approved PFDs	NA	As-Built PFDs
9	Standard P&IDs	System P&IDs	Reviewed System P&IDs	Approved system P&IDs	NA	As-Built system P&IDs
10	Layout Drawings for different rooms	Layouts and Initial Solid works or equivalent that can be used for BIM Development systems piping, electrical and control networks	Reviewed Layouts and stages of BIM showing all conflicts systems piping, electrical and control networks	Approved Layouts and stages of BIM with existing conflicts systems piping, electrical and control networks	NA	Soft copy of As built interactive (appropriate for future use) layouts and BIM softcopies.
11	Standard System Design Features	System Design Features	Reviewed design features and Shop Drawings before starting fabrication	Approved Shop Drawings before starting fabrication	NA	As built system drawings
12	Technical proposal that include: - Design Basis and calculations - Performance guarantee - Description of major processes Equipment supply and specifications Piping specifications	Technical information with respect to design	Reviewed technical information	Approved technical information.	NA	Final Technical Information that include: Design Basis and calculations Description of major processes Equipment supply and specifications Piping specifications Utility Interfaces and actual utility consumptions Final Performance and gaps

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S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
	Utility Interfaces and requirements					
13	Standard Component List submittals	Component list submittals: - Equipment list - Instrumentation list - Mechanical Components list - Electrical Panel/ Component list - PLC/ SCADA/ Network related Hardware/ Software Components list - I/O list	Reviewed component list submittals: - Equipment list - Instrumentation list - Mechanical Components list - Electrical Panel/ Component list - PLC/ SCADA/ Network related Hardware/ Software Components list - I/O list	Approved component list submittals: - Equipment list - Instrumentation list - Mechanical Components list - Electrical Panel/ Component list - PLC/ SCADA/ Network related Hardware/ Software Components list - I/O list	NA	Final Component List - Equipment list - Instrumentation list - Mechanical Components list - Electrical Panel/ Component list - PLC/ SCADA/ Network related Hardware/ Software Components list - I/O list
14	Software that will be used for development of the system – CFD, AFT Fathom or PIPEFLO, 3D Modelling, BIM, etc.	Software that are used for design calculations and for development of the system – CFD, AFT Fathom or PIPEFLO, 3D Modelling, etc.	- Computational Fluid Dynamics (CFD) results for required processes. - Results of hydraulic modeling based on AFT Fathom or PIPEFLO - 3D Modelling for system manufacturing	Approved - Computational Fluid Dynamics (CFD) results for required processes. - Results of hydraulic modeling based on AFT Fathom or PIPEFLO - 3D Modelling for system manufacturing	NA	Soft copy of As built and interactive (appropriate for future use) modelling information of CFD, AFT Fathom or PIPEFLO, 3D Modelling for systems, piping, etc.
15	Planned Control Network Architecture	Control Network Architecture	Control Network Architecture with - Graphic, Loop check documents - I/O loop check list - Instrument – HMI - SCADA loop test with	Approved Control Network Architecture and related documents. - Graphic, Loop check documents - I/O loop check list	Progressive check lists	As-built Control Network Architecture with signed related documents - Graphic, Loop check documents - I/O loop check list

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S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation and Commissioning Phase	Project Completion
			alarm and failure test list - Instrument - HMI - SCADA test forms	- Instrument – HMI - SCADA loop test with alarm and failure test list Instrument - HMI - SCADA test forms		- Instrument – HMI - SCADA loop test with alarm and failure test list Instrument - HMI - SCADA test forms
16	Electrical plans	General arrangement electrical drawings	Electrical drawings, I/O panel drawings and single line diagrams	Approved electrical & I/O panel drawings and single line diagrams	Progressive check lists	As-Built electrical drawings, electrical panel layout drawings, I/O panel drawings, single line diagrams.
17	Standard Sequence of Operation	Sequence of Operation – Theory of Operation shall be included	Reviewed SOO - Theory of Operation shall be included. Functional test with interlocks and scenarios shall be included	Approved SOO - Theory of Operation shall be included. Approved Functional test with interlocks and scenarios shall be included	Progressive check lists	Final SOO with the with signed functional test reports.
18	Standard VMB and MHB/MHEB drawings	VMB and MHB/MHEB Drawings		Approved VMB and MHB/MHEB Drawings	Progressive check lists	As-Built VMB and MHB/MHEB Layout Drawing
19	Standard HAZOP and FMEA	HAZOP and FMEA – This should lead safe and consistent system operation	HAZOP and FMEA outcome and report for necessary actions	Verification checklist if HAZOP and FMEA conclusions are followed in the systems and installations	HAZOP and FMEA related Approved check list for verification of systems and installations	Final FMEA and HAZOP analysis reports
20	Standard installation guidelines	Reviewed installation guidelines suitable for chem/slurry systems.	Approved installation guidelines suitable for chem/slurry systems.	Progressive check lists	NA	Final installation guidelines used during installations.
21	Standard commissioning plan	Commissioning plan	Reviewed commissioning plan	Progressive commissioning plan with change of any schedule and mitigation plan.	Progressive commissioning plan with change of any schedule and	Completion Certificates

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S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
					mitigation plan for completion.	
22	Standard qualification procedure	Qualification procedure for systems and other installations	Reviewed qualification procedure for systems and other installations	Approved qualification procedure for systems and other installations	Progressive check lists	Signed copy of actual usage of approved qualification procedures.
23	Standard alarm list with alarm category	Alarm list and alarm category with reasons for any particular alarm.	Reviewed alarm List and alarm category with reasons for any particular alarm.	Approved alarm list and alarm category with reasons for any particular alarm.	Progressive check lists	Final alarm list with alarm category and priority
24	Standard pressure test procedure	Applicable pressure test procedures	Reviewed pressure test procedures	Approved pressure test procedures	Progressive pressure test reports	Final pressure test reports with relevant signatures
25	Standard FAT Information	FAT procedure and check list	Reviewed FAT procedure and check list with date for FAT	Approved FAT procedure and Checklist shall be used for FAT.	NA	Signed FAT Check lists.
26	Standard SAT Information	SAT procedure and check list	Reviewed SAT procedure and check list with date for SAT	Approved SAT procedure and Checklist.	Progressive SAT check lists	Signed SAT Check lists.
27	Standard installation inspection information	Installation inspection procedure and check list	Reviewed installation inspection procedure and check list	Approved installation inspection procedure and checklist.	Progressive installation inspection reports. Field service engineer's report on installation compliance	Signed installation inspection check lists.
28	Standard Leach Test Procedure	Applicable Leach Test Procedure – Based on chemistry	Reviewed Leach Test Procedure – Based on chemistry	Approved Leach Test Procedure – Based on chemistry	Progressive leach test reports	Leach test reports
29	Standard sampling procedure analytical methods	Applicable sampling procedure analytical methods	Reviewed sampling procedure analytical methods	Approved Sampling procedure and analytical methods used	Progressive analytical reports	Results of samples tested to be incorporated in commissioning reports

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S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
30	All other standard testing and commissioning information	All other testing and commissioning procedures and check lists	All other reviewed testing and commissioning procedures and check lists	All other approved testing and commissioning procedures and check lists including Functional test with interlocks and scenarios shall be included	Progressive related testing and commissioning reports.	All other signed testing and commissioning check lists and commissioning report
31	Standard calibration, verification and mapping information	Calibration, verification and mapping procedures and other information for the systems	Reviewed calibration, verification and mapping procedures and other information for the systems	Approved calibration, verification and mapping procedures and other information for the systems	Progressive reports and certificates	Final calibration, verification and mapping procedures and certificates
32	Standard analytical and physical parameters of the systems	Analytical and physical parameters of the systems to be critically monitored	Reviewed analytical and physical parameters of the systems to be critically monitored	Approved analytical and physical parameters of the systems to be critically monitored	Progressive reports on SCADA trends	All analytical and physical parameter test reports with SCADA trends for approval
33	Standard training program that includes schedule, categories of training and appropriate training contents	List of trainings to be provided for different categories	Reviewed list of trainings to be provided for different categories	Approved list of trainings to be provided for different categories	Progressive Training information	Training certificates and attendance List
34	Standard required test reports for tanks	Required test reports for tanks for appropriate chemistries	Reviewed required test reports for tanks for appropriate chemistries	Approved required test reports and PE certifications (based on regulatory requirements) for tanks for appropriate chemistries	Progressive certificates and reports	Final required test reports and PE certifications (based on regulatory requirements) for tanks for appropriate chemistries
35	Standard QA/QC plan and QA/QC procedures	Applicable QA/QC plan and QA/QC procedures	NA	NA	NA	NA

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S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation and Commissioning Phase	Project Completion
36	Standard required PE (or Regulatory Authority) certification for the required codes and regulations for systems, rooms and other related areas.	Applicable required PE (or Regulatory Authority) certification for the required codes and regulations for systems, rooms and other related areas.	Required PE (or Regulatory Authority) certification for the required codes and regulations for systems, rooms and other related areas.	PE (or Regulatory Authority) certification for the required codes and regulations for systems, rooms and other related areas before charging chemicals.	Progressive approvals and reports	Approval from PE or regulatory authority with relevant stamps and signatures.
37	Standard Change Control Forms	Applicable Change Control Forms	NA	NA	NA	Deviation List with approved deviation forms
38	Standard start-up and shut-down procedures	Applicable start-up and shut-down procedures	Reviewed start-up and shut-down procedures	Approved start-up and shut-down procedures		Final start-up and shut-down procedures
39	NA	NA	NA	NA	NA	Critical spare part list PLC, HMI and SCADA back-up programs PLC, HMI and SCADA licenses for the appropriate versions PLC, HMI and SCADA passwords for different access levels – Operator to Programming. Relevant on-site and off-site test reports and QA/QC documents

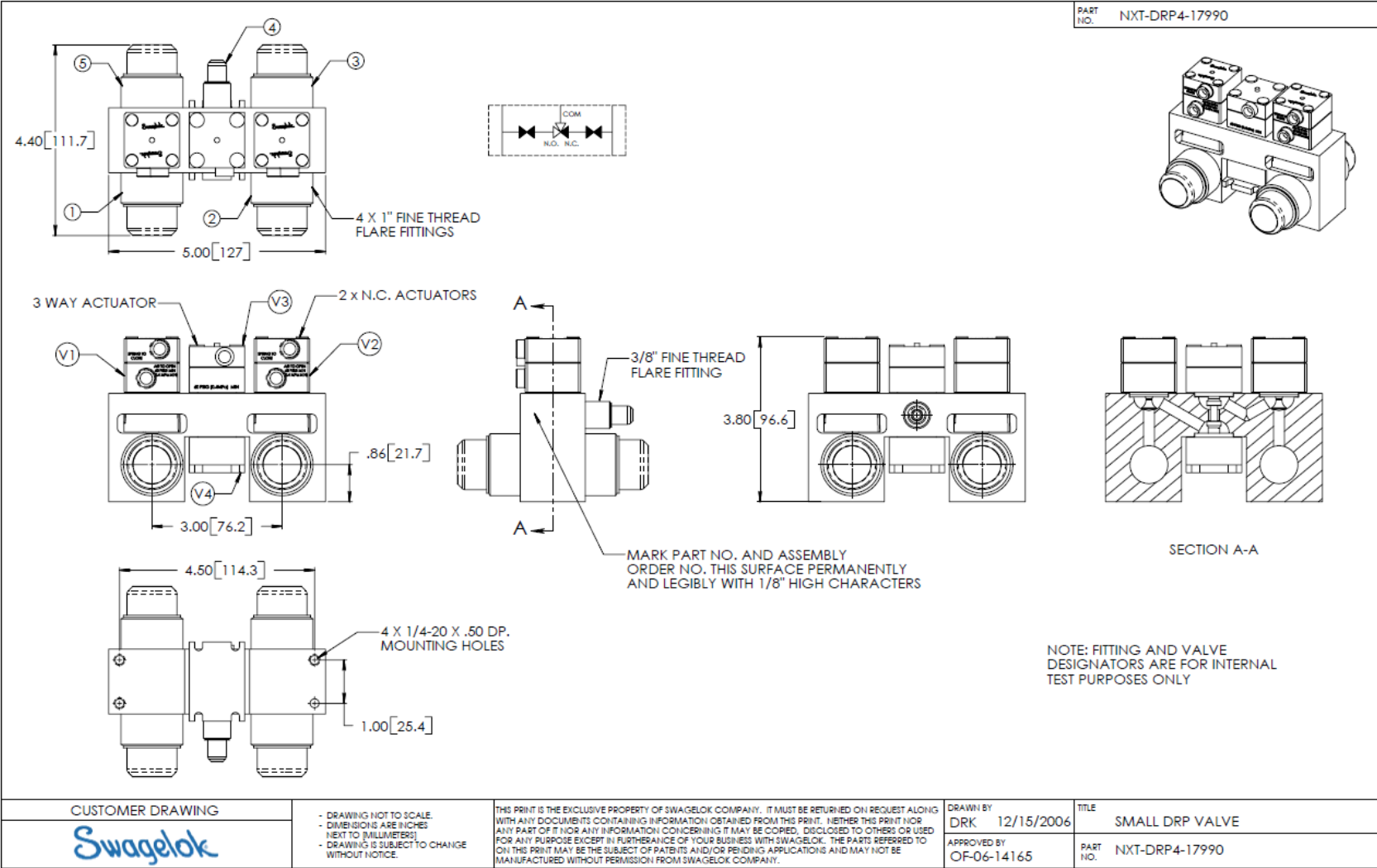
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S/N	Submissions for RFP	Upon Award and During Design	Upon End of Design	During Construction (including FAT) and Installation Phase	During, SAT, Installation Inspection and Commissioning Phase	Project Completion
40	Standard table of content for system operation and maintenance manual	Applicable table of content for system operation and maintenance manual	Reviewed table of content for system operation and maintenance manual	Approved table of content for system operation and maintenance manual	Final draft of O&M Manual as per approved table of content	System operation manual as per approved table of content
41	Submission of any other standard documents related to the project	Highlight any missing documents related codes, standards, regulations, etc.	Verification of any missing documents related codes, standards, regulations, etc.	Submissions of any missing documents related codes, standards, regulations, etc. to relevant authorities	NA	Approvals of any missing documents related codes, standards, regulations, etc.
42	Commercial proposal with required breakdown	Submission of any other Documents related to Issue For Construction (IFC)	NA	NA	NA	Approved completion certificate to start warranty of systems and components
43	Document submission plan	Document submission plan and applicable route.	NA	NA	NA	NA

Appendix 7 – Swagelok Block Valve



Appendix 8 - Desired Design Requirements for Different Slurry System Type

Requirement	STU	SBDU	STBU	SDU	Remarks
Room Requirements					
Room EMO shall be placed at the appropriate locations outside the room – Room EMO shall be hard wired to individual systems inside the room	Yes	Yes	Yes	Yes	
Remote EMO in TGCM room shall be installed – it can be either hard wired or enabled by software					
Room Exhaust shall be separate from system exhaust					
If Room Utilities are available in the room with hook-up point, it can be used for connecting to the systems					
If Room Utilities are to be installed, Contractor shall take up the responsibility within the room from hook-up points of Main Headers with appropriate isolation and non-return valves based on the utility					
Power supply for systems shall be from dual sources and if one source fails, system operation should not have any impact					
Dual DC power supply unit shall be provided to eliminate single point of failure					
General Requirements					
- Height, width, and depth of the cabinets are to be approved by Micron Staff before construction. - Cabinet design should support the weight of two 110kg workers standing on the cabinets during installation.					Room height, facility piping, Move-in doorways, docks, corridors, etc. to be considered by the vendor.

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Cabinet needs to be under negative pressure with a minimum of 2~3 air change/min. Therefore, when there is no exhaust, system shut down may be required.					
- Exhaust Pressure (flow) switch with low alarm/system shutdown					
- If any wet area door is opened when system is in operation, system shut down is needed.					
- Maintenance bypass switch with by-pass key					Door Sensors and Leak sensors can be bypassed
- System needs to be shut off with Emergency shut-off button (EMO)					
- Alarm buzzer and three-color light tower for system status indication					
- System shall be hard wired to Room EMO					
- System EMO shall be controlled from TGCM office – It can be either hard wired or programmed.					
- Leak detection shall be provided within the Slurry Distribution Unit (SDU) or separated cabinets. Response strategies shall be identified based on site-specific safety requirements, chemical types, and local/national codes and regulations.					
- Programmable time feature on door safety interlocks needed to do necessary work when system is in operation. Thus, double protection of Manual by-pass key with time setting on HMI shall be in place for bypassing door sensor and leak sensor.					
- No sedimentation is allowed in the tanks or piping.					
- Depending on the slurry properties, to prevent any sedimentation, recirculation shall be present in loops, day tanks, drums, totes, etc. If flowmeters are not present, this requirement shall be met using other available instruments.					
- Fail safe pneumatic controlled valves with on/off indication					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- PFA 451 HP tubing shall be used for all slurry and Blend lines					
- Sweep Elbows and Tees shall be used to reduce shear and pressure drops.					
- All Fittings - If unions are used, Primelok / Super pillar fittings shall be used.					
- UPW Water spray shall be available					
- Blending shall be generally performed without manual intervention					
Utility Lines					
- PFA tubing for internal DI Water piping. HP-PVDF pipe can be converted to PFA tubing outside the cabinet if UPW loop return does not enter the cabinet.					Depending on site requirements UPW loop can take place within the cabinet.
- Install check valve on UPW lines to prevent any back contaminations					
- PFA Tubing with PP header for drain piping					
- Process Cooling water supply/return assemblies consisting of isolation valves, pressure regulator, temperature sensors, flow meter, flow regulating valve, etc. shall be kept outside the cabinet. Pipes shall be converted to PFA tubing before the entry of the cabinet.					Depending on temperature control requirement
Drum or Tote Cabinet					
- Ensure proper barcode (and Key code) practice. System barcode and drum/tote barcode need to match.					
- Proper holder shall be present for dispense heads - holder shall be such that wetted part of the dispense head should not touch any part of the cabinet when drum/tote change takes place - Drip from dispense head shall be funneled to a container					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- UPW Spray gun shall be available					
Blender Units					
- Blending shall be performed based on given specification					Refer specification sheet
- Blending shall be performed using load cell or mass flowmeters or quality parameter depending on the blending feasibility					
- Blending basis (using mass flow rates or product quality parameter) shall be user configurable. - When selected as "Mass Flow Basis", blending shall be done on mass flow basis - When Selected as "Control Parameter Basis", blending shall be performed on control parameter feedback loop.					Under both circumstances, both control parameter and mass flow rates shall be used for counter checking the other.
- Coarse and fine addition of UPW, Slurry and Oxidizer shall be in place based on requirement of the quantity and quality. If coarse is not necessary, it can be omitted.					
- When automatic blending is performed, qualification of the blend to be done with online instruments					
- Automatic zeroing (or taring) to be done for load cell and mass flowmeters shall be present. However, programming to be done such that auto zeroing (or taring) shall not lead to erroneous blending.					
- For slurries having H2O2 in the blend shall monitor H2O2 concentration continuously or intermittently based on the instrument used for concentration measurement.					
- Density Shall be a process control parameter					
- Conductivity Shall be a process control parameter					Dual conductivity sensors for STBU is required
- Concentration Shall be a process control parameter					
- Particle (LPC) measurement shall be a process control Parameter					If desired

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Slurry and UPW wastage shall be very minimal					
- Maximum time to qualify the blend during each blending shall be specified by the vendor					
- Return loop (or supply loop for STBUs) or Tank cooling coil as heat exchangers as part of the SBDU/STBU/SDU equipment. Heat Exchanger for local loop re-circulation.					Depending on temperature control requirement
- Heel Transfer feature					
Sampling Cabinet					
- Sample valves to enable sampling after Drum hook-up / Blend / Dispense					Detailed locations for sampling will be discussed in P&ID Review
- Proper air change rate shall be taken place in sampling cabinet. Make sure no contamination takes place due to bad exhaust design for sample cabinet					
- UPW Water spray coil shall be long enough to clean sampling cabinet after sampling					
N2 Blanketing and N2 Humidifier					
- N2 Blanketing shall be provided to avoid any contamination. Based on requirement, Humidified N2 shall be used for slurries.					
- If humidified N2 is used, humidifier water level shall be automatically controlled to prevent the followings: - Dry N2 enters into slurry system - Overflow of water causing flooding in the system					
- If humidified N2 is used, N2 humidifier shall be constructed with transparent acrylic material. Proper mist separator has to be in place to avoid any water droplets been carried away into humidified N2 stream.					For Additive chemistries N2 humidifier requirement has to be verified
- If humidified N2 is used, UPW water shall be used for humidification.					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- N2 blanketing pressure and flow shall be such that bulging or collapsing of tanks/drums/tote should not take place.					
- Humidifier unit for each Blend/Dispense Engine/Station					
Tank Requirements					
- Day tanks to be designed to retain 1.2x the 12hour average daily consumption as a minimum.					
- Humidified Nitrogen Blanket nozzle					
- Slurry inlet/outlet nozzles					
- Slurry recirculation nozzles					
- Tank rinsing spray ball					
- Tank Cleaning Solution (such as KOH) nozzles					
- Exhaust or venting through SBDU					
- Visual indication of tank volume					
- Based on requirements, Intrusive or Non-Intrusive Digital and Analog sensors shall be in place					
- Discrete Level sensors and analog level transmitters shall be present. Dual Discrete sensor assemblies shall be used to avoid SPOF.					
- Analog load cell shall be in place to accurately measure mass of the content of the tank.					
- Low-Low level of the tank or tank empty shall be in place to make sure of the media content in the tank.					
- Overfill sensor shall be in place					
- Level sensors and level transmitter with hard-wired communication for overfill and empty levels					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Low-Low sensor or Tank empty sensor shall be placed that no vortex formation taking place.					
- Tank Isolation Valve shall be present					
- Drain valve shall be in place and there shall not be any stagnation at the drain valve port which may cause contamination after a period of time.					
- Appropriate clear PVC enclosures for day tank top and bottom fixtures/tank entry points.					
- Slurry compatible seals and gaskets					
- Spare nozzles for future expansion					
HMI					
- HMI has to be user friendly and terminologies used on HMI shall be common to Micron operators.					Unnecessary and unfamiliar jargons to be avoided
- Availability of removable SD card/memory card port that program need not to be downloaded during HMI replacement					
- Power supply shall be 24 VDC					Refer Electrical and I&C specification for further information
- Minimum 12" or 15" Colour Display					
- Manual intervention and Control of the system shall be only through local HMI unless otherwise specified					
- Levels for Password - Administrator - Engineer - Supervisor - Operator					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Modes of Operation for HMI <u>Critical Actions</u> - "Pause", "Manual" "Stop", "Abort batch", "auto correction", "Offline" and "Disable" shall have pop-up confirmation button at HMI for re-confirmation of action and shall have pre-warning alarm at HMI/SCADA. <u>Normal Actions</u> - "Resume", "Auto" and "Online" shall have pop-up confirmation button at HMI for re-confirmation of action and shall have Event log at HMI/SCADA.					
When Auto Mode is selected, system shall continuously run and supply media to manufacturing tools according to what is stated in Automatic Sequence of Operation					
- Mode of Operation for Different Engines - Stop - Pause - Resume					
- Colour Code in HMI / SCADA - Green - Open - Red - Close - Colour code shall be used for chemical flow path as well.					In Japan & Taiwan: Green - Close Red - Open
- Cycle count and (or) Hours of operation shall be available - Diaphragm valves - Diaphragm pumps - Centrifugal pumps - Filters - Dampener					
The HMI shall be capable of providing access to all operational and alarm parameters.					
Parameters and Alarm Configuration settings in HMI					
User configurable parameters and alarms to be provided for review and approval					
Description of each parameter and alarm to be provided for review and approval					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Significance of each parameter and alarm to be provided for review and approval					
- All the hard-coded timer information to be provided					
- Visual appearance of all HMI Pages need approval from Owner/Purchaser					
PLC and Associated Cards					
- Requirements and Brand of PLC shall be specified based on control architecture requirements					
- Availability of removable SD card/memory card port that program need not to be downloaded during PLC replacement					
- Available of Dual PLC or solution for Dual PLC					
- Communication cards shall have dual communication ports					
- When Separate engines are used Dual Network Ring concept has to be in place that each engine shall have dual communication					
- 25% spares for all I/O					
SCADA					
- Event log and active alarm page to be separate					
- All Alarms (Shut down, Critical, warning and process) shall appear in HMI and SCADA.					
- Prioritizing of alarms and their colour coding shall be available					
- OEM need to ensure what is alarm and what is event					
- Page Displays					
- Trends and change of time along with trends					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Parameters such as Density, Conductivity, concentration, temperature, pressure, weight, tank levels, Flowrates, engine online status, particle count, etc. shall be monitored and trends shall be available.					
- Visual appearance of all SCADA Pages need approval from Owner/Purchaser					
Engine / Station					
- Engine Redundancy shall be observed					
- If an engine or any redundant component in the system is stopped for maintenance activities, automatic mode of operation should ensure seamless operation using other engines or other respective components in the system.					
- Communication cards, I/Os, Pneumatic cluster, terminal blocks, etc. shall be dedicated for each engine					
Pumps					
- Pump specification and brands to be approved by Micron					
- Pump configuration (parallel or series) to be approved by Micron. If pumps are used in series, proper bypass valves shall be present to ensure the good operation of the engine.					
- Any malfunction of the pump and associated components to be detected by some means and to be communicated.					
- Pump or motor temperature shall not rise above certain value. Temperature can also be monitored by media temperature.					
- Chemical transfer pumps from drums/totes to the day tank shall be capable to flow 1.5 times the daily average consumption rate, as a minimum.					
- NPSH requirement to be met including suction pipeline sizing					
Pneumatic Pump					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Diaphragm or Bellow Pumps shall be used for media Transfer					Diaphragm / Bellow pumps effectiveness on heel transfer is high
- Diaphragm / Bellow pumps need dampener					
- Non-Stall diaphragm/Bellow pumps shall be used					
- Diaphragm / Bellow Rupture Sensor					
Centrifugal Pump - Levitronix					
- Centrifugal Pumps shall be used for mixing and dispense					
- Temperature sensor at pump outlet					
- Shall be VFD Driven					
- Need to meet local power supply					
- Installation angle to be observed					
- CDA shall be used for cooling if such provision is given by the pump manufacturer					
Consistent Media Supply at Consistent Pressure					
- Consistent media supply shall be present for the manufacturing tools.					
- Consistent media supply shall be present for the SBDUs/STUs					
- Pump switch over, ramp-up ramp-down or engine switch over to take place appropriately					
- Readiness of stand-by tank or stand-by engine is critical					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Tank switch over or engine switch over to take place appropriately					
- Consistent pressure in the loop is critical for manufacturing tools. Feedback control system shall be in place to maintain the loop pressure with a desired bandwidth.					
- Pressure transducer shall be placed at pump outlet, mid loop and end loop					
- Automatic back-pressure regulation shall be on the return side of the Distribution Piping Loop for each engine/station.					
- Requirement for Dual Global Loop					
Filter					
- Filter specification to be followed for each chemistry					Refer Filter Specification Section
- Filter choking has to be monitored through appropriate parameter and when filter choking is realized, it has to be communicated to PLC and displayed in HMI & SCADA					
- Filter choking shall be monitored using online differential pressure reading ($P1 - P2 = \Delta P$).					
- Filter changing provision even when the system is supplying media to manufacturing tools to be provided. Filter cartridges of Redundant or idling filter banks should able to be changed based on necessity.					
- Area of filtration shall be maximized					
- Auto venting for filters shall be in place					
- Unnecessary chemical/slurry lost to be prevented.					
Electrical, Control and Instrumentation Requirements					
- Remote system stop shall be available in TGCM Room					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- On-line and off-line metrology shall be part of SBDU and contained in a separate compartment based on requirements.					
- "Optical" or Vertical Float" type leak switches					
- Magnehelic/Photohelic gauge on the exhaust connection					
- Barcode reader shall be wireless.					
- VFD failure signals to be communicated to PLC and displayed in HMI & SCADA					
- Cabinet cooling provision for VFDs shall be in place					
- If pump/motor has temperature sensors, information needs to be communicated to PLC. Otherwise, media temperature can be used for this purpose.					
- Auto venting shall be optimized using appropriate sensors and PLC program.					
- Flow meter shall be in place at supply line					
- Flow meter shall be in place at return line					
- IP Requirements for Electrical Panels shall be strictly followed					
- Earthing requirements shall be according to code of practice					
- Earthing for power and instrument earthing shall be separated					
- Quality or quantity measuring instruments have to meet the accuracy requirements in the designed parameter range.					
Programming Concerns					
- Latest version of the required software shall be used unless it is specified.					
- Approved License for PLC/HMI/SCADA programming versions					

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Requirement	STU	SBDU	STBU	SDU	Remarks
- Debounce timer or delay timer has to be defined properly and incorporated into the control actions.					
- All interlocks shall be discussed under SOO or Control Matrix					
- All Alarms shall have explanations in SOO					
Instruments					
- Discrete Sensors for Level detection					
- Load Cells and Transmitters					
- Radar Sensors and Transmitters for Level					
- Flow controller/meter and Transmitters					
- Pressure sensor and Transmitters					
- Concentration Meter and Transmitter					
- Density Meter and Transmitter					
- Conductivity Sensors and Transmitters					

Note:

1. Day tank used between STBU and SDU is considered to be under SDU
2. If the vendor can propose better solution or additional features with respect to the above table, vendor can propose using a deviation form.
3. If changes are requested, Vendors cannot proceed without getting Micron approval.

Appendix 9 – Deviation Form

Site: _____ Name of the Project: _____ Deviation Request Date: _____ Name of Requester: _____ Micron Dept./ External Vendor: _____ Reason: (Technology / Material / Process / schedule / Cost / Space)	Name of the Standard Deviating from: Section(s) of the Standard Deviating from: Deviation Description:																				
Impacted Facilities Systems and Processes/ Workstations:																					
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr style="background-color: #d3d3d3;"> <th style="width: 25%;">Impacted Facilities System</th> <th style="width: 25%;">Impacted Processes/ Workstations (if applicable)</th> <th style="width: 50%;">Impact Description</th> </tr> </thead> <tbody> <tr> <td style="height: 100px;"></td> <td></td> <td></td> </tr> </tbody> </table>		Impacted Facilities System	Impacted Processes/ Workstations (if applicable)	Impact Description																	
Impacted Facilities System	Impacted Processes/ Workstations (if applicable)	Impact Description																			
Justification Summary: (Includes links to data and backup slides as necessary)	Long term direction for permanent deviation:																				
Mitigation plan for short term deviation:	Critical lessons learnt through this deviation if applicable:																				
Approval:																					
<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr style="background-color: #d3d3d3;"> <th style="width: 25%;">Department</th> <th style="width: 25%;">Position</th> <th style="width: 25%;">Name</th> <th style="width: 25%;">Date</th> </tr> </thead> <tbody> <tr> <td>Site Facilities</td> <td>Manager</td> <td></td> <td></td> </tr> <tr> <td>FCT</td> <td>Manager</td> <td></td> <td></td> </tr> <tr> <td>GFTT</td> <td>Manager</td> <td></td> <td></td> </tr> <tr> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>		Department	Position	Name	Date	Site Facilities	Manager			FCT	Manager			GFTT	Manager						
Department	Position	Name	Date																		
Site Facilities	Manager																				
FCT	Manager																				
GFTT	Manager																				

Note: Any deviation from the given standard shall be submitted for approval. If a temporary deviation/exception is requested, the time needed for rectification shall be mentioned in the relevant box(es).